

Research Update: OCR-Derived Egyptian Medicine Search

Meeting 10 through 24 July 2026

Generated 24 July 2026

Abstract

This report preserves the Algorithm 4 and Algorithm 5 studies and evaluates Algorithm 6 as the current evidence-gated successor. The evidence base contains a 53-system comparison plus Algorithm 6, 464 collision-free unique OCR query-target pairs, and 66,257 unique synthetic clean-core pairs, all ranked against the same 17,476-family Egyptian medicine catalog. Algorithm 6 preserves Algorithm 5's 50.4310% OCR Hit@1 and raises Hit@20 from 72.8448% to 73.2759% by recovering two additional candidates with zero paired losses. It preserves 98.1889% Hit@1 and 99.9985% Hit@20 on synthetic data. A proposed rank-1 consensus gate is rejected because its 27 synthetic gains accompany 138 losses. Automated label audit assigns 138 OCR pairs to P0, 56 to P1, and 270 to provisional P2; source-image references are absent, so the historical 93-pair holdout remains retrospective rather than fresh blind evidence. A cross-fitted correctness gate accepts 99 of 211 P2 development pairs and 27 of 59 P2 retrospective holdout pairs with no observed error, but every accepted result still requires user confirmation. The pharmacist study has 1,125 balanced Algorithm 6 assignments and no participant outcomes.

Contents

1	Key Findings	2
2	Experiment Log	3
3	Data and Evaluation Contract	3
3.1	Input, Process, Output	3
3.2	Source Inventory	3
3.3	Data Dictionary	3
3.4	Denominators	7
3.5	Metrics and Statistical Tests	7
4	OCR Dataset Composition	7
5	OCR Error Severity and Evidence	10
6	Experiment 1: Retrieval-System Comparison	13
7	Algorithm 4 Evaluation	23
8	Experiment 2: Algorithm 4 Ablation	23
9	Equal Edit-Distance Ordering	27
10	Synthetic 115,000-Case Benchmark	30

11 Expanded GitHub Competitor and Algorithm 5 Benchmark	37
11.1 Scope and Counting Rules	37
11.2 Full Leaderboard: 464 fair OCR query-target pairs	40
11.3 Full Leaderboard: 66,257 synthetic clean-core query-target pairs	42
11.4 System Distribution and Efficiency Frontier	44
11.5 Algorithm 5 Source-Revision Audit	46
11.6 Algorithm 5 Failure Taxonomy	57
11.7 External Composite Ablations and Controlled Variants	63
11.8 Algorithm 5 Ablation	71
11.9 The 0.60 Extreme OCR Cohort	75
11.10 Remaining Algorithm 5 Failures and Complementarity	78
11.11 Complete External Repository Disposition	83
11.12 Reproducible Artifacts	86
12 Algorithm 6: Evidence-Gated Consensus Retrieval	86
12.1 Input, Process, and Output	86
12.2 Label Audit and Evaluation Contract	86
12.3 Candidate Sources	86
12.4 Paired Performance and Deployment Decision	87
12.5 End-to-End Nested Component Ablation	87
12.6 Candidate-Source Ablation	87
12.7 Learned OCR Confusion Evidence	87
12.8 Learned Reranker Ablation	94
12.9 Selective Correctness Gate	94
12.10 Remaining Failures and Runtime Fidelity	94
13 Experiment 7: Pharmacist Study Status	97
14 Failures and Dead Ends	98
15 Open Questions and Next Steps	99
16 Analysis Coverage and Reproducibility	99

1 Key Findings

- **Algorithm 6 is the current system at 50.4310% Hit@1 and 73.2759% Hit@20 on 464 OCR pairs.** It preserves every Algorithm 5 rank-1 outcome and adds two top-20 recoveries with zero paired losses.
- **The broad rank-1 consensus rule was not deployed.** It gains three and loses zero OCR rank-1 cases, but gains 27 and loses 138 synthetic rank-1 cases; the apparent OCR gain does not transfer.
- **Algorithm 5 remains the strongest base ranker in the 53-system comparison.** Its 50.4310% Hit@1 and 72.8448% Hit@20 improve frozen Algorithm 4 by 2.8017 and 1.7241 percentage points on the same 464 pairs.
- **No external method closes the observed-OCR gap.** The best external Hit@1 is 38.5776%, while the best external Hit@20 is 67.4569%; paired case switches and Holm-adjusted McNemar tests identify whether each gap is consistent across queries.

- **Synthetic ranking does not transfer directly to OCR.** Algorithm 5 reaches 98.1889% Hit@1 and 99.9985% Hit@20 on 66,257 clean-core pairs, while edit-distance systems approach that synthetic ceiling but remain materially weaker on the OCR pairs.
- **The current source differs from the historical clean-core artifact.** Forty-four pairs gain rank 1, 55 lose rank 1, and 187 ranks change; the current result is lower by 11 Hit@1 cases and has one top-20 miss, `aclox`→`ADOX`.
- **Candidate rescue, not one tie rule, supplies the main gain.** Removing Algorithm 5 family rescue lowers OCR Hit@1 by 13.7931 points and Hit@20 by 20.0431 points; removing all reranking has a much larger Hit@1 effect than Hit@20 effect.
- **The remaining failures separate into demonstrated and unsupported fixes.** Algorithm 6 reduces Algorithm 5’s 126 OCR top-20 misses to 124, but 116 remaining misses are P0 label/evidence conflicts and eight are P1; no P2 row is a top-20 miss under the audit rule.
- **Label reliability limits the next accuracy claim.** No source image is linked to any of the 464 OCR pairs, and both new top-20 recoveries are P0/P1 cases. Candidate recovery is measured; target validity is not yet adjudicated.
- **The 0.60 extreme cohort remains a low-evidence stress test.** It contains 113 fair OCR pairs after deduplication and collision removal; complete per-system results, operation distributions, and concrete rows are reported rather than discarded.
- **Human-study claims are not available.** The counterbalanced protocol contains 15 participants, 75 crops, and 1,125 assignments, but accuracy, decision time, and safety cannot be reported until real responses and adjudicated labels exist.

2 Experiment Log

The log separates completed measurements from prepared work so design artifacts cannot be mistaken for experimental outcomes.

3 Data and Evaluation Contract

This section fixes the inputs, units, labels, and denominators before comparing systems.

3.1 Input, Process, Output

3.2 Source Inventory

3.3 Data Dictionary

Raw names appear here once; later sections use human labels. The generated companion dictionary contains 356 dataset-field entries, including fields not needed in the report body.

Compact form lowercases text and removes non-alphanumeric characters. For `CLEX A NE` → `CLEXANE`, source alignment counts two spaces as edits, while both compact strings are `cllexane` and compact distance is zero. With compact query q_c , target y_c , and Levenshtein distance L ,

$$d = L(q_c, y_c), \quad r = \frac{d}{\max(|y_c|, 1)}.$$

An extreme prediction has $r > 0.60$. For `GHONT` → `CALAMINE`, $d = 7$, $|y_c| = 8$, and $r = 0.875$. The 0.60 threshold is an operational review band, not a probability or clinical-risk estimate.

Experiment	Evaluation unit	Cases	Primary output	Status
OCR source audit	One model prediction for one verified family	595	Error severity, operation mix, evidence, source-model case mix	Complete
Algorithm 6 label audit	One primary OCR query-target pair	464	P0/P1/P2 priority, nearest-catalog conflict, union coverage, split contract	Automated; human image review pending
Primary OCR search	One collision-free unique compact query-target pair	464	Hit@1, Hit@20, MRR@20, safety, latency	Complete
Retrieval comparison	Same primary pair under each system	4,640	Ten-system metrics, mistake profiles, examples, and paired effects	Complete
Expanded OCR comparison	Same fair pair under each of 53 systems	24,592	Full leaderboard, paired tests, all-system error slices, latency, and failures	Complete
Expanded clean-core comparison	Same synthetic pair under each of 53 systems	3,511,621	Full leaderboard, operation and distance trends, latency, and failures	Complete
Algorithm 4 ablation	Same primary pair under each configuration	11,136	Complete system plus 23 one-component removals	Complete
Algorithm 5 OCR ablation	Same fair pair under each configuration	45,008	Complete system plus 96 grouped and atomic removals	Complete
Algorithm 5 synthetic ablation	Same clean-core pair under each configuration	861,341	Complete system plus 12 broad component removals	Complete
Algorithm 6 paired successor	Same OCR and clean-core pair under Algorithm 5 and Algorithm 6	66,721	Paired top-k switches, source ablations, runtime parity, remaining failures	Complete
Algorithm 6 learned evidence	Target-family cross-fit development plus retrospective holdout	464	OCR confusion costs, LightGBM ranker ablations, logistic selective gate	Complete; learned ranker advisory only
Equal-distance policy test	Same primary pair under each policy	3,712	Eight all/development/holdout comparisons	Complete
Synthetic benchmark	One deterministic generated search case	115,000	Four algorithms, two denominators, 34 categories, six mistake mechanisms	Complete
Enhancement regression	Locked manual, OCR, browser, sample, and full suites	122,109	Before/after retrieval and paired switches where stored	Complete
Pharmacist study	Participant-case-condition decision	1,125 assigned	No tool, DrugEye, Algorithm 6	Prepared and validated, not executed

Table 1: All source tables lack run timestamps. “Meeting 10 through 24 July 2026” therefore denotes the artifact lineage and regeneration date, not a row-level time interval.

Stage	Concrete example	Meaning
Input	OCR text MYMLAX	Text already produced by an OCR model; no image enters this search benchmark.
Process	Search 25,066 catalog medicines	Generate candidates, group commercial variants by family, rank evidence, and apply clarification gates.
Output	MYOLAX at rank 1	Ranked commercial family plus response state and evidence.
Hidden label	Verified family MYOLAX	Used only after retrieval to calculate metrics.
Score	Hit@1 = 1; Hit@20 = 1	The verified family appears first and therefore also within the first 20.

Table 2: The verified target never enters candidate generation or ranking. The benchmark measures search recovery after OCR, not prescription segmentation or OCR recognition accuracy.

Human-readable source	Rows	Fields	Role
OCR predictions	595	15	OCR text, verified target, source alignment, and model identity.
OCR search cases	595	43	Fair-score labels, evidence counts, severity cohorts, and split.
Latest OCR Algorithm 4 results	595	44	Ranked families, decision state, safety, and latency.
Retrieval and ablation case results	20,230	29	Ten retrieval systems and 24 Algorithm 4 configurations.
Retrieval aggregate metrics	612	18	Metrics by denominator, split, cohort, distance, and mistake type.
Retrieval error profiles	260	10	Ten systems by distance, operation, evidence, length, cohort, and mistake type.
Retrieval paired comparisons	32	12	Exact gains, losses, and McNemar tests.
Synthetic test cases	115,000	18	Deterministic generated inputs with category and provenance.
Synthetic Algorithms 1–4	460,000	29–39	Four aligned row-level outcome tables.
Synthetic category metrics	156	16	Four algorithms over all scopes and 34 categories.
Equal-distance evidence	595	43	Candidate evidence, first choice, and expected rank under eight policies.
Pharmacist assignments	1,125	5	Prepared participant-case-condition schedule, no responses.

Table 3: Every source file was read in full. One-to-one keys reconcile the OCR tables, and all four synthetic result tables share the same 115,000 source rows.

Raw field	Human meaning
edited_name	OCR-predicted medicine text supplied to search.
matched_canonical_name_norm	Human-verified commercial medicine family.
edit_distance_count	Source alignment edits, which can retain spaces.
edit_distance	Compact Levenshtein edits after case, punctuation, and spaces are normalized.
normalized_edit_distance	Compact edit distance divided by compact target length.
shared_ngram_count	Count of adjacent-character pairs shared by compact query and target.
analysis_cohort	Mutually exclusive severity or ambiguity cohort.
scored_case	Whether a row belongs to the fair single-answer denominator.
hit_at_1, hit_at_20	Verified-family recovery by rank 1 or rank 20.
unsafe_confident_top1	Wrong first family presented confidently without clarification.
latency_ms	Warm query time in milliseconds, excluding index preparation.
mistake_type	Search failure mechanism, independent of mutation category.

Table 4: The complete generated dictionary prevents raw implementation names from leaking into figure titles, axes, legends, or analytical claims.

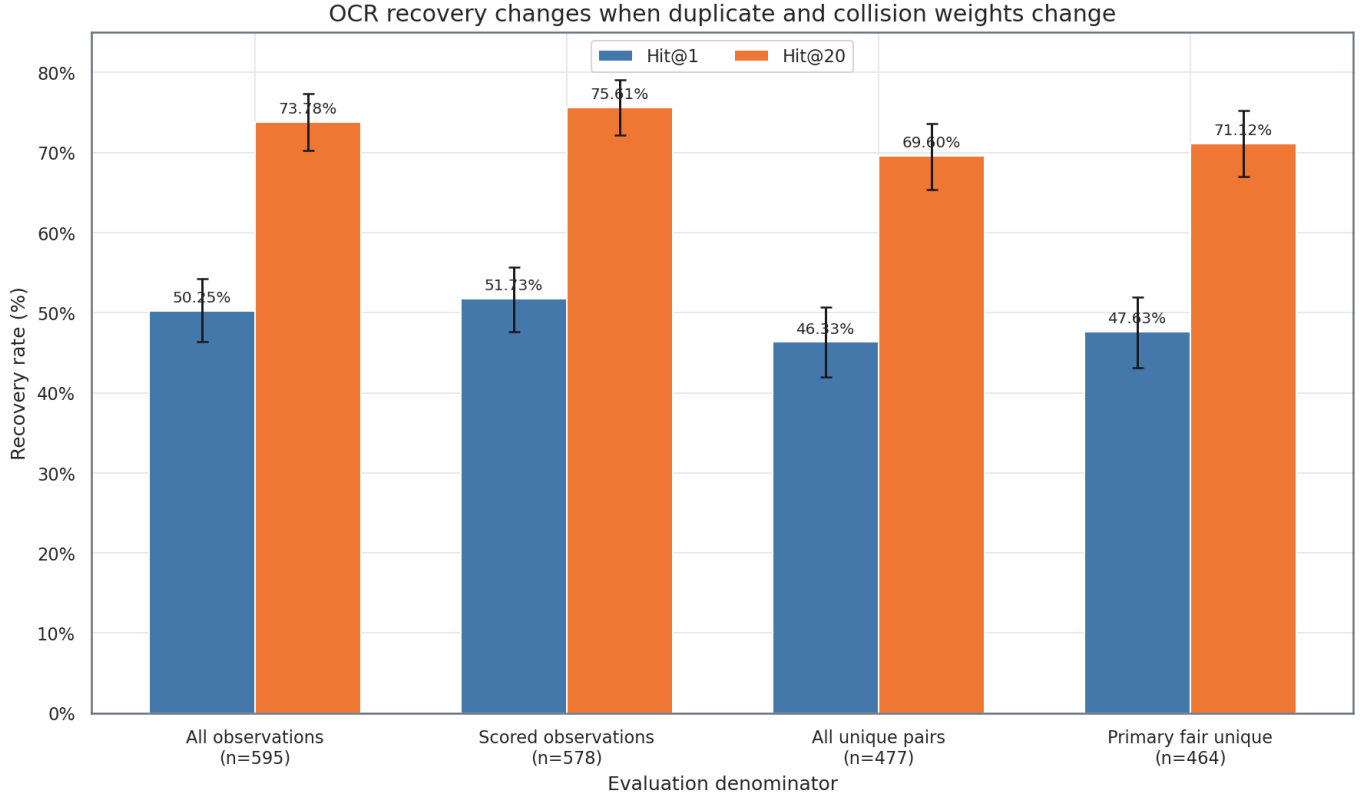


Figure 1: Each row is one denominator; the x-axis reports Hit@1 and Hit@20. Inclusive observations count every OCR-model output, scored observations remove 17 real-name collisions, unique pairs remove repeated compact query-target pairs, and the primary denominator applies both rules.

Insights

1. The number to cite for search comparison is 47.6293% Hit@1 and 71.1207% Hit@20 over 464 primary pairs. (Each distinct fair query-target pair receives one vote.)
2. The inclusive 595-row result is higher, 50.2521% Hit@1 and 73.7815% Hit@20. (Repeated OCR outputs receive repeated votes, so it answers a data-audit question rather than a pair-balanced comparison.)
3. A collision such as RIVOTRIL → RIVO stays in diagnostics but not fair single-answer accuracy. (The entered text is already another exact catalog family.)

3.4 Denominators

3.5 Metrics and Statistical Tests

For query q , $R_k(q)$ is the first k returned families and $Y(q)$ is the verified relevant-family set:

$$\text{Hit@k}(q) = \mathbf{1}[R_k(q) \cap Y(q) \neq \emptyset].$$

- Hit@1 failure with Hit@20 success is a ranking failure; Hit@20 failure is a candidate-retrieval failure.
- MRR@20 uses reciprocal rank for the first relevant result through rank 20 and zero beyond rank 20.
- Paired risk differences use 10,000 bootstrap resamples with seed 20260719. Error bars are 95% percentile intervals.
- Exact two-sided McNemar tests use pairwise gains and losses. Matched odds ratios describe effect direction. Exploratory p-values are not adjusted for multiple comparisons and are not treated as confirmatory evidence.
- Spearman correlation measures monotonic association. It does not assign causality to edit distance, evidence counts, or recovery.

4 OCR Dataset Composition

The 595 observations mix unequal model coverage, target-family-disjoint splits, and error severities, so raw model percentages require case-mix context.

The 595-row OCR benchmark mixes severity, split, and outcome

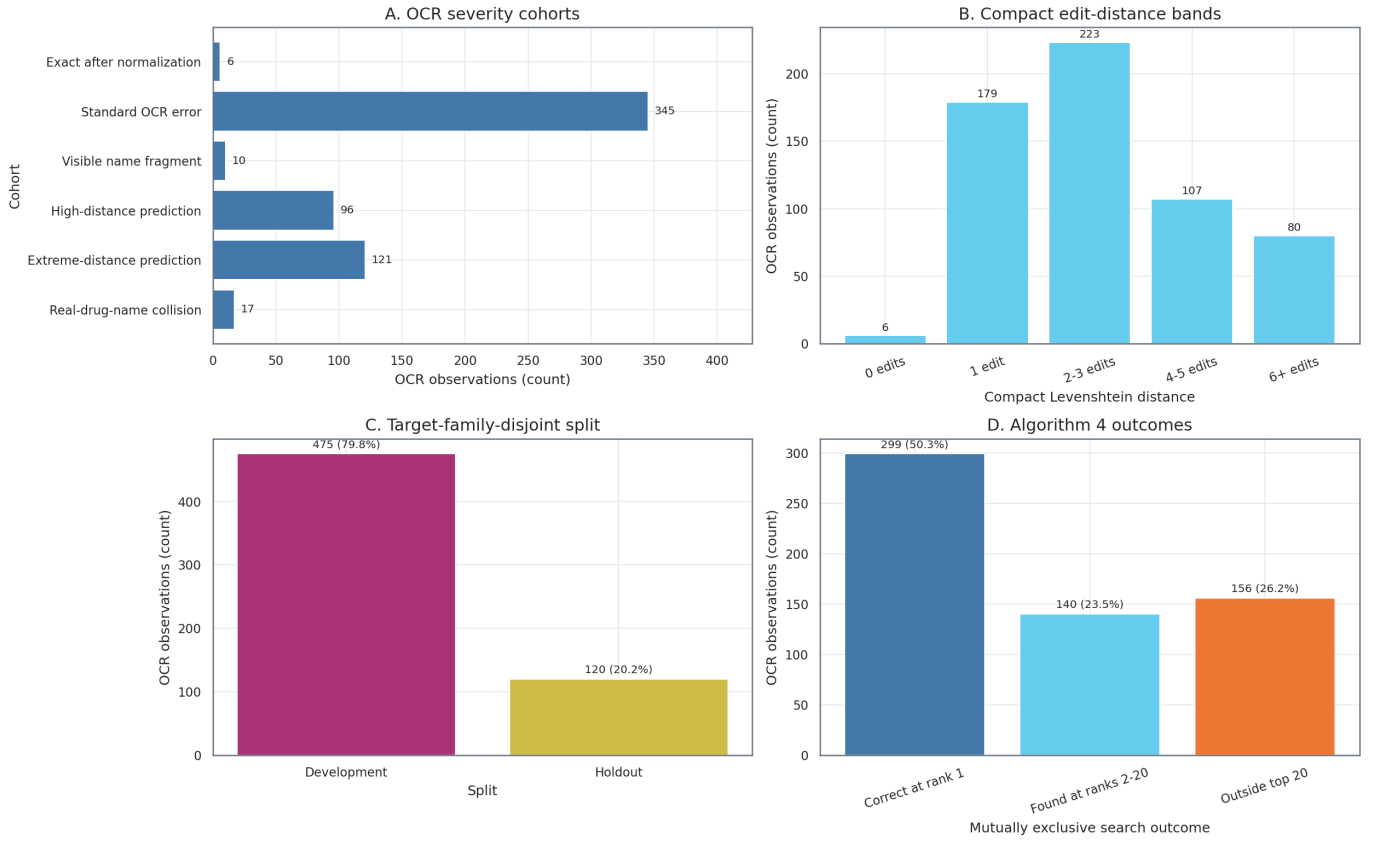


Figure 2: Panel A counts severity cohorts, panel B counts compact edit-distance bands, panel C counts development and holdout observations, and panel D partitions Algorithm 4 outcomes. Every panel uses the same 595-row inclusive denominator.

Insights

1. Extreme predictions account for 121 rows, or 20.34%, while standard OCR errors account for 345, or 57.98%. (The dataset is not dominated by ordinary one-edit spelling errors.)
2. Algorithm 4 places 299 observations at rank 1, 140 at ranks 2–20, and leaves 156 outside the top 20. (The middle group needs reranking; the last group needs broader candidate generation.)
3. The split contains 475 development and 120 holdout observations, with zero target-family overlap. (All KETOROLAC rows are development; all VIGOREX rows are holdout.)

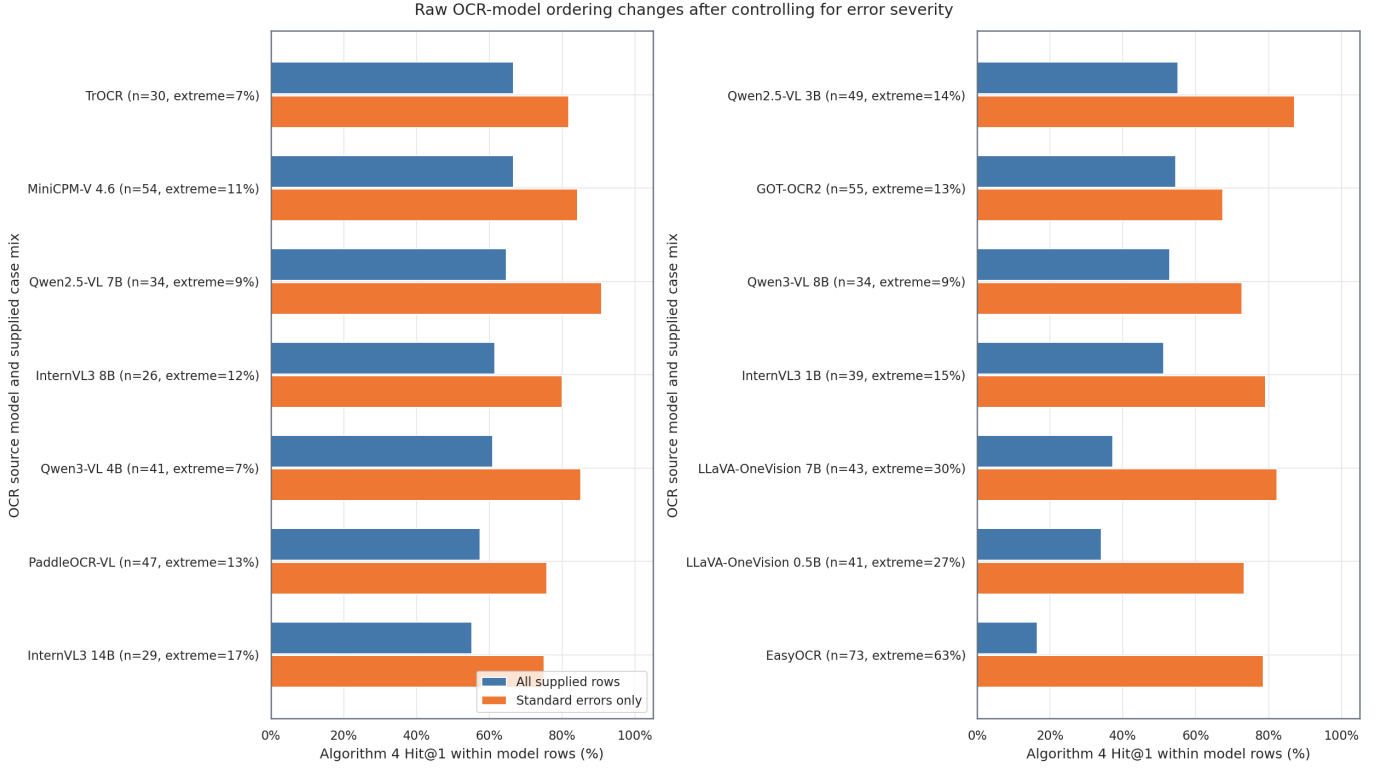


Figure 3: For each OCR source model, blue is Algorithm 4 Hit@1 on all supplied rows and orange is Hit@1 on standard errors only. Labels show model row count and its extreme-row share; models did not receive a common image set.

Insights

1. Raw downstream search rates are not an OCR leaderboard. (The 14 models contribute 26–73 observations with different targets and corruption.)
2. EasyOCR rises from 16.44% raw Hit@1 to 78.57% on its 14 standard rows. (Forty-six of its 73 supplied rows are extreme, so severity explains a large part of the raw score.)
3. A fair OCR comparison requires every model to process the same crops with the same preprocessing and target labels. (This dataset supports search analysis, not matched OCR-model ranking.)

5 OCR Error Severity and Evidence

This section tests which measurable error properties are associated with candidate recovery on the 464 primary pairs.

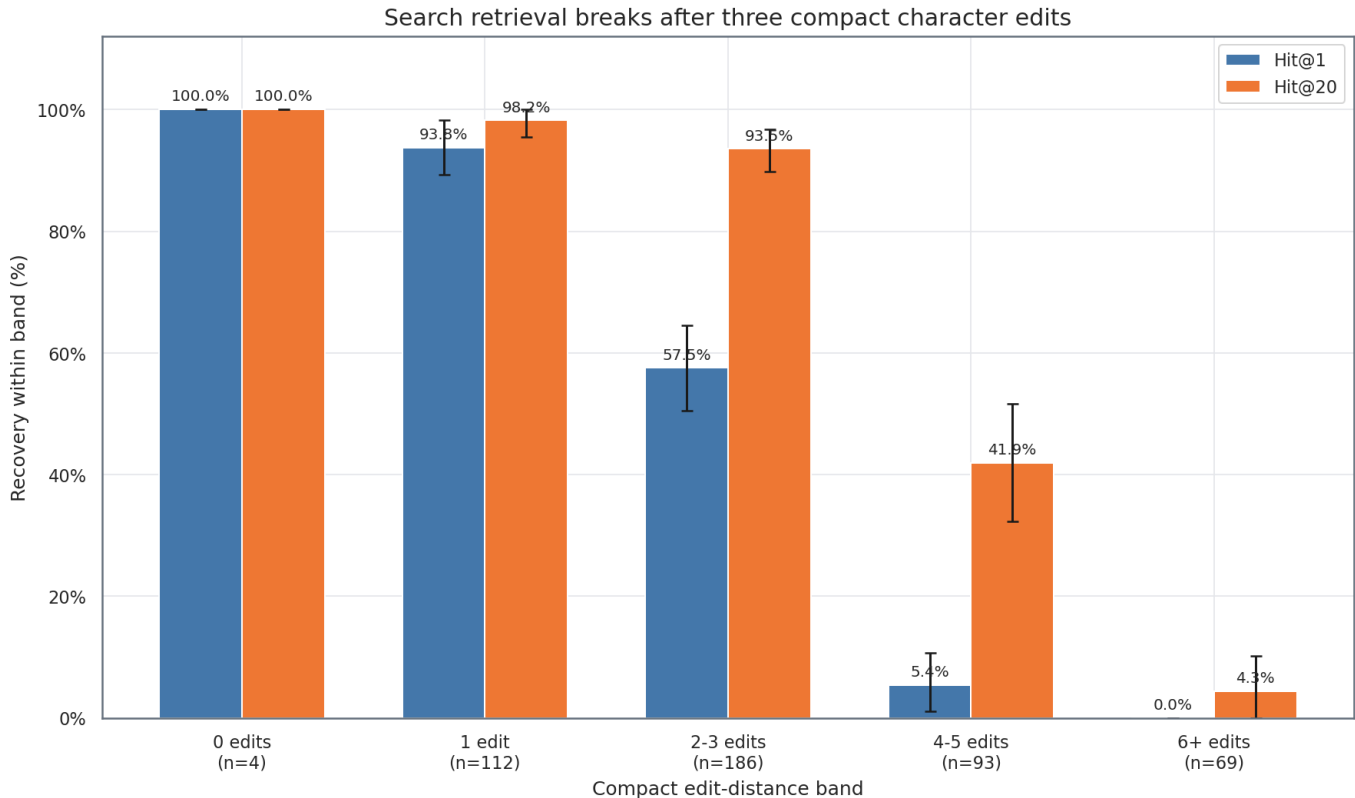


Figure 4: The x-axis groups compact Levenshtein edit counts; blue is Hit@1 and orange is Hit@20 within each group. Error bars are 95% bootstrap intervals and labels print the group denominator.

Insights

1. One-edit queries remain recoverable at 93.75% Hit@1 and 98.21% Hit@20. (The catalog and current signals handle ordinary OCR corruption.)
2. Four-to-five edits fall to 5.38% Hit@1 and 41.94% Hit@20; six or more fall to 0% and 4.35%. (Distance four is the observed retrieval break.)
3. Distance four or above creates 120 of 134 primary top-20 misses, 89.55%. (Work on severe candidate generation has greater reach than tuning already-correct one-edit cases.)

OCR input	Verified family	A4 first family	Rank	Failure interpretation
MYMLAX	MYOLAX	MYOLAX	1	One replacement; complete recovery.
KELOLAC	KETOROLAC	KETOLAC	2	Candidate exists; ranking error.
RIV	RIVOTRIL	RIVO	10	Visible fragment retains weak evidence.
GHONT	CALAMINE	GEOCAND	> 20	Seven edits remove target evidence.
H	RIVO	None	> 20	One character cannot identify a family.
RIVOTRIL	RIVO	RIVOTRIL	> 20	Exact different real-name collision.

Table 5: Worst rows are not one homogeneous error class. Ranking, retrieval, insufficient evidence, and label ambiguity require different product behavior.

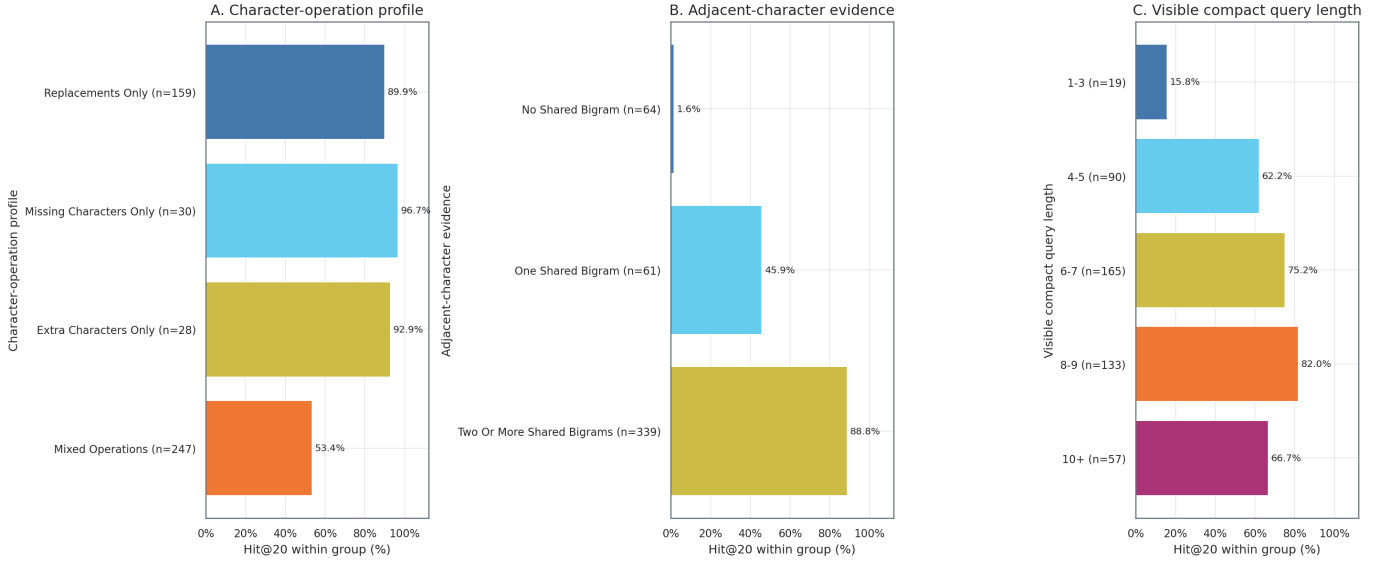


Figure 5: All panels report Hit@20 within the printed group denominator. Panel A groups source alignment operations, panel B groups shared compact bigrams, and panel C groups visible compact query length.

Insights

1. Mixed insertion, deletion, and replacement rows reach 53.44% Hit@20 and create 115 misses. (They account for 85.82% of all primary misses.)
2. No shared bigram reaches 1.56% Hit@20, one bigram 45.90%, and two or more 88.79%. (Adjacent-character evidence separates recoverable and unrecoverable queries more sharply than length alone.)
3. One-to-three-character inputs have the worst rate, 15.79% Hit@20, but six-to-seven-character inputs create more misses, 41, because that group is larger. (Failure rate and failure count answer different questions.)

Extreme >0.60 predictions are mixed, low-bigram, and usually outside top 20

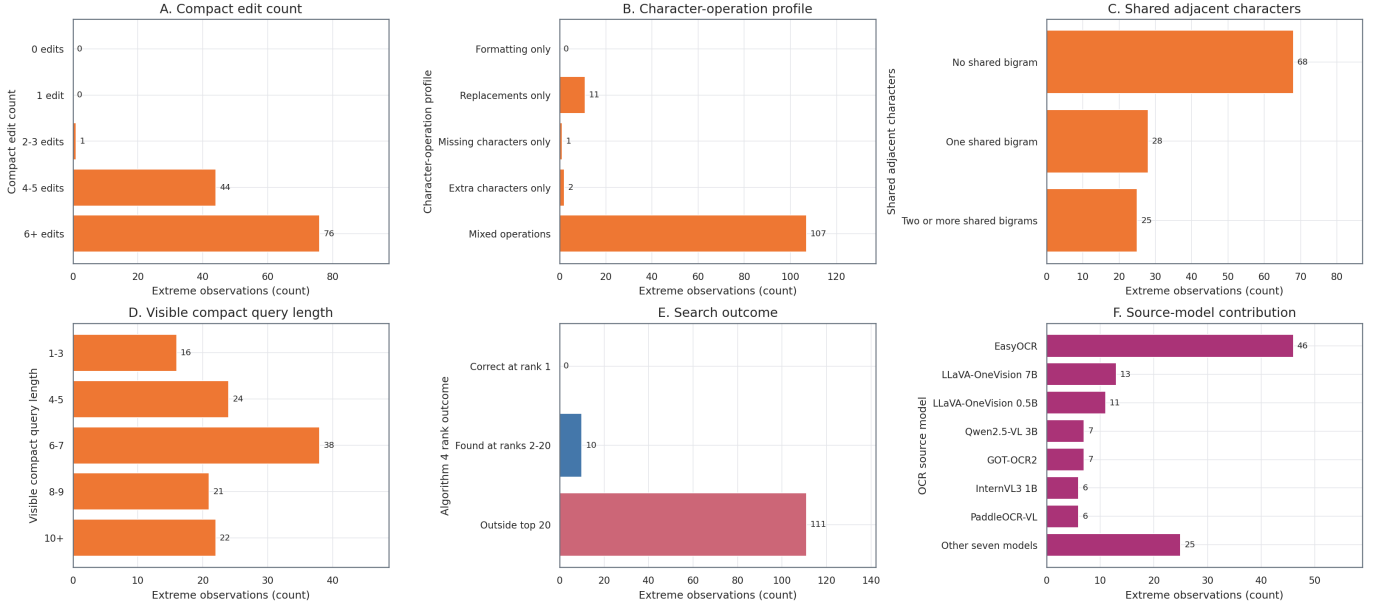


Figure 6: Each panel counts the 121 extreme observations, defined as compact distance divided by target length above 0.60 after collision and exact-match priority. Panels cover edit count, operation profile, shared bigrams, and source model.

Insights

1. The extreme cohort contains 76 six-plus-edit rows and 44 four-to-five edit rows. (It is a severe tail, not an extension of one-edit typos.)
2. Mixed operations account for 107 of 121 extreme rows, and 96 have at most one shared bigram. (Most rows lose both simple edit and local sequence evidence.)
3. Zero extreme rows reach rank 1, 10 reach ranks 2–20, and 111 miss the top 20. (Ordinary lexical retrieval rarely survives this evidence level.)

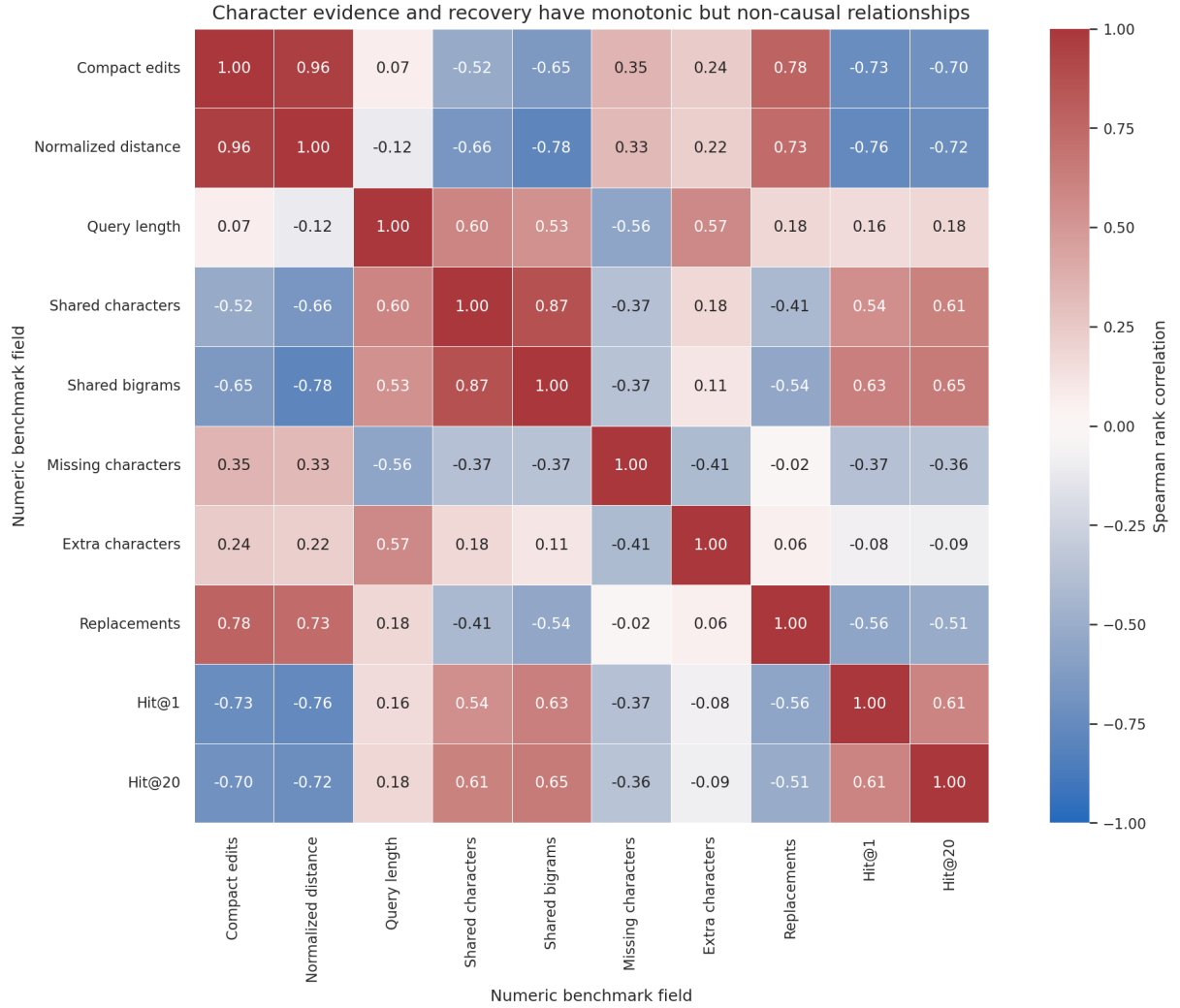


Figure 7: Cells are Spearman rank correlations over 464 primary pairs. Positive values mean two numeric fields tend to rise together; negative values mean one tends to fall as the other rises. Correlation does not prove causation.

Insights

1. Normalized distance has the strongest negative association with Hit@1, $\rho = -0.7592$, and Hit@20, $\rho = -0.7202$. (Larger corruption tracks lower recovery across the observed pairs.)
2. Shared bigrams correlate positively with Hit@1, $\rho = 0.6279$, and Hit@20, $\rho = 0.6520$. (Local sequence evidence is a useful candidate expansion and confidence feature.)
3. Replacement count correlates with Hit@20 at $\rho = -0.5100$. (The association includes overlap with total distance and mixed errors, so it is not an isolated causal effect.)

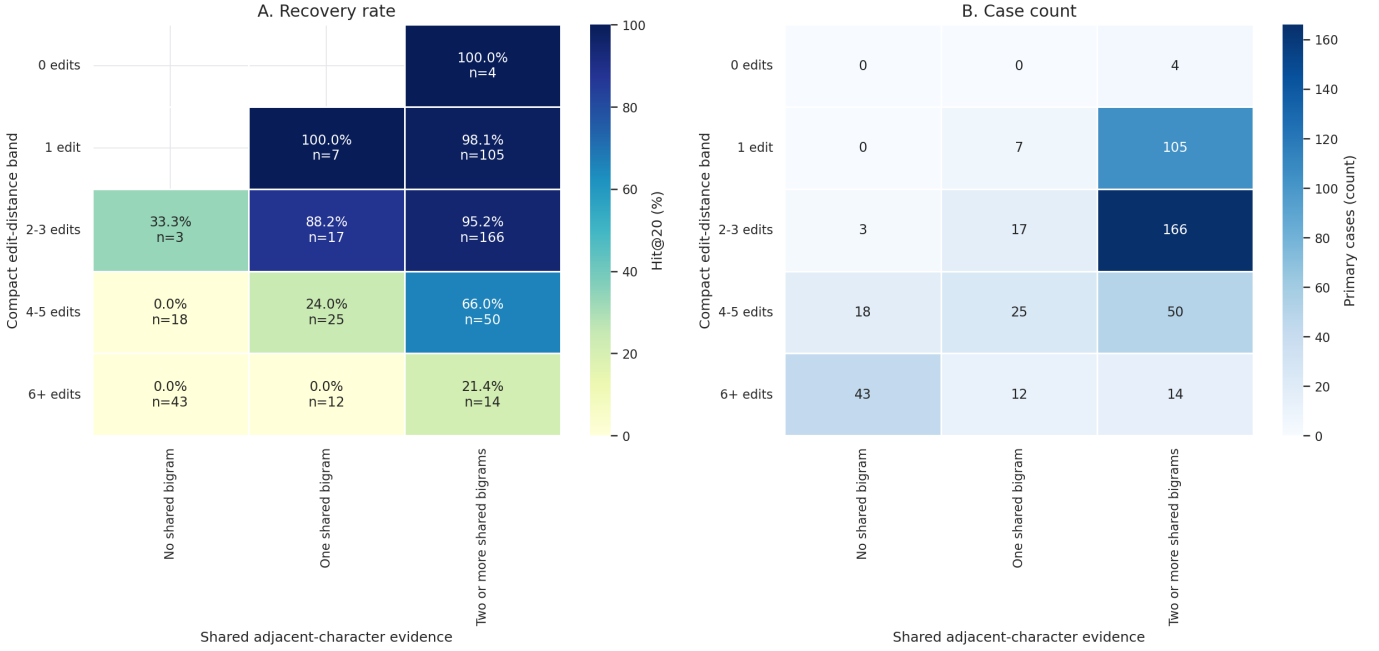


Figure 8: Rows are compact edit-distance bands and columns are shared-bigram bands. The left heatmap reports Hit@20 and case count in each cell; the right heatmap reports the corresponding denominator.

Insights

1. At four-to-five edits, Hit@20 is 0% with no shared bigram, 24% with one, and 66% with two or more. (Bigram evidence materially separates cases inside the same edit band.)
2. At six or more edits, even two-plus bigrams reach only 21.4% Hit@20. (Bigrams moderate bounded corruption but cannot replace missing lexical evidence in the severe tail.)

6 Experiment 1: Retrieval-System Comparison

This section preserves the original ten-system Algorithm 4 comparison. The expanded 53-system Algorithm 5 study later in the report supersedes it for the current headline result while retaining these rows as historical evidence.

Ten systems receive the same 464 primary pairs, so differences are paired by query rather than compared across unrelated case sets.

System	Hit@1	Hit@5	Hit@20	MRR@20	Median ms
Algorithm 4, family rescue	47.6293%	61.8534%	71.1207%	0.537476	14.0704
Exhaustive Levenshtein	34.2672%	50.8621%	59.2672%	0.416955	0.7039
Jaro-Winkler	32.1121%	55.1724%	67.4569%	0.425369	0.6713
RapidFuzz token ratio	31.4655%	48.9224%	59.0517%	0.394378	8.0152
Algorithm 2, external fast	25.8621%	43.7500%	51.2931%	0.329160	3.8593
Algorithm 3, rank fusion	25.6466%	40.9483%	53.0172%	0.326419	6.1017
Algorithm 1, current app	23.7069%	31.4655%	38.1466%	0.272602	0.7372
Character 3-gram TF-IDF	18.3190%	35.7759%	47.4138%	0.257995	0.5042
Phonetic baseline	6.8966%	16.8103%	23.2759%	0.110216	0.4722
Exact or prefix match	2.5862%	2.8017%	3.0172%	0.027155	0.7060

Table 6: Best retrieval values are bold. Query latency excludes preparation; Algorithm 4 prepares its combined index in 8,731.705 ms before warm queries.

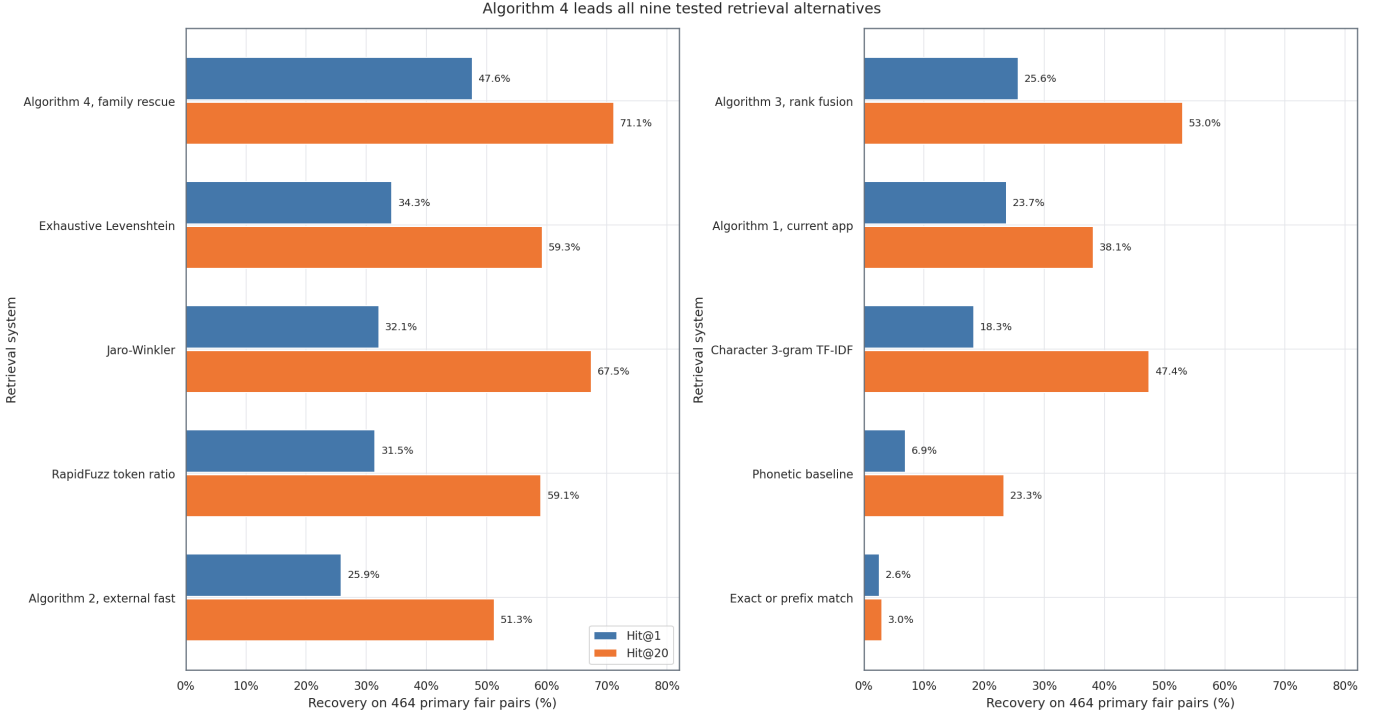


Figure 9: Blue bars are Hit@1 and orange bars Hit@20 over the same 464 primary pairs. Systems are split into two panels only to preserve readable labels; the x-axis scale is shared.

Insights

1. Algorithm 4 leads all nine alternatives at both cutoffs. (Its 47.6293% Hit@1 is 13.3621 points above exhaustive Levenshtein.)
2. Jaro-Winkler is the strongest classical Hit@20 baseline at 67.4569%. (Algorithm 4 adds 3.6638 points, so its main advantage over Jaro-Winkler is not unlimited candidate recall.)
3. Exact/prefix and phonetic-only systems fail on this OCR error mix. (Their Hit@20 values are 3.0172% and 23.2759%.)

System	Overall	1 edit	2–3	4–5	6+	Mixed
Exact/prefix	3.0%	6.2%	1.1%	1.1%	0.0%	0.0%
Levenshtein	59.3%	85.7%	82.8%	23.7%	0.0%	40.9%
Jaro-Winkler	67.5%	100.0%	86.0%	32.3%	10.1%	49.8%
Character 3-gram TF-IDF	47.4%	79.5%	57.0%	20.4%	2.9%	34.4%
RapidFuzz token ratio	59.1%	92.0%	80.6%	19.4%	1.4%	42.5%
Phonetic baseline	23.3%	56.2%	19.4%	6.5%	0.0%	12.1%
Algorithm 1	38.1%	83.0%	40.9%	4.3%	0.0%	16.2%
Algorithm 2	51.3%	96.4%	59.1%	17.2%	0.0%	31.2%
Algorithm 3	53.0%	95.5%	62.9%	18.3%	1.4%	32.4%
Algorithm 4	71.1%	98.2%	93.5%	41.9%	4.3%	53.4%

Table 7: Hit@20 for Experiment 1 by compact edit count and OCR operation shape. Overall uses 464 primary pairs; the one-edit, two-to-three, four-to-five, six-plus, and mixed-operation groups contain 112, 186, 93, 69, and 247 pairs. Bold marks the best result in each column.

Experiment 1: every retrieval method degrades as character evidence disappears

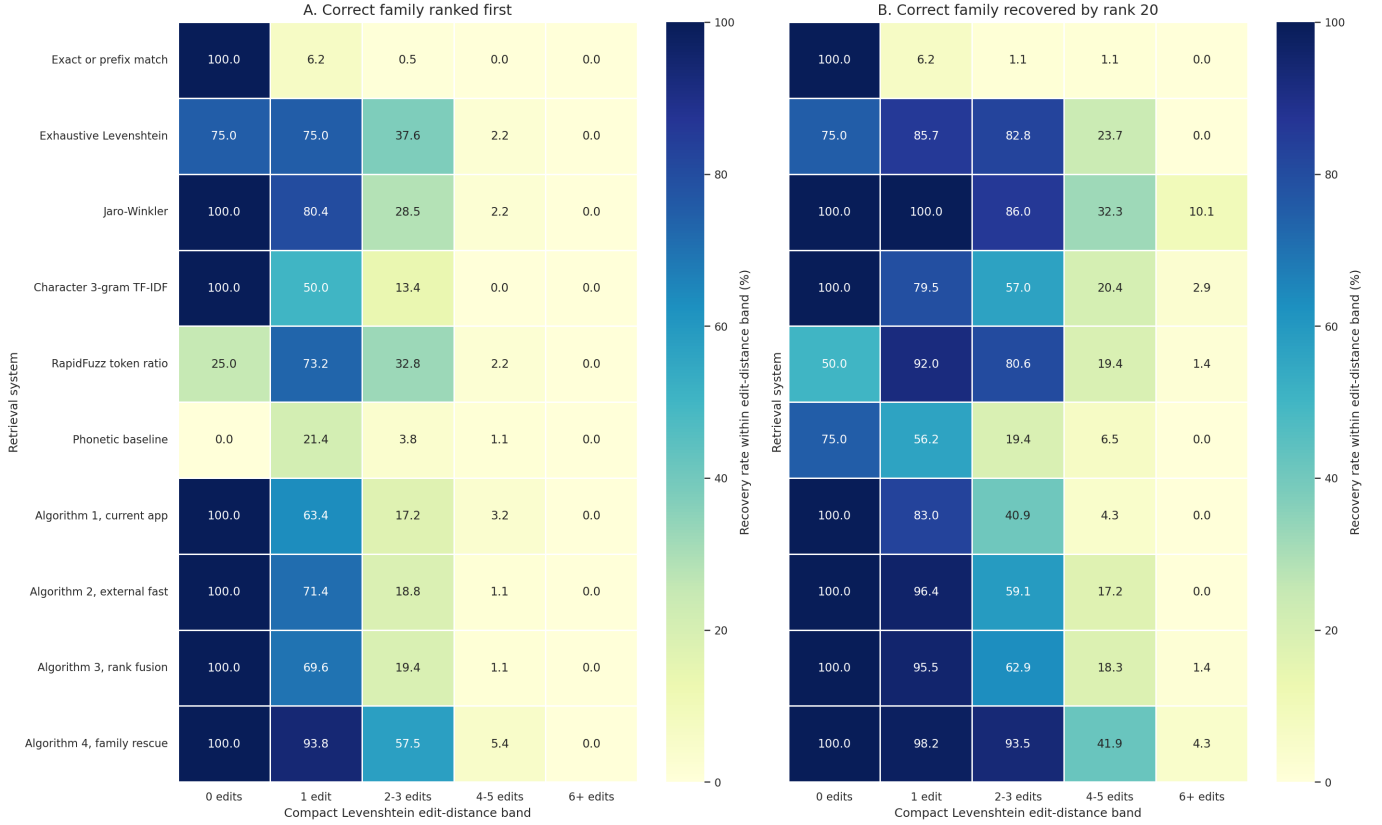


Figure 10: Rows are the ten Experiment 1 systems and columns are compact Levenshtein bands. Panel A reports Hit@1 and panel B Hit@20 within each band; all systems use the same 464 primary pairs.

Insights

1. Algorithm 4 creates its main advantage at bounded corruption: 93.5% Hit@20 for two-to-three edits and 41.9% for four-to-five. (The next values are Jaro-Winkler at 86.0% and 32.3%.)
2. Jaro-Winkler is the only method with 100% Hit@20 on all 112 one-edit pairs and the strongest six-plus result at 10.1%. (Its prefix-weighted similarity is broad enough to retain candidates but does not place them first as often as Algorithm 4.)
3. No method solves severe corruption. (At six-plus edits every Hit@1 is zero; the best Hit@20 is 10.1%, so this band needs more image or user evidence rather than another small rank bonus.)

Mixed edits and missing bigrams expose different baseline assumptions

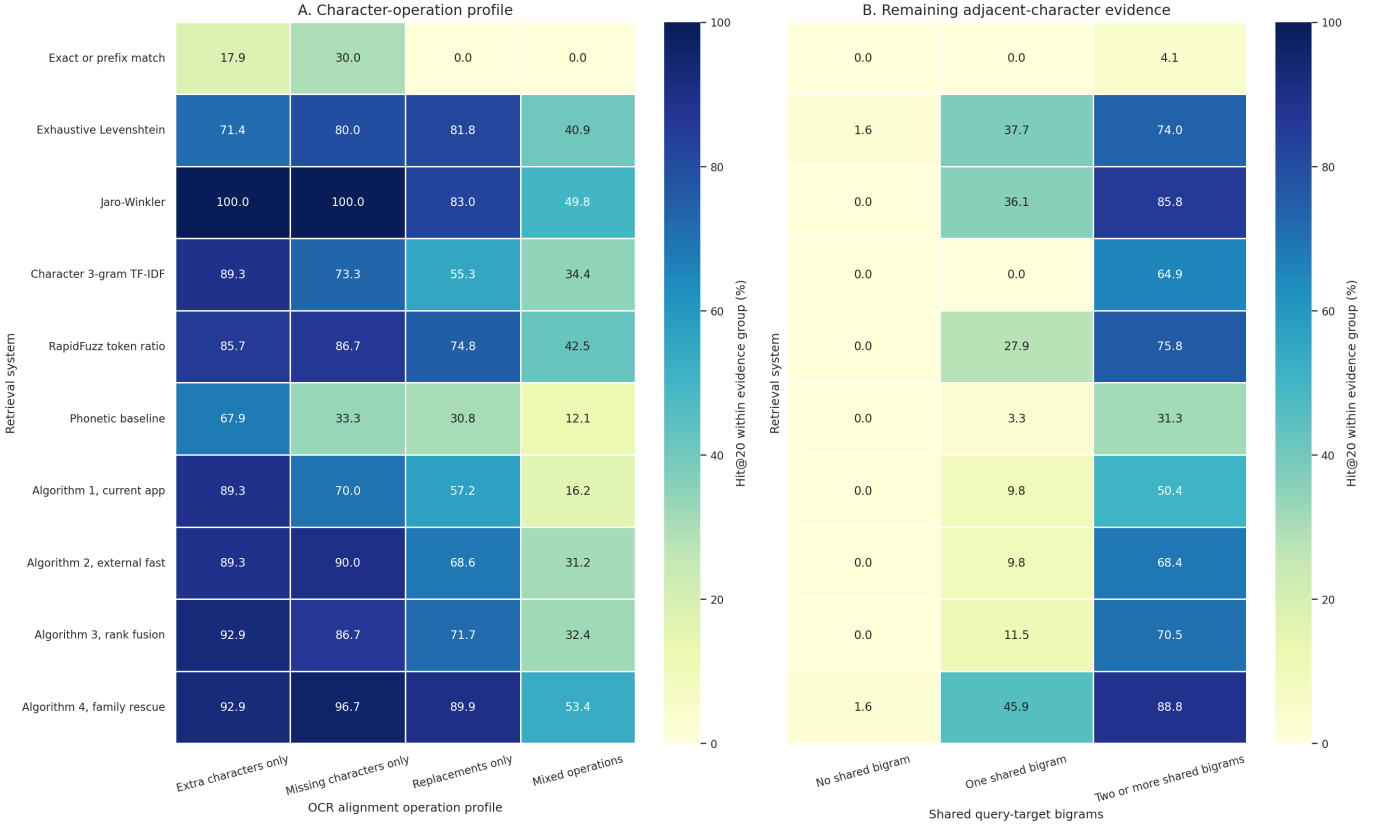


Figure 11: Rows are Experiment 1 systems. Panel A reports Hit@20 for extra-only, missing-only, replacement-only, and mixed OCR operations; panel B reports Hit@20 for zero, one, or at least two shared query-target bigrams. Group sizes are identical across systems.

System-by-system interpretation

1. Exact/prefix handles literal fragments but almost no corruption: mixed and replacement-only Hit@20 are both 0%. (*KETONOLAC* → *KETOROLAC* returns no candidate.)
2. Levenshtein handles isolated replacements at 81.8% Hit@20 but falls to 40.9% on mixed operations. (Uniform edit cost cannot distinguish medically plausible confusions from equally close catalog names.)
3. Jaro-Winkler is the strongest classical profile, 100% on isolated additions and deletions and 49.8% on mixed edits, but reaches 0% when no bigram remains. (Sequence evidence helps until local order vanishes.)
4. Character 3-gram TF-IDF reaches 64.9% with at least two bigrams and 0% with zero or one. (A 3-gram retriever cannot use a single shared bigram.)
5. RapidFuzz reaches 75.8% with two-plus bigrams and 42.5% on mixed edits, but spacing can change token scoring. (*T0 C0* should normalize to *T0C0*, yet the baseline returns *EC0T0* outside the top 20.)
6. The phonetic baseline reaches only 31.3% with two-plus bigrams and 12.1% on mixed edits. (A phonetic key is complementary evidence, not a stand-alone medicine-family ranker.)
7. Algorithm 1 reaches 57.2% on replacement-only rows but 16.2% on mixed rows. (Its conservative application retrieval does not expand enough once several corruption mechanisms combine.)
8. Algorithm 2 raises mixed-operation Hit@20 to 31.2% and one-edit Hit@20 to 96.4%. (Its typo retrieval improves breadth, but the aggregate safety analysis still requires clarification gates.)
9. Algorithm 3 reaches 32.4% on mixed rows and 70.5% with two-plus bigrams. (Fusion improves Algorithm 2 slightly but does not add the A4 family-rescue candidates.)
10. Algorithm 4 leads mixed rows at 53.4% and one-bigram rows at 45.9%, but zero-bigram Hit@20 is only 1.6%. (Family rescue extends bounded evidence; it does not infer a family from unrelated text.)

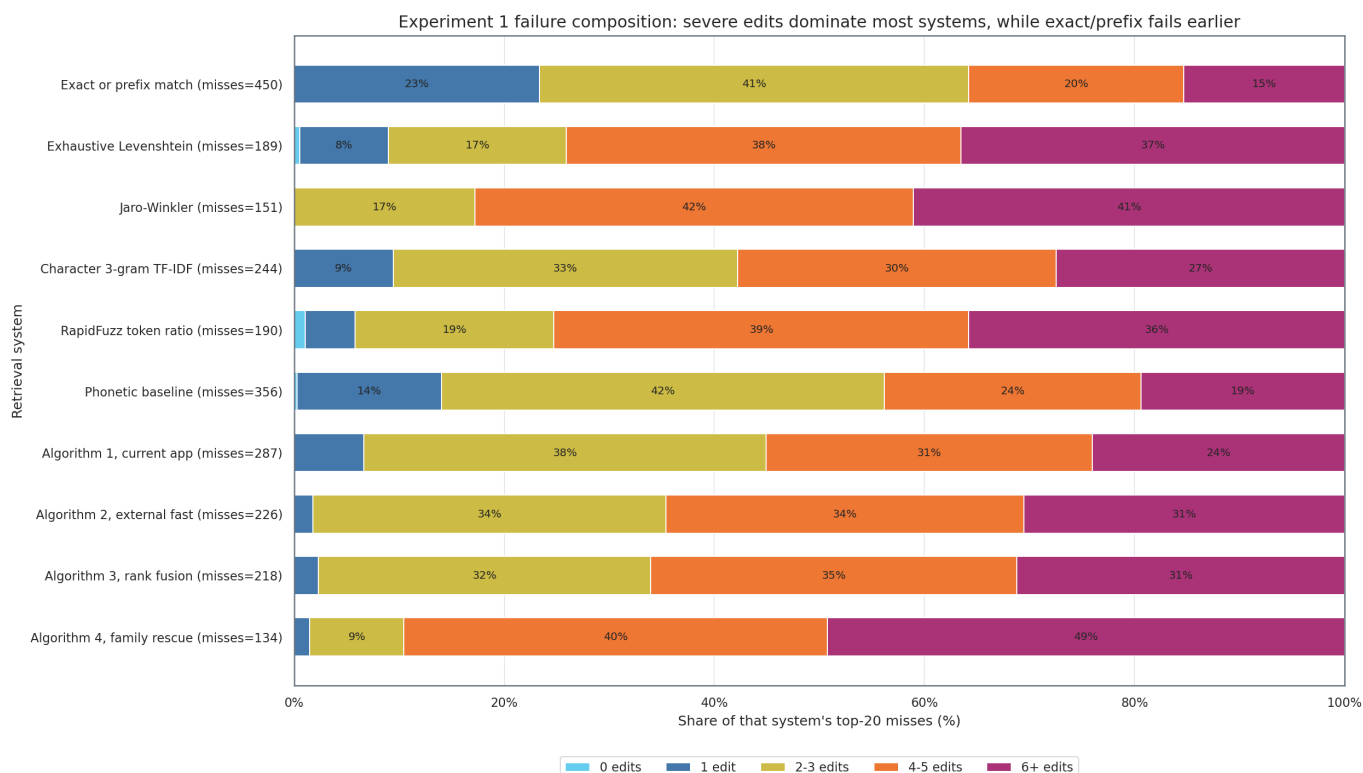


Figure 12: Each row partitions one system’s top-20 misses by compact edit band; the x-axis is the share of that system’s misses and the label prints the miss count. Segment width measures failure composition, not recovery rate.

Insights

1. Algorithm 4 has the fewest misses, 134; four-or-more edits create 120, or 89.55%. (The residual is concentrated after ordinary typo recovery has already succeeded.)
2. Exact/prefix misses 450 pairs and fails before the severe tail: 105 one-edit and 184 two-to-three-edit rows are absent by rank 20. (Its 2.59% Hit@1 is a method limitation, not a difficult-case-only result.)
3. Jaro-Winkler misses 151 pairs, only 17 more than Algorithm 4, but 33 of its misses are recoverable by A4 at rank 20 after accounting for the 16 reverse wins. (The two methods retain complementary candidates.)

Table 8: One deterministic top-20 miss per Experiment 1 system. These are the lowest compact-distance retrieval misses selected from each system’s row-level output, not invented examples. The companion artifact also records one rank-only failure per system.

System	Input	Expected	First	Rank	Why it fails
Exact/prefix	KETONOLAC	KETOROLAC	None	> 20	One substitution is outside the method.
Levenshtein	ABA SAGLAR	ABASAGLAR	ADALAT LA	> 20	Compact-exact text is lost by its ranking and normalization path.
Jaro-Winkler	ABISAGLOR	ABASAGLAR	ABITAL	> 20	Two edits retain misleading prefix similarity.
3-gram TF-IDF	PROSEST	PROGEST	PRO S	> 20	One replacement breaks a decisive 3-gram.
RapidFuzz	TO CO	TOCO	ECOTO	> 20	Token spacing changes the score despite compact equality.
Phonetic	ABA SAGLAR	ABASAGLAR	PASCOLAR	> 20	Broad phonetic collision outranks literal family evidence.
Algorithm 1	KETORCOLAC	KETOROLAC	KIRA	> 20	Candidate generation misses a one-edit insertion.
Algorithm 2	ABRASAGLAR	ABASAGLAR	BRANDAGLOB	> 20	One extra character defeats the external candidate set.
Algorithm 3	PROSEST	PROGEST	PROSTAT	> 20	Neither fused child contributes the target.
Algorithm 4	A BAA G LAR	ABASAGLAR	EFFACLAR H	> 20	Source spacing plus one edit bypasses current rescue.

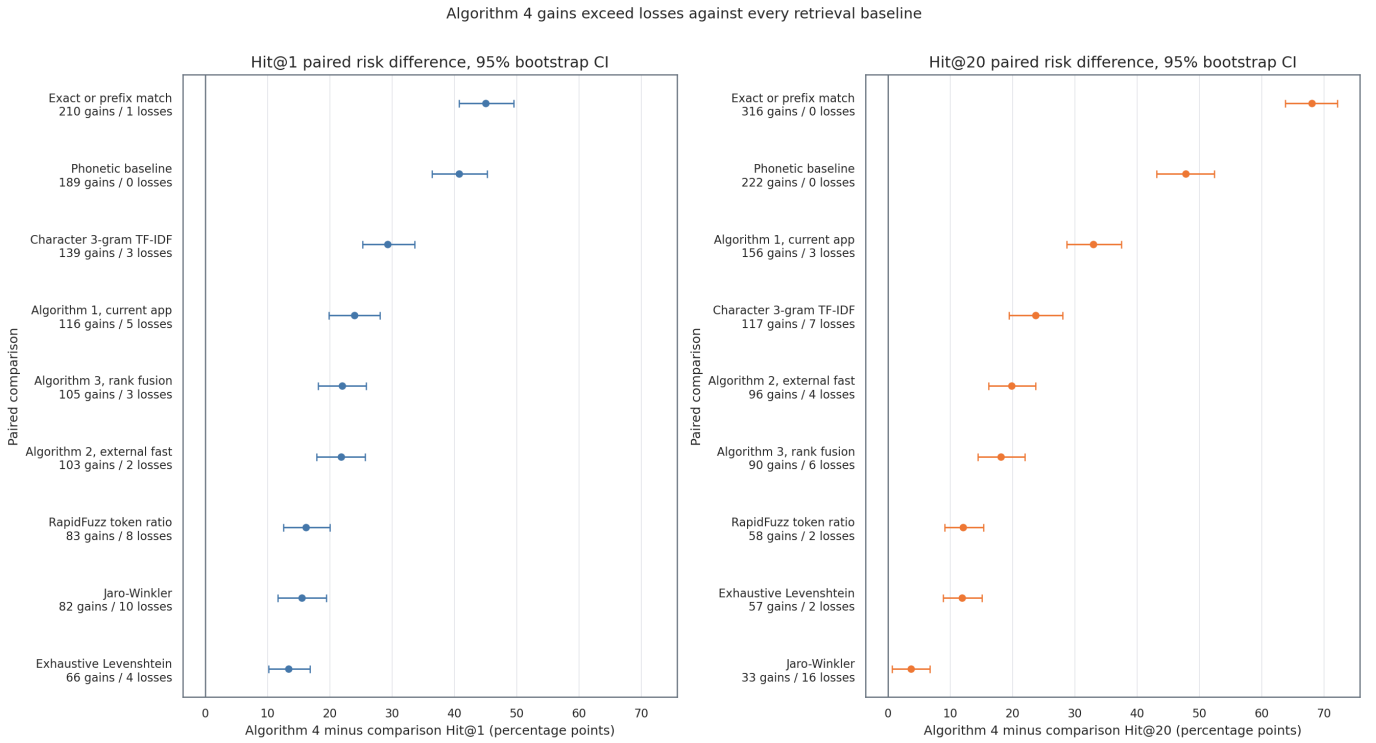


Figure 13: Each point is Algorithm 4 minus one comparison on the same 464 pairs; horizontal bars are 95% paired bootstrap intervals. Labels show cases won by Algorithm 4 and cases won by the comparison.

Insights

1. Against exhaustive Levenshtein, Algorithm 4 gains 66 and loses four Hit@1 pairs. (Difference +13.3621 points, 95% CI [10.1293, 16.8103], exact McNemar $p = 1.65 \times 10^{-15}$.)
2. Against Jaro-Winkler at Hit@20, Algorithm 4 gains 33 and loses 16. (Difference +3.6638 points, 95% CI [0.6466, 6.6810], $p = 0.0213$, matched odds ratio 2.03.)
3. Against Algorithm 3, Algorithm 4 gains 105 and loses three Hit@1 pairs. (The OCR ranking architectures are not equivalent even though both use multiple retrieval signals.)

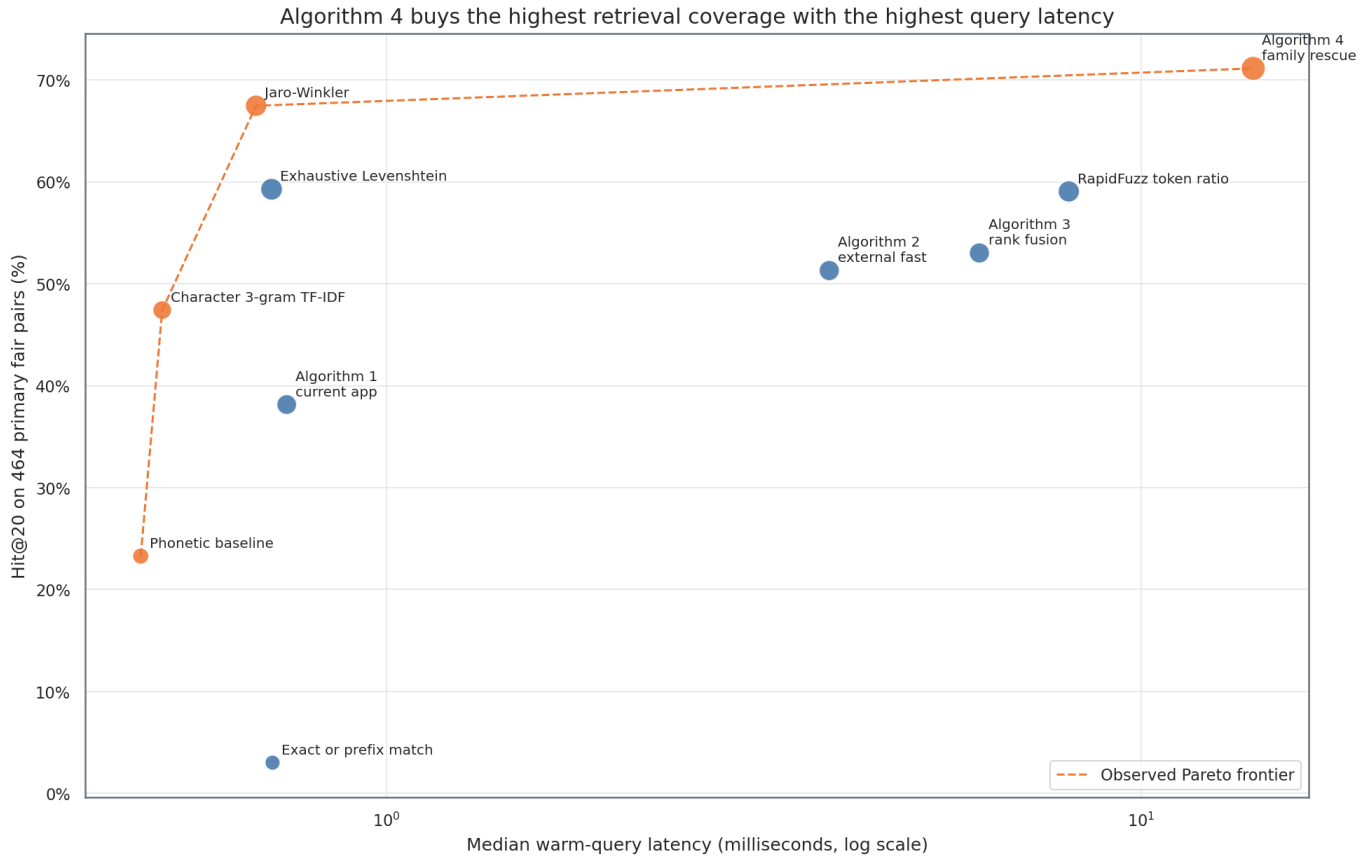


Figure 14: The x-axis is median warm-query latency on a logarithmic scale; the y-axis is Hit@20. Point size encodes Hit@1, and orange points form the observed latency-recovery Pareto frontier.

Insights

1. Algorithm 4 has the highest Hit@20 and the highest median query latency, 14.0704 ms. (It trades roughly 20 times Levenshtein latency for 11.8535 additional Hit@20 points.)
2. Jaro-Winkler, character 3-gram TF-IDF, the phonetic baseline, and Algorithm 4 lie on the observed frontier. (Each occupies a distinct speed-coverage point among the tested implementations.)
3. The frontier is machine- and implementation-specific. (It measures this single-process benchmark, not production throughput under concurrency.)

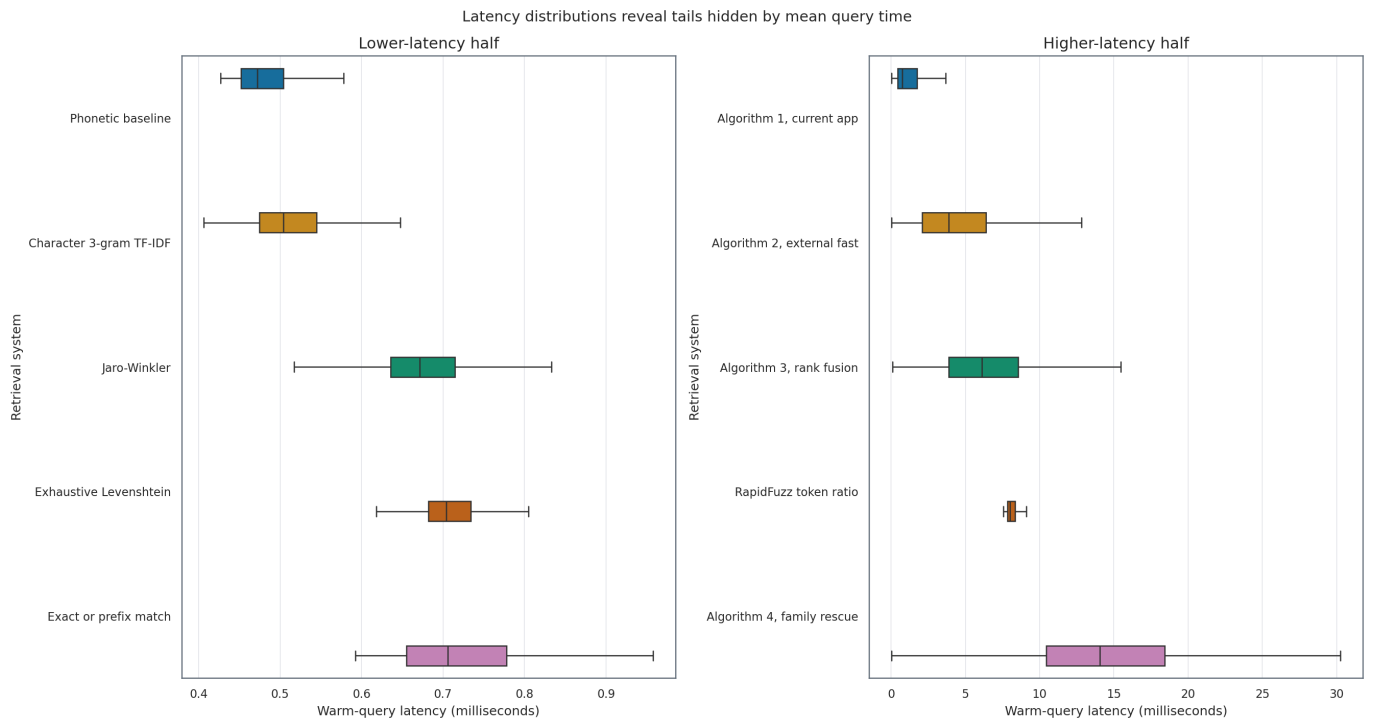


Figure 15: Each box plot shows warm-query latency over 464 primary pairs; the center line is the median, the box spans the middle 50%, and whiskers exclude displayed outliers. Systems are split into lower- and higher-latency panels.

Insights

1. Median and mean are insufficient for deployment sizing because query latency has visible tails. (Algorithm 4 mean is 14.8852 ms and median 14.0704 ms; its measured 95th percentile is 30.1791 ms.)
2. RapidFuzz's median is 8.0152 ms despite weaker retrieval than sub-1-ms Levenshtein and Jaro-Winkler. (Library name alone does not guarantee a faster implementation under this workload.)

Retrieval systems occupy different accuracy and efficiency profiles

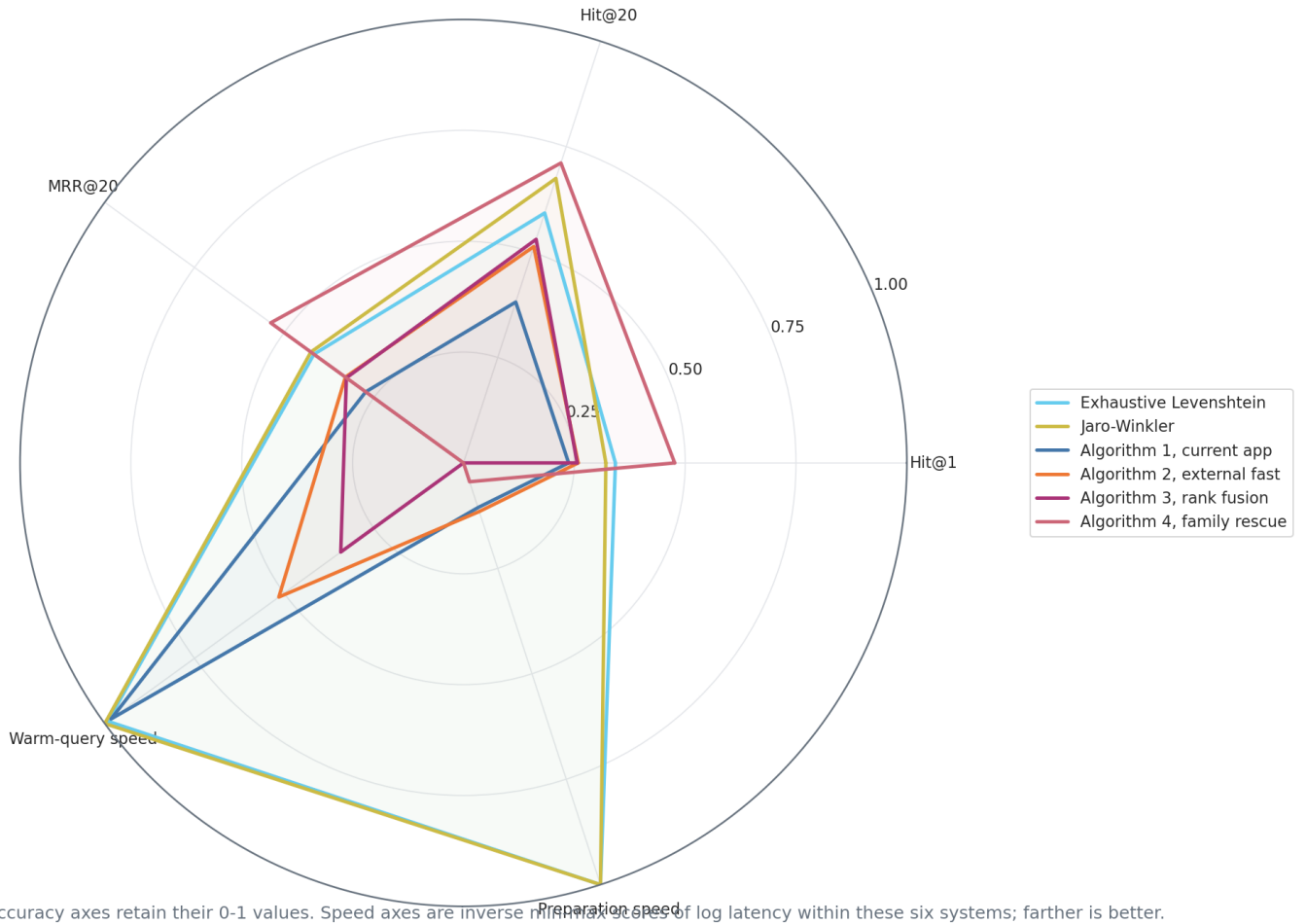


Figure 16: Six representative systems are compared on Hit@1, Hit@20, MRR@20, warm-query speed, and preparation speed. Accuracy axes keep their 0–1 values; speed axes are inverse min-max scores of log latency within these six systems, so farther from the center is better but relative.

Insights

1. Algorithm 4 dominates the three recovery axes but not either efficiency axis. (The chart makes the accuracy-cost trade-off explicit rather than compressing it into one score.)
2. The radial speed values are not percentages and cannot be compared outside this six-system set. (They are visualization-only min-max transforms; Table 6 contains actual milliseconds.)

Algorithm 4 recovers many cases missed by strong alternatives, but not all

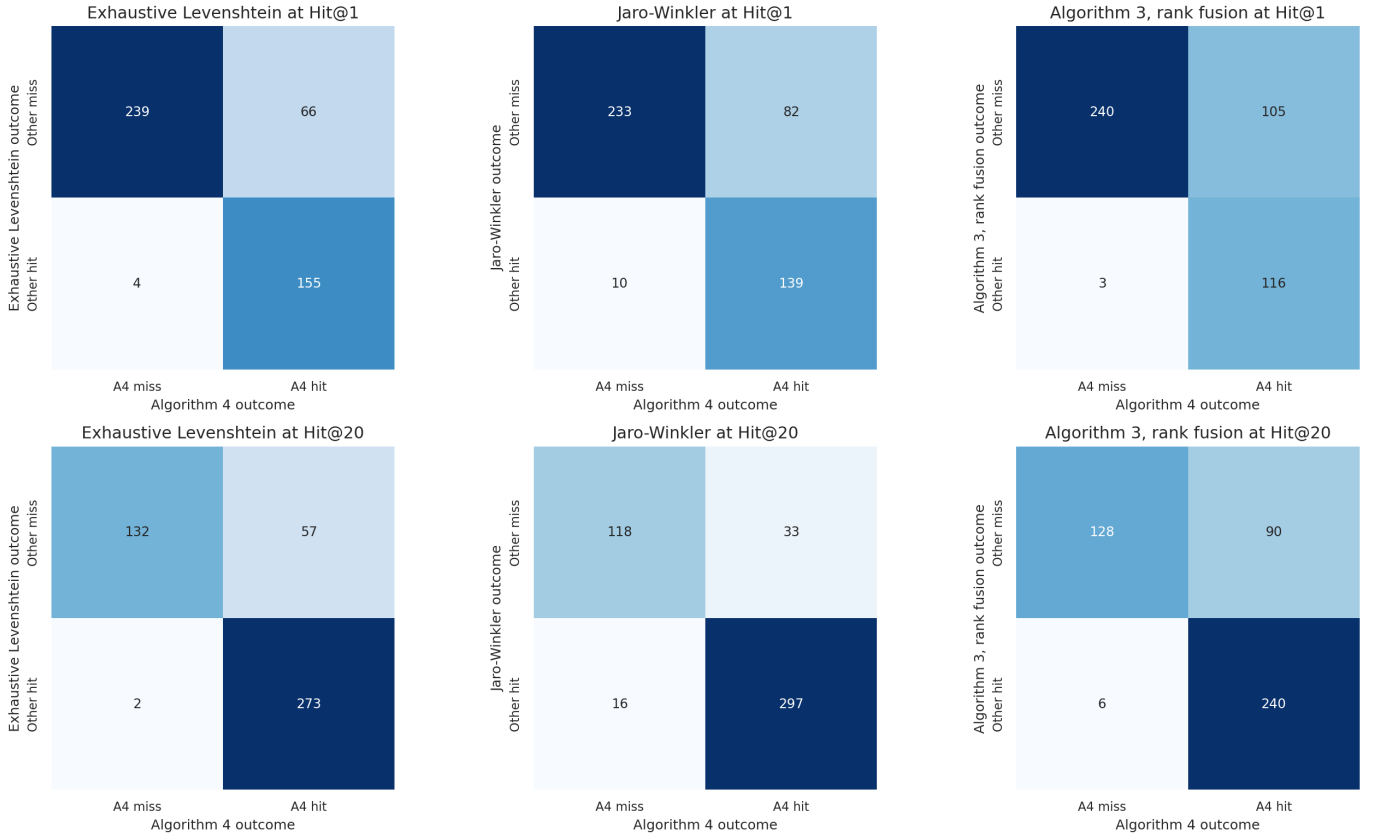


Figure 17: Each matrix cross-tabulates another system’s hit or miss against Algorithm 4 on the same pair. Columns are Algorithm 4 outcomes, rows are the comparison outcomes, and cells are pair counts.

Insights

1. Algorithm 4 uniquely recovers 66 Hit@1 pairs missed by Levenshtein, while Levenshtein uniquely recovers four. (Most of the aggregate gain comes from real complementary cases, not reshuffling shared successes.)
2. At Hit@20, Jaro-Winkler uniquely recovers 16 pairs that Algorithm 4 misses. (The strongest classical retriever still contains useful recall not captured by the combined system.)
3. Algorithm 4 uniquely recovers 90 Hit@20 pairs over Algorithm 3 and loses six. (Family rescue changes candidate presence, not only top-rank score.)

7 Algorithm 4 Evaluation

This section reports the frozen Algorithm 4 implementation. Algorithm 5 and its complete component study appear in the expanded benchmark section.

The enhancement converts observed failure patterns into bounded, general rules without storing query-specific answers or benchmark medicine names.

Rule	General behavior	Concrete recovery
Three-character delete-key retrieval	Permit bounded deletion-index candidates for compact queries of length three.	RIO $\rightarrow$ RIVO.
Short-edge retrieval	Search conservative prefix and suffix indexes with fixed core and rescue caps.	Retains evidence when only an edge is readable.
Family-head rescue	Compare a corrupted query with the shared head of commercial variants.	MYOLANA $\rightarrow$ MYOLAX.
Compatible-length scan	Expand only to catalog families in plausible query-length buckets.	LGCMU $\rightarrow$ LACTO.
Strict full-name correction	Allow a rescue result to replace first only for length at least five, raw distance at most three, strictly better raw distance, no worse weighted distance, and score gap at most 0.35.	NIGROHEX $\rightarrow$ VIGOREX; NEITROSET $\rightarrow$ NEUROCET.
Clarification state	Keep equal evidence, family variants, real-name collisions, and low-evidence strings non-confident.	CONAL compares COBAL and CONIL.

Table 9: The rules depend on query shape and catalog evidence. No literal test query or expected medicine was inserted into Algorithm 4.

Locked suite	Cases	H@1 before	H@1 after	H@20 before/after	Paired guardrail
Manual failed cases	50	72.00%	74.00%	92.00% / 94.00%	No unsafe confident result.
Primary OCR pairs	464	46.34%	47.63%	70.91% / 71.12%	H@1 gains/losses 6/0; H@20 1/0.
Browser primary pairs	464	37.93%	40.09%	62.07% / 63.58%	H@1 gains/losses 10/0; H@20 7/0.
Proportional synthetic sample	6,000	81.12%	81.40%	92.83% / 93.28%	Behavior after 93.45%.
Full synthetic benchmark	115,000	81.86%	82.12%	92.99% / 93.41%	H@1 gains/losses 180/95; H@20 132/5.

Table 10: The final rules were retained only after broader suites showed net gains without unsafe confident first-result regression. Browser values remain separate because JavaScript and Python retrieval implementations differ.

8 Experiment 2: Algorithm 4 Ablation

These 24 configurations explain the predecessor. The expanded section reports the current 97-configuration Algorithm 5 OCR ablation and 13 synthetic confirmation runs under the same paired-case contract.

Each ablation removes one named query-time component while holding the 464 primary pairs, catalog, and scoring contract fixed.

Table 11: Complete 24-configuration ablation result on 464 primary pairs. Deltas are percentage points relative to complete Algorithm 4.

Configuration	Hit@1	Hit@20	Δ H@1	Δ H@20
Complete Algorithm 4	47.63%	71.12%	0.00	0.00
Without external retriever	47.20%	71.55%	-0.43	+0.43

Configuration	Hit@1	Hit@20	Δ H@1	Δ H@20
Without context cleanup	47.63%	71.34%	0.00	+0.22
Without family rescue	27.37%	51.08%	-20.26	-20.04
Without raw edit similarity	46.77%	65.30%	-0.86	-5.82
Without weighted edit similarity	46.34%	63.79%	-1.29	-7.33
Without prefix evidence	46.55%	68.53%	-1.08	-2.59
Without suffix evidence	47.63%	71.12%	0.00	0.00
Without character n-grams	46.77%	70.69%	-0.86	-0.43
Without phonetic evidence	47.20%	71.12%	-0.43	0.00
Without consonant skeleton	46.98%	70.69%	-0.65	-0.43
Without subsequence evidence	47.20%	70.91%	-0.43	-0.22
Without position evidence	46.12%	69.40%	-1.51	-1.72
Without length coverage	47.20%	67.67%	-0.43	-3.45
Without delete-key retrieval	47.63%	70.91%	0.00	-0.22
Without short-edge retrieval	46.77%	67.46%	-0.86	-3.66
Without first-character expansion	47.63%	71.12%	0.00	0.00
Without compatible-length scan	46.98%	68.97%	-0.65	-2.16
Without family-head rescue	44.61%	69.61%	-3.02	-1.51
Without weighted confusion costs	46.77%	70.69%	-0.86	-0.43
Without retriever agreement	47.20%	70.91%	-0.43	-0.22
Without strict full-name correction	46.55%	71.12%	-1.08	0.00
Without conservative reranker	41.16%	70.47%	-6.47	-0.65
Without safety clarification gate	47.63%	71.12%	0.00	0.00

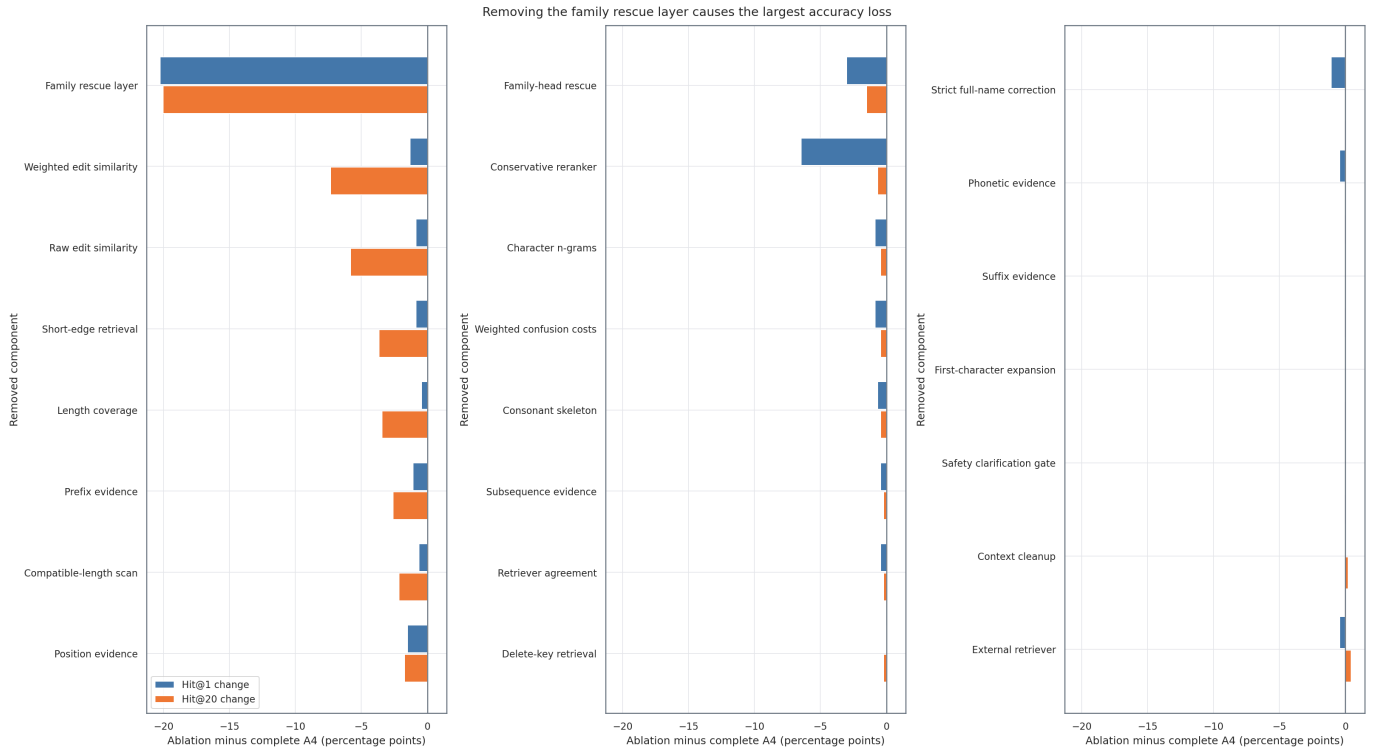


Figure 18: Each row removes one Algorithm 4 component. Blue is Hit@1 change and orange Hit@20 change relative to complete Algorithm 4; negative values mean the complete system performs better.

Insights

1. Removing family rescue produces the largest loss, -20.2586 Hit@1 points and -20.0431 Hit@20 points. (It is the main source of candidate recovery beyond the child algorithms.)
2. Weighted edit similarity causes the next-largest Hit@20 loss, -7.3276 points; raw edit similarity loses -5.8190 points. (Both distance views contribute non-redundant retrieval evidence.)
3. Removing the external retriever or context cleanup raises Hit@20 by only 0.4310 and 0.2155 points. (Paired tests are not significant, so these small reversals require larger confirmation before deletion.)

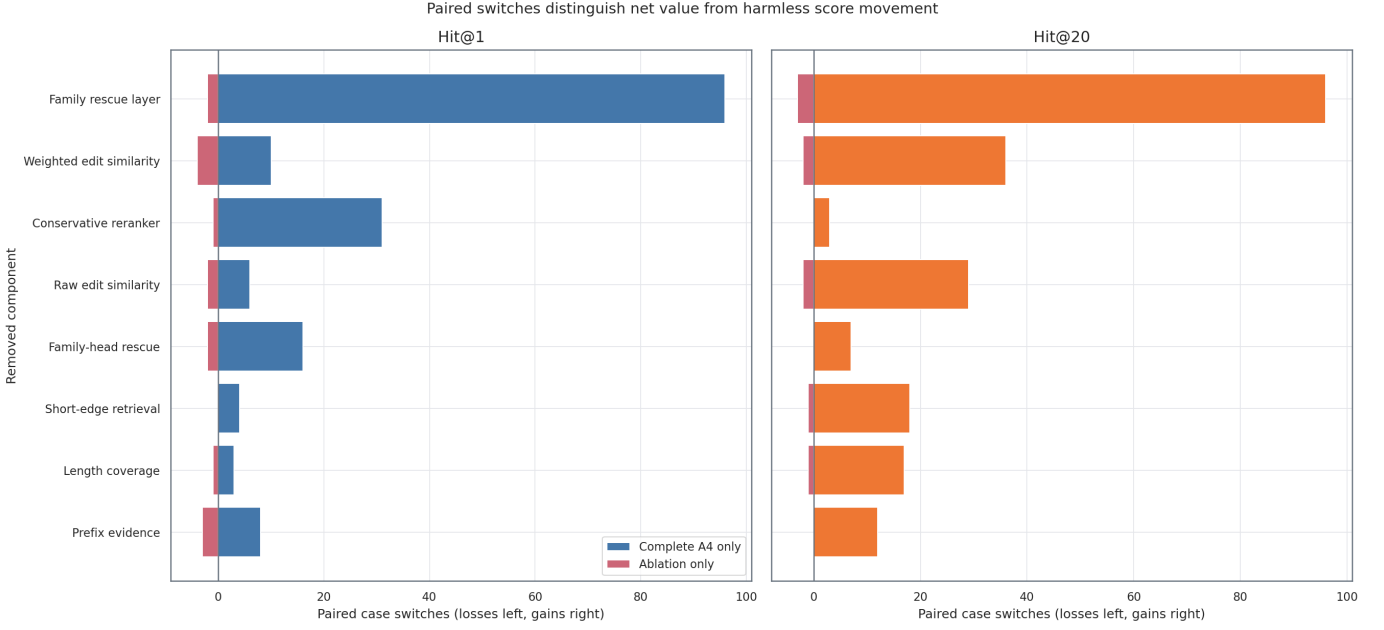


Figure 19: For each removed component, blue counts primary pairs where complete Algorithm 4 succeeds and the ablation fails; orange counts the reverse. Panels separate Hit@1 and Hit@20 switches.

Insights

1. Complete Algorithm 4 beats no-rescue on 96 pairs and loses on two at Hit@1; at Hit@20 the counts are 96 and three. (Exact McNemar p -values are below 10^{-24} .)
2. The conservative reranker changes first rank far more than candidate recall: 31 gains versus one loss at Hit@1, but only three gains at Hit@20. (Its role is ordering, not broad retrieval.)
3. Strict full-name correction yields five Hit@1 gains and zero losses, with exact $p = 0.0625$. (Direction is consistent across development and holdout, but this sample is too small for a 0.05 claim.)

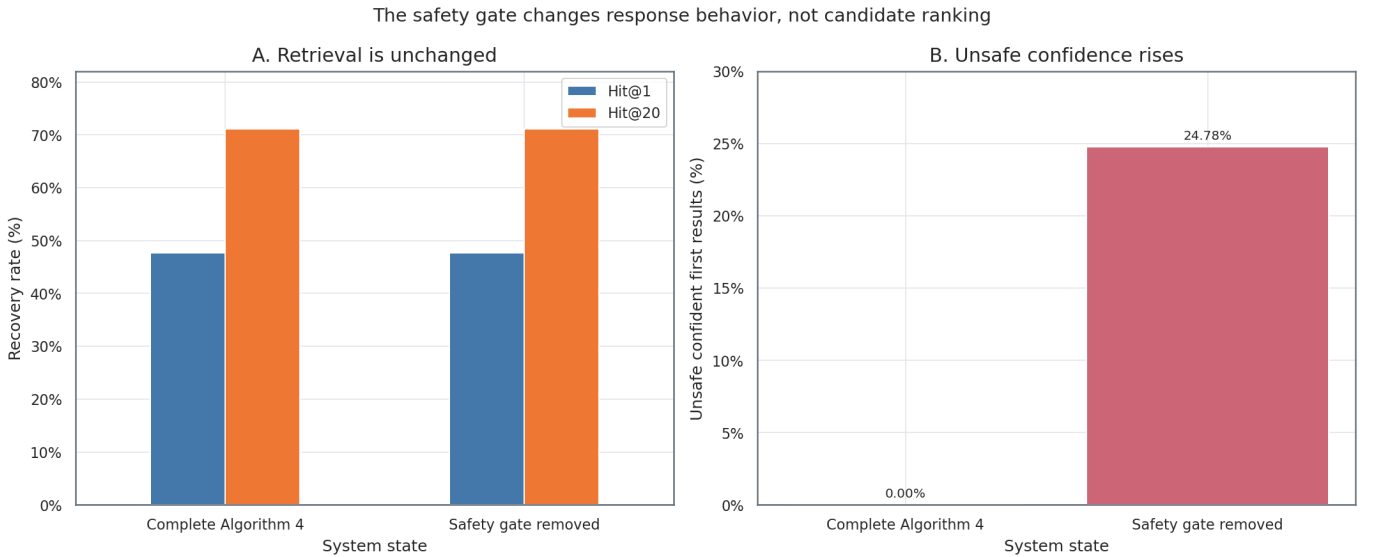


Figure 20: The left panel compares Hit@1 and Hit@20 with and without the safety clarification gate; the right panel compares unsafe confident first-result rate. Both variants use identical ranked candidates.

Insights

1. Retrieval metrics remain exactly 47.6293% Hit@1 and 71.1207% Hit@20 after gate removal. (The gate operates on response confidence, not rank.)
2. Unsafe confident first results rise from 0% to 24.7845%. (A retrieval ablation alone would falsely label the safety gate as unnecessary.)

9 Equal Edit-Distance Ordering

Equal raw edit distance leaves 37 primary ranking failures unresolved, so eight generic evidence policies were tested without privileging a specific character position.

A generic rule is allowed to act only when the query is brand-like, the current first candidate has nonzero raw distance, another candidate has the same raw distance, and its full-model score is within the configured gap. Lower weighted distance is better; higher position, edge, agreement, and full score are better.

Tested choice	Exact ordered criteria
Current full-evidence order	Keep Algorithm 4 score order, including retriever, rescue, lexical, family, context, and safety evidence.
Weighted then position	Lowest confusion-weighted edit cost; highest position evidence; highest edge evidence; dual-retriever agreement; full score.
Position then weighted	Highest position evidence; lowest weighted edit cost; highest edge evidence; agreement; full score.
Edge then weighted	Highest prefix/suffix edge evidence; lowest weighted edit cost; highest position evidence; agreement; full score.
Composite evidence	Maximize $-d_w + 0.35p + 0.25e + 0.10a$; then full score, where d_w is weighted cost, p position, e edge, and $a \in \{0, 1\}$ agreement.
Pareto, gap 0.25	Switch only if the candidate is no worse on weighted cost, position, edge, and agreement, better on at least one, and within 0.25 full-score units.
Pareto, gap 0.15	Same dominance condition with a 0.15 full-score gap.
Pareto, gap 0.10	Same dominance condition with a 0.10 full-score gap.

Table 12: The eight tested choices. Original candidate order is the final deterministic fallback. No policy gives the first character, last character, or alphabetic order automatic priority.

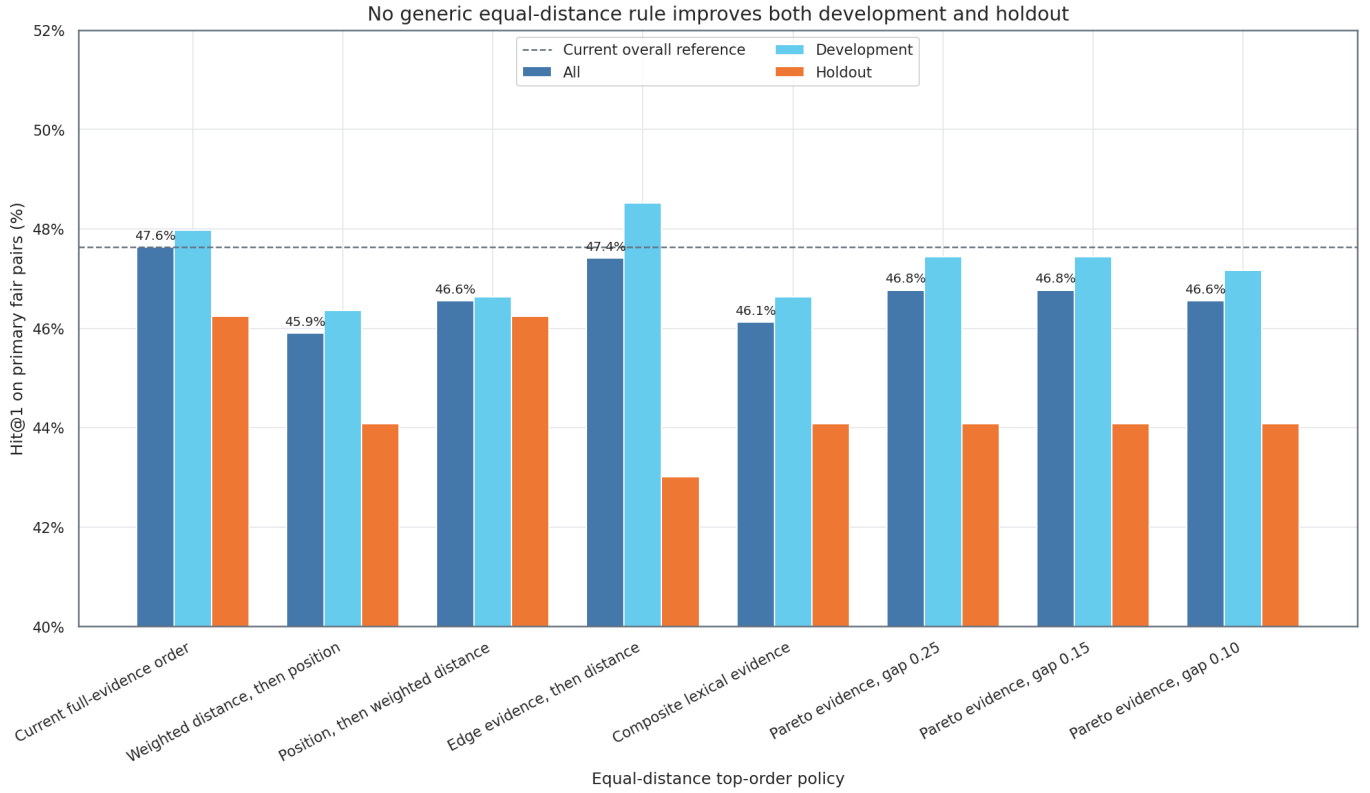


Figure 21: The y-axis is Hit@1; each policy has all, development, and holdout bars. The dashed line is current overall ordering. Target-family hashing gives 371 development pairs and 93 disjoint holdout pairs.

Insights

1. No replacement policy improves both development and holdout. (Current ordering is 47.63% overall, 47.98% development, and 46.24% holdout.)
2. Edge-first ordering rises to 48.52% on development but falls to 43.01% on holdout. (Choosing it from development would overfit the selected target families.)
3. Position-first ties current holdout but reduces development and overall accuracy. (Matching the first character is useful evidence, not a safe universal decision chain.)

Input	Verified	Current first	Policy behavior	What the example proves
LAMIX	LASIX	VAMIX	All tested rules retain VAMIX; expected rank is 3.	Raw distance, weighted distance, and position all tie; edge evidence favors the wrong candidate 0.80 to 0.40.
ANYOLAX	MYOLAX	AGIOLAX	Edge-first selects MYOLAX; the other generic rules retain AGIOLAX.	One evidence source can fix a development row while conflicting with stronger position evidence.
RIVOFN2	RIVOTRIL	RIOPAN	Position-first and edge-first select RIVOTRIL; weighted and Pareto choices select other families.	The rules disagree even with the same four-edit tie.
CEFAXIME	CEFIXIME	CEFAXIM	Weighted, position, and composite select CEFIXIME; edge and Pareto retain CEFAXIM.	A holdout win under one rule does not justify that rule when other holdout rows are lost.
CLARANE	CLEXANE	CLARA	Weighted, position, and composite select CLEXANE; edge and Pareto retain CLARA.	Weighted cost can resolve one row, but its overall holdout policy still loses accuracy.
INDEVIO	INDERAL	ENTYVIO	Every rule retains ENTYVIO; expected rank is 2.	The expected family does not dominate the first candidate on available evidence.
CONAL	COBAL	CONIL	Both are one raw edit away in the synthetic benchmark.	The first character matches both; the safe action is to compare medicines, not invent a positional chain.

Table 13: Six real OCR policy cases plus the synthetic CONAL example. The product response for unresolved evidence is candidate comparison and clarification, not a hidden alphabetic or first-character rule.

10 Synthetic 115,000-Case Benchmark

This diagnostic benchmark includes behavior, ambiguity, and collision rows. The later 66,257-pair clean-core benchmark is the current positive-retrieval denominator and must not be numerically mixed with this 115,000-row result.

The deterministic synthetic benchmark measures four algorithms over 34 named mutation categories, four scopes, safety behavior, and failure mechanism.

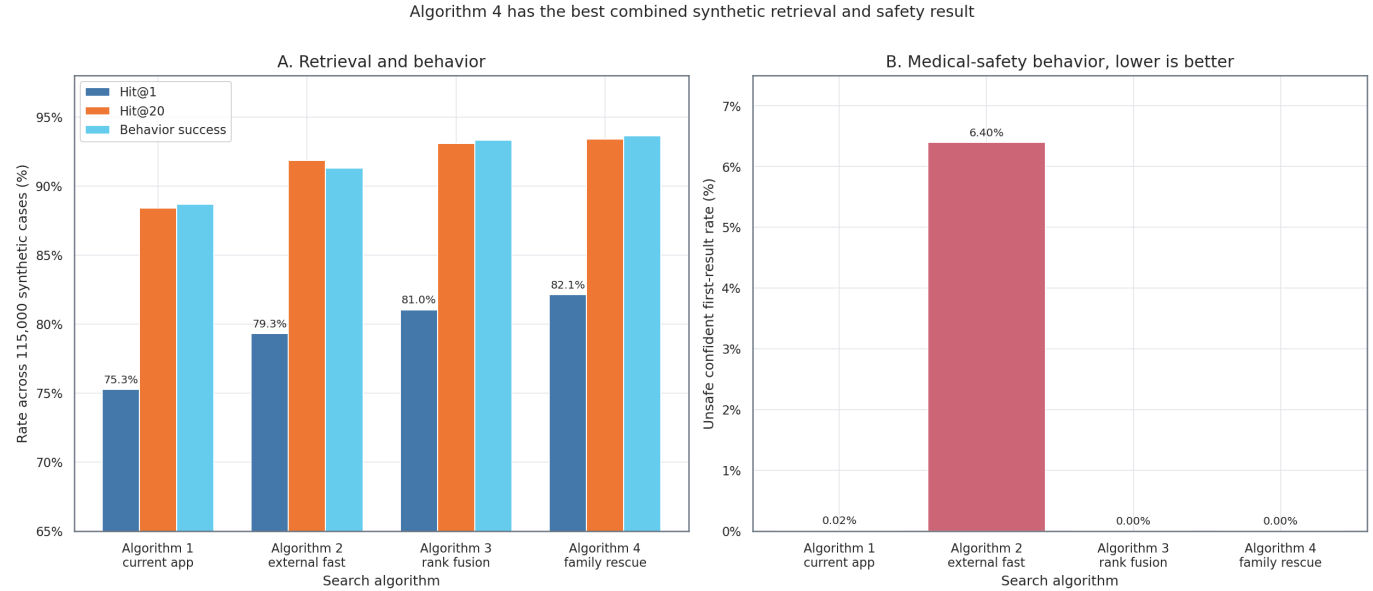


Figure 22: Panel A compares Hit@1, Hit@20, and behavior success over all 115,000 cases; panel B reports unsafe confident first-result rate, where lower is better. Algorithms 1–4 use aligned source rows.

Insights

1. Algorithm 4 reaches 82.1165% Hit@1, 93.4122% Hit@20, and 93.6374% behavior success. (All are the best aggregate values among the four.)
2. Algorithm 2 reaches 79.3009% Hit@1 but produces 6.3957% unsafe confident first results. (High retrieval without clarification control is unacceptable for the intended interface.)
3. Algorithm 4 versus Algorithm 3 gains 6,701 and loses 5,457 Hit@1 rows. (Net gain 1,244 rows, +1.0817 points, 95% CI [0.8913, 1.2739].)

Algorithm rankings differ by query scope, especially at rank 1

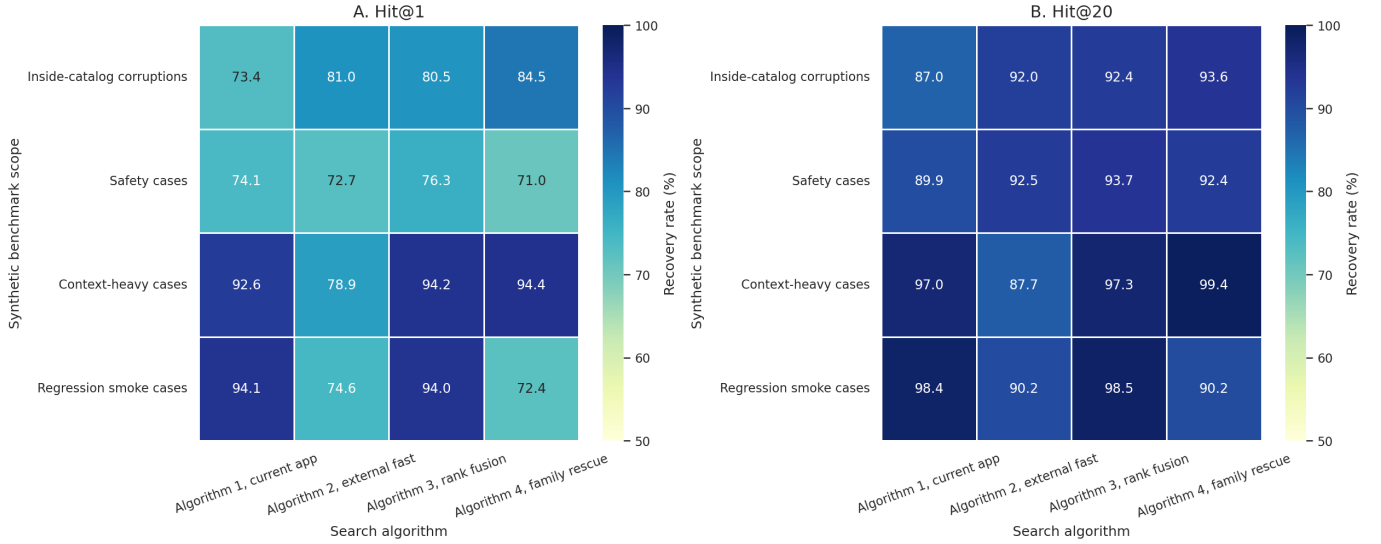


Figure 23: Rows are inside-catalog corruptions, safety cases, context-heavy queries, and regression smoke tests; columns are Algorithms 1–4. The left heatmap is Hit@1 and the right is Hit@20.

Insights

1. Algorithm 4 is strongest on inside-catalog Hit@1 at 84.5% and context-heavy Hit@20 at 99.4%. (Family rescue and context cleanup match those query shapes.)
2. Algorithm 4 does not lead safety Hit@20 or smoke Hit@20. (Algorithm 3 reaches 93.7% and 98.5% respectively, while Algorithm 4 reaches 92.4% and 90.2%.)
3. Aggregate leadership therefore hides regressions in specific scopes. (Release criteria must keep scope-level floors, not only one overall target.)

Type	Input → expected	A4 result	Measured state	Correct treatment
1	epigent → APIGENT	Exact EPIGENT first	Real-name collision	Exclude from fair single-answer accuracy; keep safety diagnostic.
2	conal → COBAL	CONIL first; expected rank 3	Equal edit evidence	Compare candidates; do not force one position rule.
3	acc plus unreadable-after → ACCELAFEN	Expected rank 4	Known missing continuation	Use user/image location evidence.
4	candeblockb → CANDEBLOCK D	Family head first	Family-variant mismatch	Recover family, then choose form or variant.
5	akc → ACC	No target in top 20	Candidate-generation failure	Expand bounded retrieval or ask for more evidence.
6	tidanhair → TITAN HAIR	Expected rank 2	Candidate-ranking failure	Improve rank evidence while retaining clarification.

Table 14: The six mechanisms are orthogonal to the 34 mutation categories and map directly to different algorithm or interface actions.

Algorithm	Incl. H@1	Fair H@1	Incl. H@20	Fair H@20	Fair behavior	Fair unsafe
Algorithm 1	75.2565%	78.6841%	88.4043%	90.6578%	90.9260%	0.0173%
Algorithm 2	79.3009%	82.9251%	91.8374%	93.9740%	93.3975%	2.8407%
Algorithm 3	81.0348%	84.7373%	93.0948%	95.3389%	95.5799%	0.0000%
Algorithm 4	82.1165%	85.8694%	93.4122%	95.6490%	95.8845%	0.0000%

Table 15: The fair-data change for every synthetic algorithm. Inclusive has 115,000 rows; fair has 109,974. The original rows are retained and identified by `scored_case=0`, so safety and ambiguity analysis still sees them.

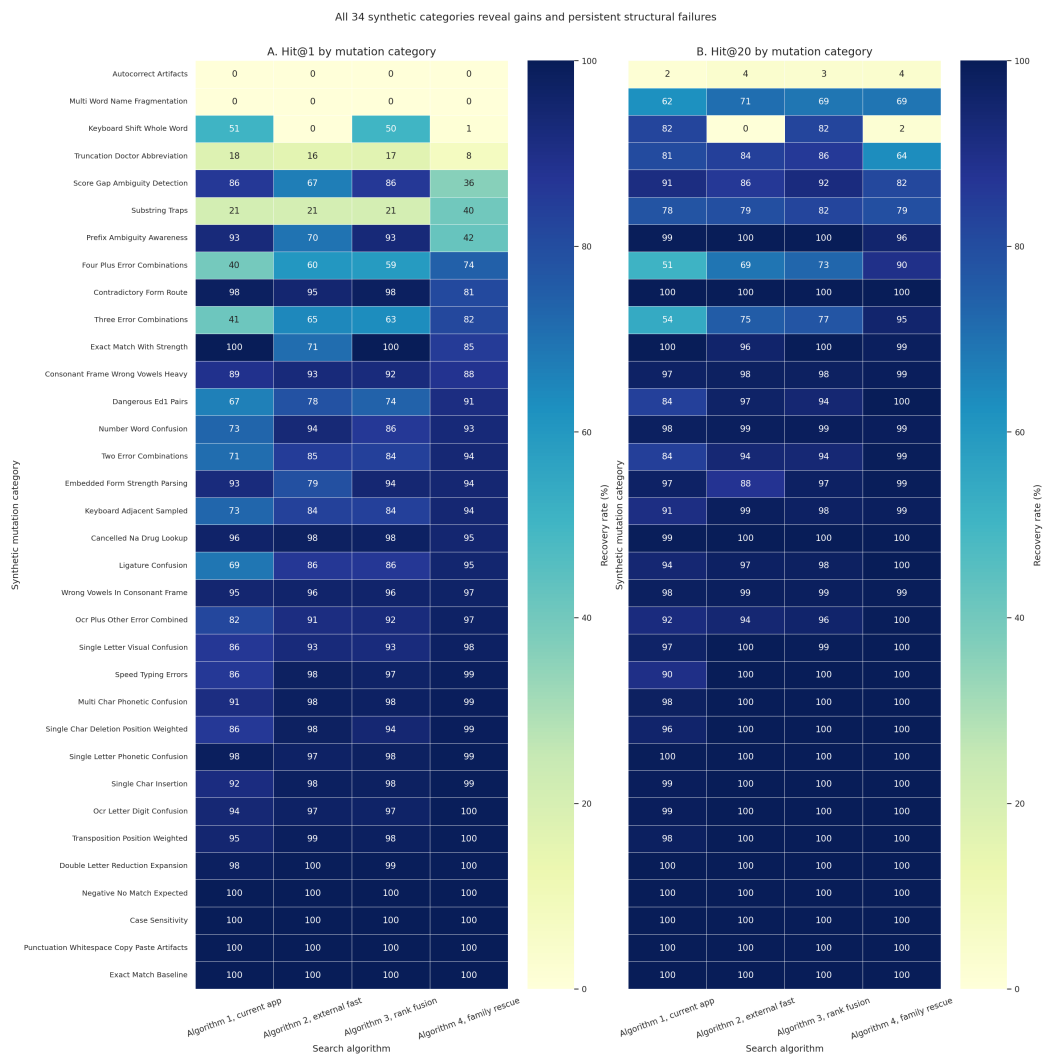


Figure 24: Each row is one of 34 mutation categories ordered by Algorithm 4 Hit@1; columns are Algorithms 1–4. The left heatmap reports Hit@1 and the right Hit@20, both from 0% to 100%.

Insights

1. Algorithm 4 reaches near-complete recovery on many one-edit, OCR, punctuation, and exact-name categories. (These rows supply character evidence compatible with its indexes and scoring signals.)
2. Autocorrect artifacts remain effectively unrecoverable, 0% Hit@1 and 3.65% Hit@20. (Inputs such as a different English word can contain no lexical evidence for the expected medicine.)
3. Keyboard shift and doctor abbreviation remain structural weaknesses. (Algorithm 4 Hit@20 is 2.0% and 63.5%; the first needs a dedicated reversible decoder, while the second needs cautious prefix retrieval.)
4. Multi-word fragmentation has 0% Hit@1 but 69.35% Hit@20. (Inputs such as PR0 or IVY can be valid shorter catalog families, so forced first-rank scoring conflicts with real ambiguity.)

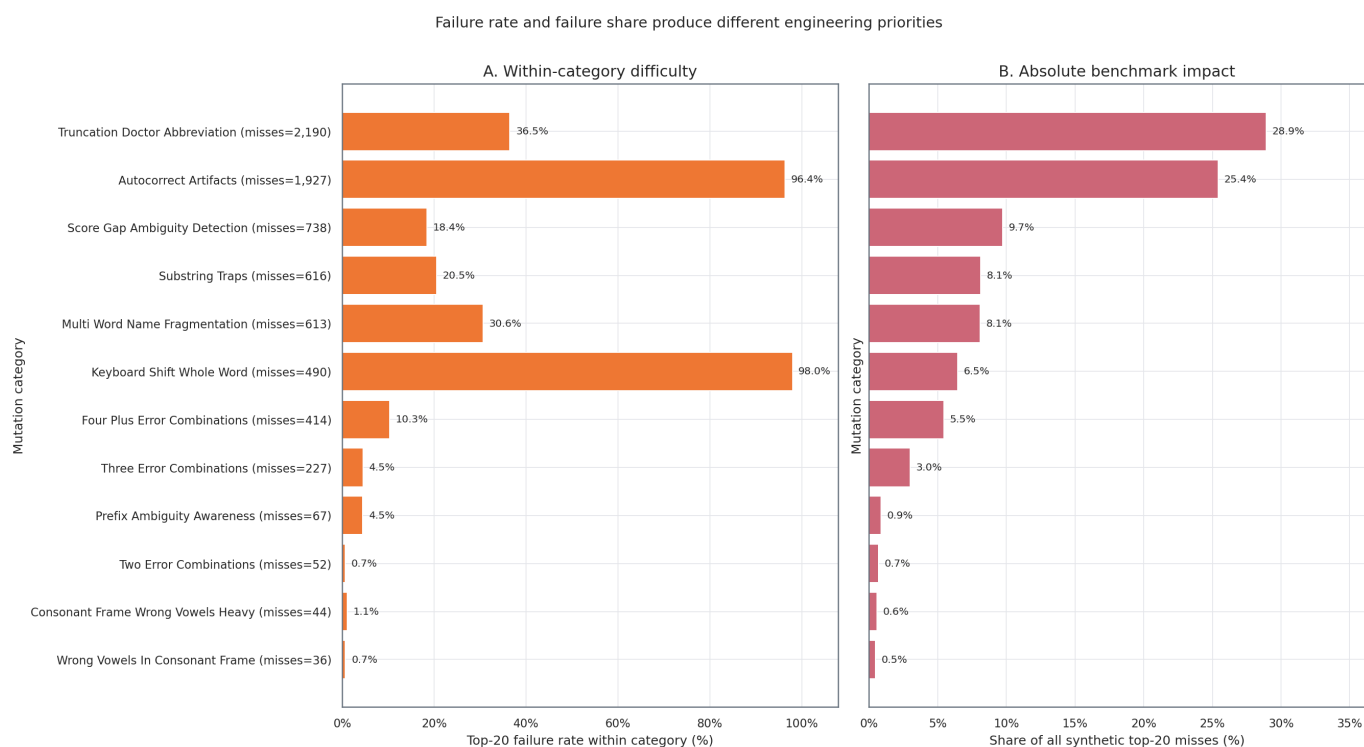


Figure 25: The left panel is top-20 failure rate inside each of the 12 largest failure-contributing categories; the right panel is each category’s share of all Algorithm 4 top-20 misses. Labels print category size or miss count.

Insights

1. Doctor abbreviations create 2,190 misses, 28.91% of all top-20 misses, while autocorrect artifacts create 1,927, 25.44%. (Together they cause more than half of the benchmark’s retrieval failures.)
2. Keyboard shift has a 98% failure rate but only 6.47% failure share because it contains 500 rows. (A high rate does not automatically make a category the largest engineering target.)
3. The six largest categories create 86.77% of all misses. (Prioritizing doctor abbreviation, autocorrect, score-gap ambiguity, substring traps, fragmentation, and keyboard shift covers most measured failure impact.)

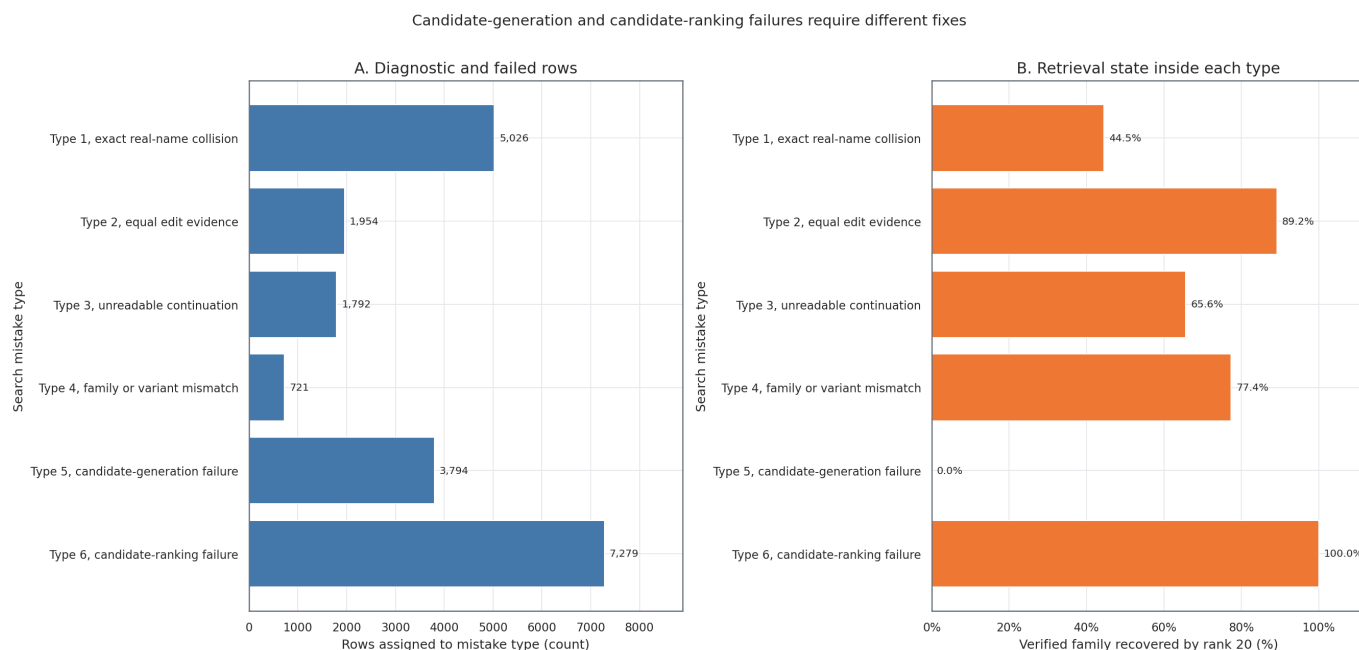


Figure 26: Panel A counts diagnostic or failed rows assigned to each search mistake mechanism; panel B reports Hit@20 inside that mechanism. Mutation category describes how input was generated, while mistake type describes why search did not return the intended first answer.

Insights

1. Candidate-ranking failure is largest at 7,279 rows and has 100% Hit@20. (The expected family already exists in the list, so reranking and family presentation can address it.)
2. Candidate-generation failure contains 3,794 rows and 0% Hit@20. (No reranker can fix a target absent from the candidate set.)
3. Known unreadable continuation reaches 65.63% Hit@20. (User-supplied position evidence before, after, or in the middle can narrow candidates, but it cannot manufacture missing characters.)

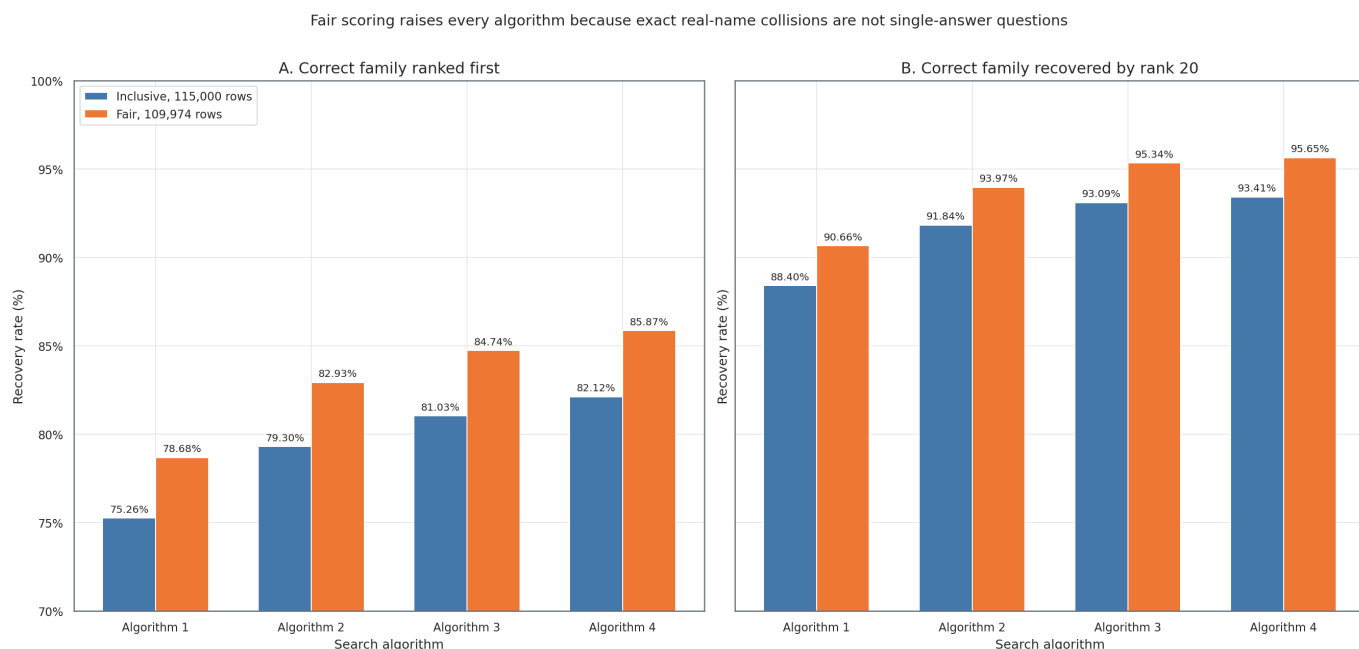


Figure 27: Panels report Hit@1 and Hit@20 for Algorithms 1–4. Blue uses all 115,000 rows; orange uses the same source table with 5,026 exact-real-name collisions marked unscored, leaving 109,974 fair single-answer rows.

Insights

1. Fair Hit@1 rises by 3.4275, 3.6242, 3.7025, and 3.7529 points for Algorithms 1–4. (The correction applies to every algorithm on the same source rows; it is not an A4-only score adjustment.)
2. Algorithm ordering does not change: A4 remains first, then A3, A2, and A1 at both cutoffs. (The fair view removes an invalid question type rather than selecting a winner.)
3. Algorithm 2 unsafe confident top-1 falls from 6.3957% inclusive to 2.8407% fair but remains nonzero. (Some unsafe behavior is caused by collisions, while the rest occurs on otherwise scored rows.)

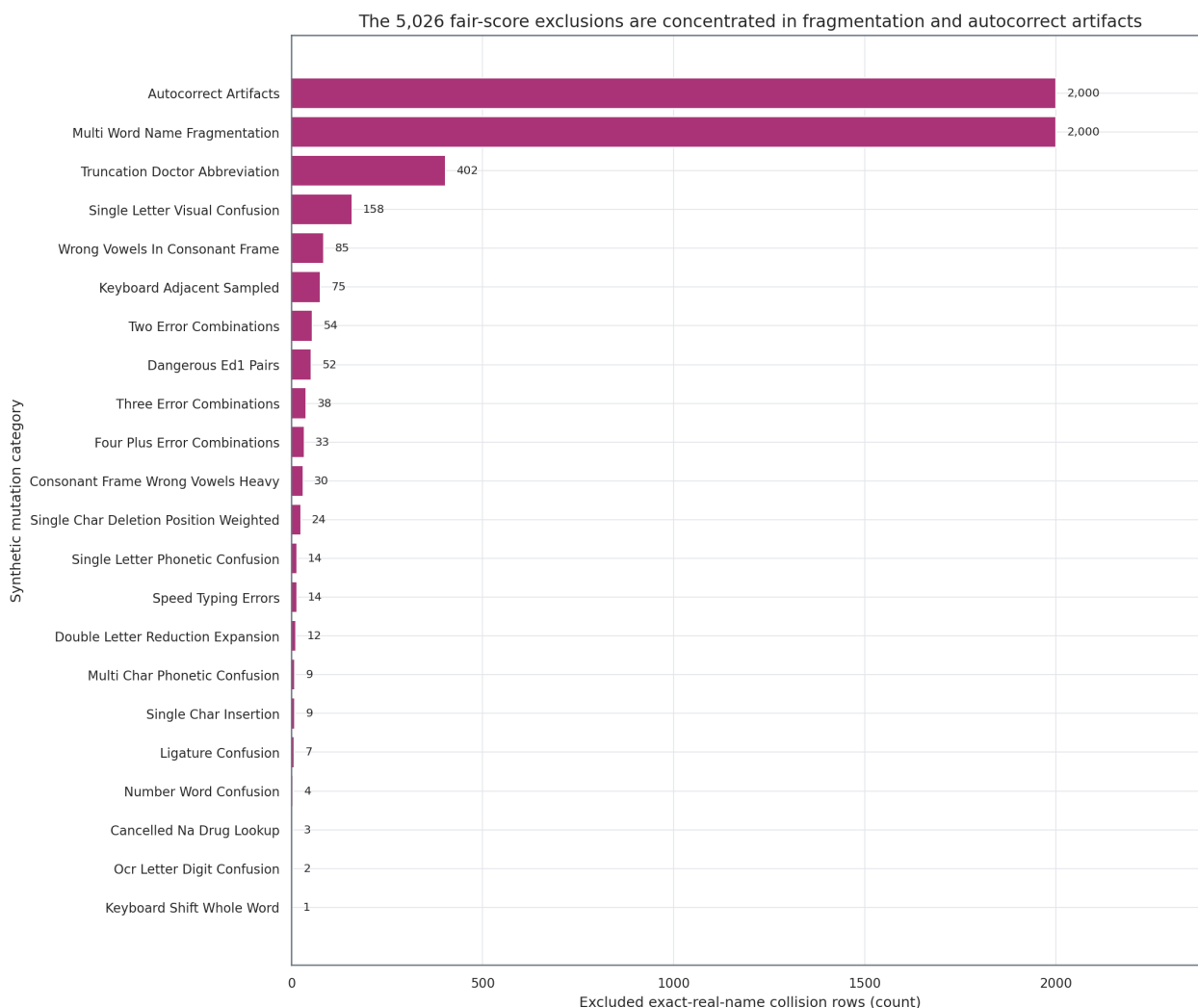


Figure 28: The y-axis lists every mutation category containing an excluded exact-real-name collision; the x-axis counts those rows. Exclusion means absent from fair single-answer accuracy, not deleted from the 115,000-row benchmark.

Insights

1. Autocorrect artifacts and multi-word fragmentation contribute 2,000 collisions each, 79.59% of all 5,026 exclusions. (Many generated inputs are exact catalog names with a different valid interpretation.)
2. Doctor abbreviation contributes another 402 rows, 8.00%. (A short entered family can be valid itself while also being a fragment of the generated target.)
3. The remaining 624 rows span 19 categories. (Fairness is concentrated but not confined to the two redesigned categories, so collision checks must remain a row-level rule.)

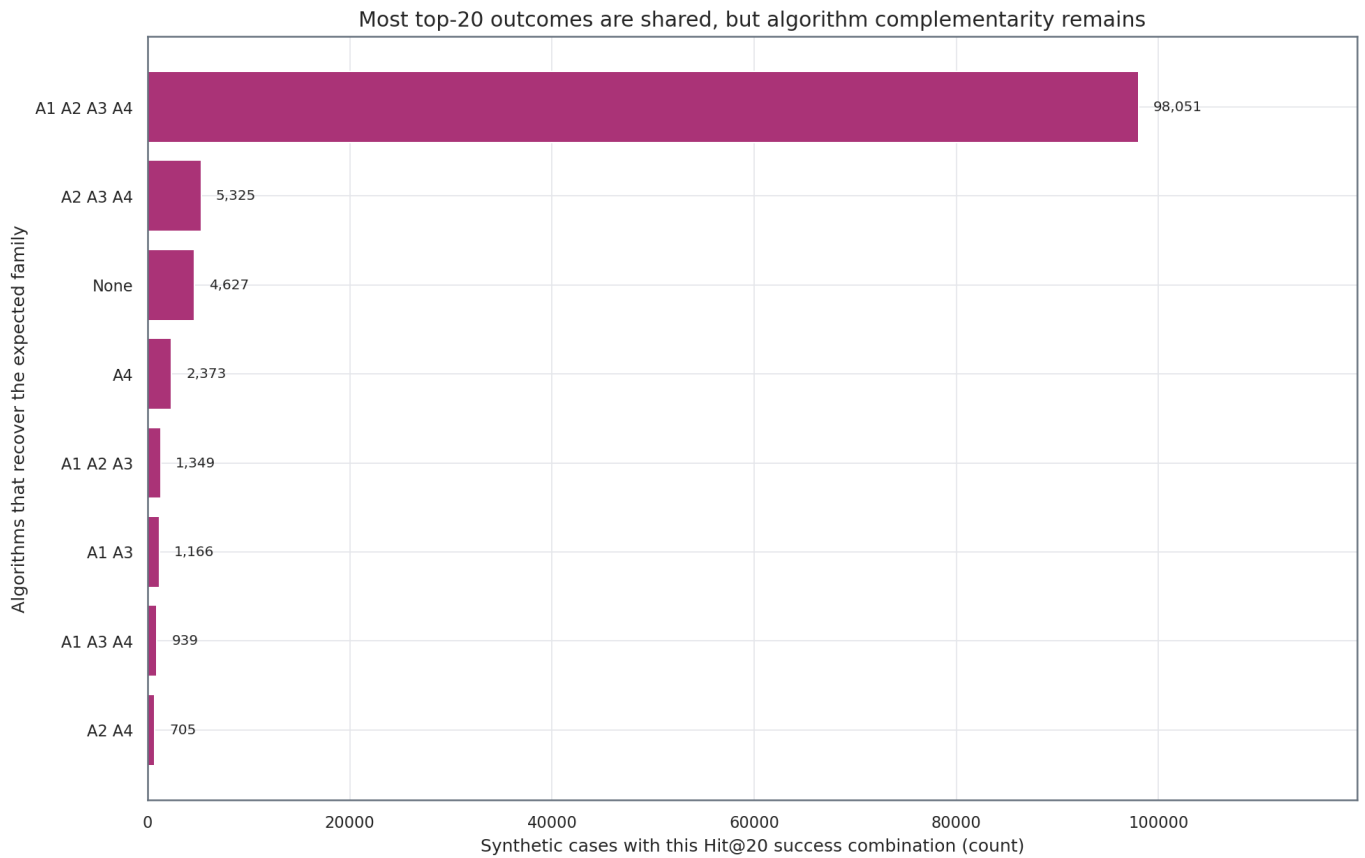


Figure 29: The y-axis names which Algorithms 1–4 recover the expected family by rank 20; the x-axis counts aligned synthetic rows with that exact success combination. The eight most frequent combinations are shown.

Insights

1. All four algorithms recover 98,051 cases, while none recovers 4,627. (Most outcomes are shared, and a hard residual remains outside every tested retrieval strategy.)
2. Algorithm 4 alone recovers 2,373 cases by rank 20. (Family rescue adds unique coverage beyond simple union behavior.)
3. Algorithms 2, 3, and 4 recover 5,325 cases missed by Algorithm 1. (The external retrieval path contributes broad complementary coverage.)

11 Expanded GitHub Competitor and Algorithm 5 Benchmark

This locked extension compares 53 retrieval configurations on the same 464 fair OCR pairs and 66,257 synthetic clean-core pairs against the same 17,476-family Egyptian medicine catalog. The external inventory contains 75 evaluated, cached, represented, duplicate, or excluded entries. Algorithms 1–3 use their cached row-level predictions; they were not executed again.

11.1 Scope and Counting Rules

A system earns one vote per distinct query-target pair. Hit@1 means the verified family is first. Hit@20 means it occurs anywhere in the first 20. Each system rate has a 95% Wilson interval. Paired system differences use matched case switches, multinomial 95% intervals, exact McNemar tests, and Holm correction. Dense biomedical baselines use their published checkpoints as frozen encoders and index the Egyptian family names directly. No OCR target labels are used for fine-tuning, so these rows measure off-the-shelf retrieval rather than a supervised upper bound. The repository search, completed on 24 July 2026, covered fuzzy and approximate string matching, spelling correction, browser and server search, drug-name normalization, biomedical entity normalization, and sparse or dense retrieval. The final inventory records method-level inclusion decisions instead of counting multiple wrappers around the same ranking rule as new evidence.

Table 16: Benchmark and open-source inventory disposition. The 42 new, six classical, four cached project, and one current project systems sum to 53. The 75-entry repository inventory covers the distinct feasible method families found in the documented search; it is not a claim that every GitHub repository ever created was enumerated.

Benchmark group	Systems or entries	Executed rows	Meaning
Newly evaluated configurations	42	2,802,282	A distinct scoring or retrieval configuration was executed on both locked datasets.
Cached classical baselines	6	400,326	Six existing classical row-level rankings were reused.
Cached project Algorithms 1–4	4	266,884	Existing paired rows were reused; Algorithms 1–3 were not executed again.
Current Algorithm 5	1	66,721	The complete evidence-guided system was evaluated on both locked datasets.
Represented by measured components	7	0	The repository wraps or serves a scoring family already measured directly.
Duplicate implementation	4	0	The implementation changes speed, not candidate ordering.
Not comparable	16	0	The system needs a different ontology, contextual training data, or a server-specific ranking contract.

Table 17: Software versions and hardware used to generate the expanded benchmark. The pinned requirements file matches the package rows; latency values are specific to this machine.

Runtime component	Recorded value
Python	3.11.15
NumPy	2.3.5
pandas	3.0.3
SciPy	1.17.1
scikit-learn	1.9.0
RapidFuzz	3.14.5
jellyfish	1.2.1
symSpellpy	6.9.0
rank-bm25	0.2.2
hnswlib	0.8.0
PyTorch	2.13.0
Transformers	4.57.6
Node.js	22.19.0

Runtime component	Recorded value
Fuse.js	7.5.0
FlexSearch	0.8.212
Operating system	macOS-26.5.2-arm64-arm-64bit
CPU	arm
Logical CPU cores	unknown
Memory	unknown

Table 18: Exact input-file fingerprints and denominator construction. The OCR source retains 595 diagnostic rows; accepted fair filtering and query-target deduplication produce 464 evaluated pairs.

Dataset file	Source rows	Evaluated pairs	Selection	SHA-256
OCR source cases	595	464	Accepted, scored, collision-free unique query-target pairs	cc823ea76a73c88157082b5889021a1d43241c634c8a2a8ed273e6f9119646c2
Synthetic clean core	66257	66257	All unique catalog-resolvable query-target pairs	65f81b58dee1e7127386652383d2f6a8db1734e832dcbc73f14a9b831678886f

Table 19: Exact frozen biomedical checkpoints and shared HNSW settings. Every encoder uses CLS pooling before the pooler, L2 normalization, top 20 retrieval, and index seed 20260723.

Dense system	Checkpoint	Snapshot revision	Dim.	HNSW M	Build ef	Search ef
BiomedBERT dense HNSW	microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext	e1354b7a3a09615f6aba48dfad4b7a613eef7062	768	32	400	500
SapBERT dense HNSW	cambridgeltl/SapBERT-from-PubMedBERT-fulltext	090663c3ae57bf35ffe4d0d468a2a88d03051a4d	768	32	400	500
CODER++ dense HNSW	GanjinZero/coder_eng_pp	3cfddf7cb7bf2f59996b7cdfcb56a121c4d9db1f	768	32	400	500

11.2 Full Leaderboard: 464 fair OCR query-target pairs

Table 20: Complete 53-system leaderboard on 464 locked pairs. Prep is one-time index construction. Median and P95 are measured scoring time per query; vectorized methods divide batch time by batch size, while project algorithms time individual calls. No result means zero returned families. Mean candidates is the implementation-reported list size; external adapters expose at most 20 while project systems can expose a larger internal list, so this column is diagnostic rather than a normalized efficiency comparison.

Rank	System	Method family	Hit@1 (95% CI)	Hit@5	Hit@20 (95% CI)	MRR@20	No result	Mean cand.	Prep. s	Median ms	P95 ms	H@20	misses
1	Algorithm 5, evidence-guided rescue	project algorithm	50.43% [45.90%, 54.96%]	63.58%	72.84% [68.62%, 76.69%]	0.5622	1.94%	29.4	9.805	23.818	66.272		126
2	Algorithm 4, family rescue	project algorithm	47.63% [43.12%, 52.17%]	61.85%	71.12% [66.84%, 75.06%]	0.5375	2.59%	28.5	8.732	14.070	30.179		134
3	SymSpell, distance 3, catalog frequency	delete-index spelling correction	38.58% [34.26%, 43.08%]	50.86%	55.82% [51.27%, 60.27%]	0.4397	12.93%	10.6	0.720	0.245	0.281		205
4	SymSpell, distance 3, uniform frequency	delete-index spelling correction	36.64% [32.38%, 41.11%]	49.14%	56.25% [51.70%, 60.69%]	0.4247	12.93%	10.6	0.861	0.233	0.257		203
5	SymSpell, distance 2, catalog frequency	delete-index spelling correction	34.91% [30.72%, 39.36%]	40.73%	41.59% [37.20%, 46.13%]	0.3755	35.13%	3.0	0.799	0.039	0.046		271
6	Exhaustive Levenshtein	edit distance	34.27% [30.09%, 38.70%]	50.86%	59.27% [54.74%, 63.64%]	0.4170	0.00%	17476.0	0.000	0.704	0.896		189
7	Optimal string alignment	edit distance	34.27% [30.09%, 38.70%]	50.65%	59.27% [54.74%, 63.64%]	0.4144	0.00%	20.0	0.001	0.320	0.476		189
8	Damerau-Levenshtein	edit distance	34.27% [30.09%, 38.70%]	50.65%	59.27% [54.74%, 63.64%]	0.4144	0.00%	20.0	0.001	0.983	1.002		189
9	Fuse.js default Bitap	client-side fuzzy search	32.97% [28.85%, 37.38%]	48.06%	58.84% [54.30%, 63.22%]	0.4016	0.00%	20.0	0.262	45.078	45.078		191
10	RapidFuzz quick ratio	normalized edit similarity	32.76% [28.64%, 37.16%]	49.35%	60.34% [55.83%, 64.69%]	0.4053	0.00%	20.0	0.001	0.269	0.280		184
11	RapidFuzz ratio	normalized edit similarity	32.76% [28.64%, 37.16%]	49.35%	60.34% [55.83%, 64.69%]	0.4053	0.00%	20.0	0.001	0.277	0.286		184
12	Jaro-Winkler	character similarity	32.11% [28.03%, 36.49%]	55.17%	67.46% [63.07%, 71.56%]	0.4254	0.00%	17476.0	0.000	0.671	0.880		151
13	Longest-common-subsequence similarity	ordered subsequence	32.11% [28.03%, 36.49%]	49.57%	56.68% [52.14%, 61.12%]	0.3929	0.00%	20.0	0.001	0.287	0.289		201
14	RapidFuzz token ratio	token-aware similarity	31.47% [27.41%, 35.83%]	48.92%	59.05% [54.52%, 63.43%]	0.3944	0.00%	17476.0	0.000	8.015	9.358		190
15	RapidFuzz token-sort ratio	token-aware edit similarity	31.47% [27.41%, 35.83%]	48.71%	59.05% [54.52%, 63.43%]	0.3933	0.00%	20.0	0.001	1.049	1.063		190
16	RapidFuzz token-set ratio	token-aware edit similarity	31.25% [27.20%, 35.61%]	48.71%	59.05% [54.52%, 63.43%]	0.3919	0.00%	20.0	0.001	2.220	2.477		190
17	Jaro similarity	character similarity	29.53% [25.56%, 33.83%]	47.63%	60.56% [56.04%, 64.90%]	0.3761	0.00%	20.0	0.001	0.331	0.401		183
18	Fuse.js token search	token-aware client-side fuzzy search	26.08% [22.29%, 30.26%]	41.16%	51.29% [46.75%, 55.81%]	0.3243	0.00%	20.0	0.175	43.940	43.940		226
19	Algorithm 2, external fast	project algorithm	25.86% [22.09%, 30.03%]	43.75%	51.29% [46.75%, 55.81%]	0.3292	2.16%	132.2	4.464	3.859	11.881		226
20	Algorithm 3, rank fusion	project algorithm	25.65% [21.88%, 29.81%]	40.95%	53.02% [48.47%, 57.52%]	0.3264	1.08%	42.3	13.347	6.102	15.394		218
21	Algorithm 1, current app	project algorithm	23.71% [20.06%, 27.78%]	31.47%	38.15% [33.84%, 42.65%]	0.2726	2.80%	116.5	4.879	0.737	3.649		287
22	BM25 Okapi over character bigrams	sparse probabilistic retrieval	22.63% [19.06%, 26.65%]	40.73%	50.43% [45.90%, 54.96%]	0.3111	0.65%	19.8	0.074	0.203	0.220		230
23	Character-bigram Dice	q-gram set similarity	21.98% [18.45%, 25.97%]	40.52%	53.02% [48.47%, 57.52%]	0.3115	0.65%	19.8	0.063	0.233	0.248		218
24	BM25L over character trigrams	sparse probabilistic retrieval	21.34% [17.85%, 25.29%]	34.48%	46.77% [42.27%, 51.32%]	0.2708	3.66%	18.6	0.101	0.066	0.072		247
25	BM25 Okapi over character trigrams	sparse probabilistic retrieval	20.69% [17.25%, 24.61%]	35.34%	47.41% [42.91%, 51.96%]	0.2700	3.66%	18.6	0.072	0.067	0.072		244
26	BioSyn-style hybrid, sparse weight 0.50	sparse-dense hybrid	20.47% [17.05%, 24.38%]	40.30%	53.02% [48.47%, 57.52%]	0.2923	0.00%	20.0	32.453	1.523	1.540		218
27	BM25Plus over character trigrams	sparse probabilistic retrieval	20.26% [16.85%, 24.15%]	35.13%	47.41% [42.91%, 51.96%]	0.2676	3.66%	18.6	0.072	0.072	0.075		244
28	BioSyn-style hybrid, sparse weight 0.75	sparse-dense hybrid	19.40% [16.06%, 23.24%]	39.66%	50.86% [46.32%, 55.39%]	0.2818	0.00%	20.0	32.453	1.601	1.622		228
29	BioSyn-style hybrid, sparse weight 0.25	sparse-dense hybrid	18.97% [15.66%, 22.78%]	35.13%	52.16% [47.61%, 56.66%]	0.2656	0.00%	20.0	32.453	1.499	1.535		222
30	QuickUMLS-style 3-gram Dice	q-gram set similarity	18.97% [15.66%, 22.78%]	35.99%	46.12% [41.64%, 50.67%]	0.2591	3.66%	18.6	0.001	0.085	0.091		250
31	QuickUMLS-style 3-gram Jaccard	q-gram set similarity	18.97% [15.66%, 22.78%]	35.99%	46.12% [41.64%, 50.67%]	0.2591	3.66%	18.6	0.067	0.091	0.093		250
32	Character 3-gram TF-IDF	sparse character retrieval	18.32% [15.06%, 22.09%]	35.78%	47.41% [42.91%, 51.96%]	0.2580	3.66%	420.1	0.072	0.504	0.685		244
33	QuickUMLS-style 3-gram cosine	q-gram set similarity	17.67% [14.47%, 21.40%]	35.78%	45.91% [41.42%, 50.45%]	0.2513	3.66%	18.6	0.000	0.087	0.094		251
34	CODER++ dense HNSW	biomedical dense retrieval	17.67% [14.47%, 21.40%]	26.08%	32.33% [28.23%, 36.71%]	0.2141	0.00%	20.0	38.797	1.677	1.712		314
35	xMEN-style SapBERT plus TF-IDF RRF	rank-fusion ensemble	16.81% [13.68%, 20.48%]	37.50%	52.37% [47.83%, 56.88%]	0.2577	0.00%	20.0	32.453	1.530	1.599		221
36	NYSIIS plus edit ranking	phonetic encoding	16.16% [13.09%, 19.79%]	28.66%	39.22% [34.89%, 43.74%]	0.2209	0.00%	20.0	0.010	0.293	0.297		282
37	Match Rating Codex plus edit ranking	phonetic encoding	15.73% [12.70%, 19.33%]	31.25%	46.34% [41.85%, 50.88%]	0.2316	0.43%	19.9	0.059	0.229	0.233		249
38	preon without token matching	staged drug-name normalization	15.30% [12.31%, 18.86%]	16.16%	16.16% [13.09%, 19.79%]	0.1573	79.96%	0.2	0.004	0.209	0.209		389
39	preon default drug normalizer	staged drug-name normalization	15.09% [12.12%, 18.63%]	15.95%	15.95% [12.90%, 19.56%]	0.1552	79.53%	0.2	0.003	0.158	0.169		390
40	Metaphone plus edit ranking	phonetic encoding	12.07% [9.41%, 15.35%]	21.12%	34.05% [29.89%, 38.48%]	0.1624	0.65%	19.9	0.007	0.255	0.293		306
41	SapBERT dense HNSW	biomedical dense retrieval	11.42% [8.84%, 14.64%]	22.63%	38.58% [34.26%, 43.08%]	0.1676	0.00%	20.0	32.928	1.374	1.380		285
42	RapidFuzz weighted ratio	adaptive edit similarity	11.21% [8.65%, 14.40%]	40.30%	56.25% [51.70%, 60.69%]	0.2356	0.00%	20.0	0.001	3.618	4.299		203
43	QuickUMLS-style 3-gram overlap	q-gram set similarity	8.41% [6.21%, 11.28%]	24.78%	37.93% [33.63%, 42.43%]	0.1595	3.66%	18.6	0.001	0.090	0.096		288
44	FlexSearch tolerant LatinAdvanced	indexed fuzzy and phonetic search	7.54% [5.47%, 10.31%]	8.84%	9.70% [7.33%, 12.73%]	0.0818	66.59%	2.4	0.456	0.027	0.027		419
45	Phonetic baseline	phonetic	6.90% [4.93%, 9.57%]	16.81%	23.28% [19.66%, 27.33%]	0.1102	0.00%	17476.0	0.000	0.472	0.533		356

Rank System	Method family	Hit@1 (95% CI)	Hit@5	Hit@20 (95% CI)	MRR@20	No result	Mean	cand. Prep. s	Median	ms P95	ms H@20	misses
46 RapidFuzz partial ratio	substring edit similarity	6.25% [4.39%, 8.83%]	30.17%	44.40% [39.94%, 48.94%]	0.1647	0.00%	20.0	0.001	1.103	1.194	258	
47 Soundex plus edit ranking	phonetic encoding	5.60% [3.85%, 8.08%]	19.18%	24.35% [20.67%, 28.46%]	0.1022	0.00%	20.0	0.006	0.181	0.195	351	
48 FlexSearch tolerant LatinExtra	indexed aggressive phonetic search	4.74% [3.15%, 7.07%]	12.72%	18.75% [15.46%, 22.55%]	0.0771	20.04%	10.1	0.292	0.024	0.024	377	
49 FlexSearch full LatinAdvanced	indexed substring and phonetic search	4.74% [3.15%, 7.07%]	5.82%	7.54% [5.47%, 10.31%]	0.0546	63.79%	4.0	0.623	0.053	0.053	429	
50 FlexSearch tolerant default	indexed fuzzy search	4.74% [3.15%, 7.07%]	4.74%	4.74% [3.15%, 7.07%]	0.0474	83.19%	0.9	0.535	0.013	0.013	442	
51 BiomedBERT dense HNSW	biomedical dense retrieval	2.80% [1.64%, 4.73%]	5.39%	7.33% [5.29%, 10.07%]	0.0412	0.00%	20.0	30.095	1.190	1.221	430	
52 Exact or prefix match	exact and prefix	2.59% [1.49%, 4.47%]	2.80%	3.02% [1.81%, 5.00%]	0.0272	93.53%	2.2	0.002	0.706	1.141	450	
53 preon without partial edit matching	staged drug-name normalization	0.65% [0.22%, 1.88%]	0.65%	0.65% [0.22%, 1.88%]	0.0065	98.71%	0.0	0.005	0.005	0.006	461	

11.3 Full Leaderboard: 66,257 synthetic clean-core query-target pairs

Table 21: Complete 53-system leaderboard on 66,257 locked pairs. Prep is one-time index construction. Median and P95 are measured scoring time per query; vectorized methods divide batch time by batch size, while project algorithms time individual calls. No result means zero returned families. Mean candidates is the implementation-reported list size; external adapters expose at most 20 while project systems can expose a larger internal list, so this column is diagnostic rather than a normalized efficiency comparison.

Rank	System	Method family	Hit@1 (95% CI)	Hit@5	Hit@20 (95% CI)	MRR@20	No result	Mean cand.	Prep. s	Median ms	P95 ms	H@20	misses
1	Algorithm 5, evidence-guided rescue	project algorithm	98.19% [98.08%, 98.29%]	99.65%	100.00% [99.99%, 100.00%]	0.9884	0.00%	24.6	15.520	21.632	46.782		1
2	Algorithm 4, family rescue	project algorithm	97.19% [97.06%, 97.31%]	99.24%	99.52% [99.46%, 99.57%]	0.9809	0.05%	24.4	8.737	14.620	31.509		321
3	Damerau-Levenshtein	edit distance	94.78% [94.61%, 94.95%]	98.69%	99.64% [99.59%, 99.68%]	0.9653	0.00%	20.0	0.001	1.152	1.533		239
4	Optimal string alignment	edit distance	94.75% [94.58%, 94.92%]	98.64%	99.58% [99.52%, 99.62%]	0.9649	0.00%	20.0	0.001	0.274	0.287		281
5	SymSpell, distance 3, uniform frequency	delete-index spelling correction	94.51% [94.33%, 94.68%]	98.18%	98.86% [98.77%, 98.93%]	0.9614	0.75%	4.0	1.118	0.166	0.222		758
6	SymSpell, distance 3, catalog frequency	delete-index spelling correction	94.22% [94.04%, 94.40%]	98.08%	98.80% [98.71%, 98.88%]	0.9592	0.75%	4.0	0.885	0.165	0.223		795
7	Exhaustive Levenshtein	edit distance	94.05% [93.87%, 94.23%]	98.40%	99.44% [99.38%, 99.50%]	0.9599	0.00%	17476.0	0.000	0.736	0.799		370
8	Longest-common-subsequence similarity	ordered subsequence	93.60% [93.41%, 93.78%]	98.01%	99.21% [99.14%, 99.28%]	0.9556	0.00%	20.0	0.001	0.258	0.267		521
9	RapidFuzz quick ratio	normalized edit similarity	92.79% [92.59%, 92.99%]	97.67%	99.00% [98.93%, 99.08%]	0.9497	0.00%	20.0	0.001	0.266	0.276		660
10	RapidFuzz ratio	normalized edit similarity	92.79% [92.59%, 92.99%]	97.67%	99.00% [98.93%, 99.08%]	0.9497	0.00%	20.0	0.001	0.270	0.280		660
11	Algorithm 2, external fast	project algorithm	92.41% [92.20%, 92.60%]	96.33%	96.88% [96.74%, 97.01%]	0.9414	0.15%	93.8	4.345	5.729	21.904		2068
12	Algorithm 3, rank fusion	project algorithm	91.84% [91.63%, 92.04%]	96.18%	97.12% [96.99%, 97.24%]	0.9381	0.04%	34.8	12.792	8.346	24.457		1908
13	Jaro-Winkler	character similarity	89.57% [89.33%, 89.80%]	96.26%	98.43% [98.33%, 98.52%]	0.9256	0.00%	17476.0	0.000	0.725	0.791		1042
14	Jaro similarity	character similarity	89.29% [89.05%, 89.52%]	96.15%	98.38% [98.28%, 98.47%]	0.9239	0.00%	20.0	0.001	0.291	0.301		1076
15	SymSpell, distance 2, catalog frequency	delete-index spelling correction	87.84% [87.59%, 88.08%]	90.39%	90.55% [90.33%, 90.78%]	0.8897	8.64%	1.3	0.903	0.032	0.046		6258
16	Character-bigram Dice	q-gram set similarity	84.54% [84.26%, 84.81%]	92.39%	95.56% [95.40%, 95.72%]	0.8808	0.00%	20.0	0.063	0.283	0.374		2941
17	Algorithm 1, current app	project algorithm	83.86% [83.58%, 84.14%]	89.72%	92.23% [92.03%, 92.43%]	0.8637	1.94%	35.9	4.722	1.225	4.575		5146
18	BM25 Okapi over character bigrams	sparse probabilistic retrieval	82.03% [81.73%, 82.32%]	90.52%	94.40% [94.22%, 94.57%]	0.8585	0.00%	20.0	0.071	0.262	0.353		3712
19	Fuse.js default Bitap	client-side fuzzy search	81.22% [80.92%, 81.52%]	90.76%	95.00% [94.83%, 95.17%]	0.8551	0.00%	19.0	0.125	64.059	155.589		3311
20	BM25L over character trigrams	sparse probabilistic retrieval	78.96% [78.65%, 79.27%]	87.48%	91.82% [91.61%, 92.02%]	0.8281	0.29%	19.8	0.098	0.079	0.100		5421
21	QuickUMLS-style 3-gram Dice	q-gram set similarity	78.34% [78.03%, 78.66%]	88.30%	92.52% [92.32%, 92.72%]	0.8286	0.29%	19.8	0.000	0.098	0.119		4954
22	QuickUMLS-style 3-gram Jaccard	q-gram set similarity	78.34% [78.03%, 78.66%]	88.30%	92.52% [92.32%, 92.72%]	0.8286	0.29%	19.8	0.067	0.099	0.119		4954
23	BM25 Okapi over character trigrams	sparse probabilistic retrieval	78.25% [77.93%, 78.56%]	87.81%	92.17% [91.97%, 92.37%]	0.8257	0.29%	19.8	0.072	0.079	0.100		5186
24	BM25Plus over character trigrams	sparse probabilistic retrieval	78.19% [77.88%, 78.51%]	87.79%	92.17% [91.96%, 92.37%]	0.8254	0.29%	19.8	0.071	0.079	0.100		5190
25	BioSyn-style hybrid, sparse weight 0.50	sparse-dense hybrid	78.17% [77.85%, 78.48%]	89.45%	93.78% [93.60%, 93.96%]	0.8327	0.00%	20.0	30.436	1.843	1.891		4119
26	Fuse.js token search	token-aware client-side fuzzy search	78.06% [77.74%, 78.37%]	88.30%	93.34% [93.15%, 93.53%]	0.8270	0.00%	19.4	0.146	63.719	150.375		4413
27	BioSyn-style hybrid, sparse weight 0.75	sparse-dense hybrid	77.88% [77.56%, 78.19%]	88.88%	93.40% [93.21%, 93.58%]	0.8283	0.00%	20.0	30.436	1.839	1.909		4375
28	QuickUMLS-style 3-gram cosine	q-gram set similarity	77.46% [77.14%, 77.78%]	88.06%	92.40% [92.19%, 92.60%]	0.8224	0.29%	19.8	0.001	0.100	0.121		5038
29	NYSIIS plus edit ranking	phonetic encoding	77.01% [76.68%, 77.33%]	88.41%	93.90% [93.72%, 94.08%]	0.8212	0.00%	20.0	0.060	0.270	0.280		4040
30	Metaphone plus edit ranking	phonetic encoding	75.55% [75.23%, 75.88%]	88.15%	94.62% [94.45%, 94.79%]	0.8123	0.00%	20.0	0.058	0.224	0.233		3564
31	Character 3-gram TF-IDF	sparse character retrieval	75.21% [74.88%, 75.53%]	86.72%	91.66% [91.45%, 91.87%]	0.8042	0.29%	921.3	0.064	0.513	0.621		5525
32	preon default drug normalizer	staged drug-name normalization	74.16% [73.82%, 74.49%]	74.37%	74.37% [74.03%, 74.70%]	0.7426	25.39%	0.8	0.004	0.142	0.161		16983
33	preon without token matching	staged drug-name normalization	74.16% [73.82%, 74.49%]	74.37%	74.37% [74.03%, 74.70%]	0.7426	25.39%	0.8	0.004	0.142	0.161		16983
34	Phonetic baseline	phonetic	69.66% [69.30%, 70.00%]	84.92%	92.07% [91.86%, 92.27%]	0.7642	0.00%	17475.7	0.000	0.692	0.784		5254
35	BioSyn-style hybrid, sparse weight 0.25	sparse-dense hybrid	66.01% [65.65%, 66.37%]	83.40%	90.20% [89.97%, 90.43%]	0.7376	0.00%	20.0	30.436	1.843	1.881		6491
36	Match Rating Codex plus edit ranking	phonetic encoding	65.82% [65.46%, 66.18%]	85.27%	94.58% [94.40%, 94.75%]	0.7452	0.00%	20.0	0.060	0.206	0.217		3592
37	RapidFuzz token ratio	token-aware similarity	65.67% [65.31%, 66.03%]	78.25%	85.47% [85.20%, 85.74%]	0.7135	0.00%	17476.0	0.000	8.141	12.214		9626
38	RapidFuzz token-set ratio	token-aware edit similarity	65.66% [65.29%, 66.02%]	78.23%	85.46% [85.18%, 85.72%]	0.7133	0.00%	20.0	0.003	2.518	2.718		9637
39	RapidFuzz token-sort ratio	token-aware edit similarity	65.64% [65.28%, 66.01%]	78.16%	85.46% [85.19%, 85.73%]	0.7130	0.00%	20.0	0.001	0.899	1.011		9634
40	RapidFuzz weighted ratio	adaptive edit similarity	55.73% [55.35%, 56.11%]	93.49%	97.20% [97.07%, 97.32%]	0.7191	0.00%	20.0	0.001	4.371	5.652		1857
41	xMEN-style SapBERT plus TF-IDF RRF	rank-fusion ensemble	48.85% [48.47%, 49.23%]	78.51%	92.30% [92.09%, 92.50%]	0.6154	0.00%	20.0	30.436	1.820	1.860		5104
42	CODER++ dense HNSW	biomedical dense retrieval	42.56% [42.18%, 42.93%]	56.93%	66.26% [65.90%, 66.62%]	0.4907	0.00%	20.0	87.875	1.686	1.999		22352
43	QuickUMLS-style 3-gram overlap	q-gram set similarity	36.74% [36.38%, 37.11%]	80.43%	88.37% [88.12%, 88.61%]	0.5531	0.29%	19.8	0.001	0.103	0.124		7706
44	SapBERT dense HNSW	biomedical dense retrieval	27.67% [27.33%, 28.01%]	44.00%	56.45% [56.08%, 56.83%]	0.3505	0.00%	20.0	95.131	1.728	1.755		28852
45	FlexSearch tolerant LatinAdvanced	indexed fuzzy and phonetic search	20.32% [20.02%, 20.63%]	21.65%	21.89% [21.58%, 22.21%]	0.2091	75.80%	0.5	0.594	0.007	0.014		51752

Rank System	Method family	Hit@1 (95% CI)	Hit@5	Hit@20 (95% CI)	MRR@20	No result	Mean cand.	Prep. s	Median ms	P95 ms	H@20	misses
46 RapidFuzz partial ratio	substring edit similarity	16.68% [16.40%, 16.97%]	80.57%	92.90% [92.70%, 93.09%]	0.4174	0.00%	20.0	0.001	1.515	2.294		4705
47 FlexSearch tolerant LatinExtra	indexed aggressive phonetic search	15.43% [15.15%, 15.70%]	24.29%	29.51% [29.16%, 29.85%]	0.1933	57.17%	3.2	0.391	0.006	0.009		46707
48 FlexSearch full LatinAdvanced	indexed substring and phonetic search	13.94% [13.68%, 14.21%]	15.27%	15.75% [15.47%, 16.03%]	0.1454	80.18%	0.7	0.425	0.005	0.016		55823
49 Soundex plus edit ranking	phonetic encoding	13.09% [12.83%, 13.35%]	41.74%	67.73% [67.37%, 68.08%]	0.2593	0.00%	20.0	0.058	0.158	0.170		21383
50 FlexSearch tolerant default	indexed fuzzy search	10.56% [10.33%, 10.79%]	10.78%	10.91% [10.68%, 11.15%]	0.1066	88.80%	0.2	0.442	0.005	0.014		59027
51 BiomedBERT dense HNSW	biomedical dense retrieval	6.22% [6.04%, 6.41%]	9.60%	13.01% [12.75%, 13.26%]	0.0780	0.00%	20.0	126.385	1.869	1.889		57640
52 Exact or prefix match	exact and prefix	3.00% [2.87%, 3.13%]	3.01%	3.01% [2.89%, 3.15%]	0.0301	96.78%	0.1	0.002	0.708	0.805		64261
53 preon without partial edit matching	staged drug-name normalization	0.00% [0.00%, 0.01%]	0.00%	0.00% [0.00%, 0.01%]	0.0000	100.00%	0.0	0.004	0.005	0.006		66257

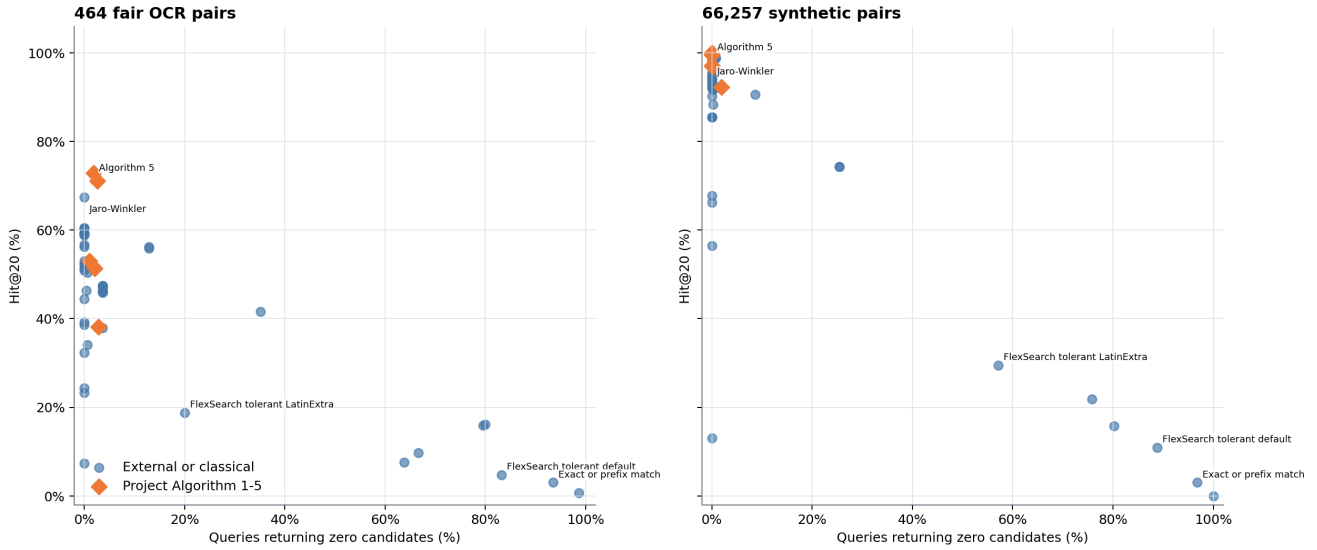


Figure 30: Each point is one retrieval system. The x-axis is the percentage of queries returning zero candidates, and the y-axis is Hit@20; upper-left therefore combines broad candidate coverage with high verified-family recovery. The left panel contains 464 fair OCR pairs and the right contains 66,257 synthetic pairs.

Figure insights

1. A zero-candidate error is a retrieval failure and cannot be fixed by reranking the returned list.
2. On OCR, FlexSearch tolerant default returns no candidate for 83.19% of queries; LatinExtra lowers that rate to 20.04% and raises Hit@20 from 4.74% to 18.75%.

11.4 System Distribution and Efficiency Frontier

Distribution statistics use all 53 systems. The Pareto table keeps a system only when no measured alternative is both at least as accurate at Hit@1 and at least as fast in median measured scoring time per query.

Table 22: Best, worst, mean, median, and sample standard deviation across the complete 53-system roster. Rate metrics are fractions; time metrics are milliseconds.

Dataset	Metric	Best system	Best Worst system	Worst	Mean	Median	Std. dev.
464 fair OCR pairs	Hit@1	Algorithm 5, evidence-guided rescue	0.5043 preon without partial edit matching	0.0065	0.2122	0.2026	0.1186
464 fair OCR pairs	Hit@5	Algorithm 5, evidence-guided rescue	0.6358 preon without partial edit matching	0.0065	0.3467	0.3599	0.1578
464 fair OCR pairs	Hit@20	Algorithm 5, evidence-guided rescue	0.7284 preon without partial edit matching	0.0065	0.4359	0.4741	0.1865
464 fair OCR pairs	MRR@20	Algorithm 5, evidence-guided rescue	0.5622 preon without partial edit matching	0.0065	0.2716	0.2700	0.1338
464 fair OCR pairs	No-result rate	Exhaustive Levenshtein	0.0000 preon without partial edit matching	0.9871	0.1299	0.0065	0.2766
464 fair OCR pairs	Preparation time (ms)	Exhaustive Levenshtein	0.0000 CODER++ dense HNSW	38797.0389	5249.2272	67.2166	11304.2950
464 fair OCR pairs	Median measured time/query (ms)	preon without partial edit matching	0.0055 Fuse.js default Bitap	45.0781	3.2545	0.3197	9.1182
464 fair OCR pairs	P95 measured time/query (ms)	preon without partial edit matching	0.0057 Algorithm 5, evidence-guided rescue	66.2715	4.8193	0.4009	12.8323
66,257 synthetic pairs	Hit@1	Algorithm 5, evidence-guided rescue	0.9819 preon without partial edit matching	0.0000	0.6639	0.7788	0.2940
66,257 synthetic pairs	Hit@5	Algorithm 5, evidence-guided rescue	0.9965 preon without partial edit matching	0.0000	0.7694	0.8815	0.2833
66,257 synthetic pairs	Hit@20	Algorithm 5, evidence-guided rescue	1.0000 preon without partial edit matching	0.0000	0.8119	0.9252	0.2831
66,257 synthetic pairs	MRR@20	Algorithm 5, evidence-guided rescue	0.9884 preon without partial edit matching	0.0000	0.7101	0.8254	0.2842
66,257 synthetic pairs	No-result rate	Algorithm 5, evidence-guided rescue	0.0000 preon without partial edit matching	1.0000	0.1064	0.0000	0.2705
66,257 synthetic pairs	Preparation time (ms)	Jaro-Winkler	0.0002 BiomedBERT dense HNSW	126385.0450	9114.1593	63.9863	25028.3117
66,257 synthetic pairs	Median measured time/query (ms)	FlexSearch full LatinAdvanced	0.0049 Fuse.js default Bitap	64.0586	4.1132	0.2827	12.5382
66,257 synthetic pairs	P95 measured time/query (ms)	preon without partial edit matching	0.0061 Fuse.js default Bitap	155.5889	9.0998	0.3527	30.0443

Table 23: Measured Hit@1 versus scoring-time Pareto frontier. Timing includes vectorized batch methods and per-query project methods, so this table identifies measured tradeoffs rather than production throughput guarantees.

Dataset	Non-dominated system	Hit@1	Hit@20	Median ms	P95 ms	Prep. s
464 fair OCR pairs	preon without partial edit matching	0.65%	0.65%	0.0055	0.0057	0.005
464 fair OCR pairs	FlexSearch tolerant default	4.74%	4.74%	0.0129	0.0129	0.535
464 fair OCR pairs	FlexSearch tolerant LatinAdvanced	7.54%	9.70%	0.0271	0.0271	0.456
464 fair OCR pairs	SymSpell, distance 2, catalog frequency	34.91%	41.59%	0.0390	0.0456	0.799
464 fair OCR pairs	SymSpell, distance 3, uniform frequency	36.64%	56.25%	0.2331	0.2570	0.861
464 fair OCR pairs	SymSpell, distance 3, catalog frequency	38.58%	55.82%	0.2452	0.2812	0.720
464 fair OCR pairs	Algorithm 4, family rescue	47.63%	71.12%	14.0704	30.1791	8.732
464 fair OCR pairs	Algorithm 5, evidence-guided rescue	50.43%	72.84%	23.8182	66.2715	9.805
66,257 synthetic pairs	FlexSearch full LatinAdvanced	13.94%	15.75%	0.0049	0.0156	0.425
66,257 synthetic pairs	FlexSearch tolerant LatinExtra	15.43%	29.51%	0.0059	0.0093	0.391
66,257 synthetic pairs	FlexSearch tolerant LatinAdvanced	20.32%	21.89%	0.0068	0.0138	0.594
66,257 synthetic pairs	SymSpell, distance 2, catalog frequency	87.84%	90.55%	0.0320	0.0455	0.903
66,257 synthetic pairs	SymSpell, distance 3, catalog frequency	94.22%	98.80%	0.1652	0.2226	0.885
66,257 synthetic pairs	SymSpell, distance 3, uniform frequency	94.51%	98.86%	0.1655	0.2218	1.118
66,257 synthetic pairs	Optimal string alignment	94.75%	99.58%	0.2742	0.2873	0.001
66,257 synthetic pairs	Damerau-Levenshtein	94.78%	99.64%	1.1520	1.5329	0.001
66,257 synthetic pairs	Algorithm 4, family rescue	97.19%	99.52%	14.6203	31.5090	8.737
66,257 synthetic pairs	Algorithm 5, evidence-guided rescue	98.19%	100.00%	21.6318	46.7816	15.520

11.5 Algorithm 5 Source-Revision Audit

The earlier clean-core acceptance artifact and the current Algorithm 5 source are paired by the same 66,257 case IDs. The current-source result is the value used everywhere in the expanded benchmark; the older value remains historical.

Table 24: Paired source-revision audit. The current OCR-oriented rules gain 44 and lose 55 clean-core Hit@1 cases overall; holdout gains one net Hit@1 case, while development loses 12 and one development target moves outside the top 20.

Split	Cases	Old H@1	Current H@1	H@1 gains	H@1 losses	Old H@20	Current H@20	H@20 gains	H@20 losses	Rank changes
All	66257	98.2055%	98.1889%	44	55	100.0000%	99.9985%	0	1	187
Development	52906	98.2157%	98.1930%	36	48	100.0000%	99.9981%	0	1	156
Holdout	13351	98.1649%	98.1724%	8	7	100.0000%	100.0000%	0	0	31

Table 25: Concrete paired changes between the historical and current Algorithm 5 source. Rank 999 means the verified family is outside the returned top 20.

Change	Input	Expected	Old top 1	Old rank	Current top 1	Current rank
Hit@20 loss	aclox	ADOX	AMOCLOX	20	AMOCLOX	999
Hit@1 gain	catoiozt	CALCI OST	CATAFAST	2	CALCI OST	1
Hit@1 gain	mokce	MEGACE	FORCE	2	MEGACE	1
			CARBAMIDE			
Hit@1 gain	intreda	ENTRADA	INTRA D	2	ENTRADA	1
Hit@1 gain	randak	RANITAK	RANDO	2	RANITAK	1
Hit@1 gain	soivinn	SOLVIN N	SOLVIN	20	SOLVIN N	1
Hit@1 loss	claremmunhair	DERMENUM	DERMENUM	1	TIA DO HAIR	2
	conditioner	HAIR	HAIR		CONDITIONER	
		CONDITIONER	CONDITIONER			
Hit@1 loss	ferroza	FERROZAL	FERROZAL	1	FERRONZA	2
Hit@1 loss	amezor	AMOSAR	AMOSAR	1	AZOR ANTI DANDRUFF	2
Hit@1 loss	alcsi	ALEKSA	ALEKSA	1	ACTI COLLA	2
Hit@1 loss	anfaartenanti	ANAFORTEN	ANAFORTEN	1	DANDRENE	2
	clandruff	ANTI DANDRUFF	ANTI DANDRUFF		ANTIDANDRUFF	
rank-only change	sorloc	SOTALOC	PERLOC	3	PERLOC	4
rank-only change	ccttex	COLDEX	CLATEX	7	CLATEX	8

Change	Input	Expected	Old top 1	Old rank	Current top 1	Current rank
rank-only change	horno	HOMO	HORMO	7	HORMO	9

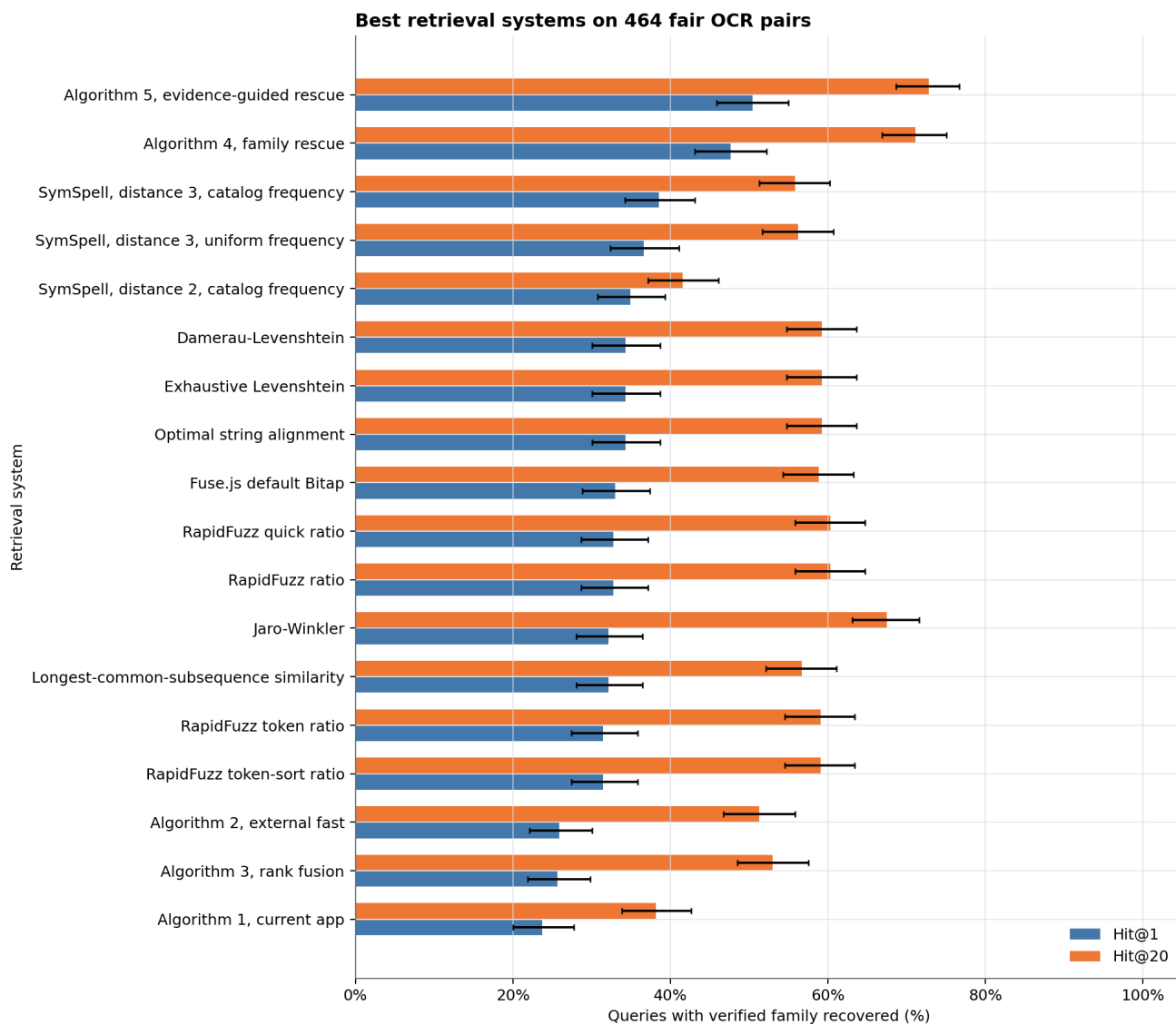


Figure 31: Bars report Hit@1 and Hit@20 on the 464 fair OCR pairs. The x-axis is the percentage of queries whose verified family is recovered; the y-axis lists the leading systems plus project algorithms. Whiskers are 95% Wilson confidence intervals.

Figure insights

1. Algorithm 5 reaches 50.43% Hit@1 and 72.84% Hit@20.
2. The strongest external Hit@1 system is SymSpell, distance 3, catalog frequency at 38.58%; the strongest external Hit@20 system is Jaro-Winkler at 67.46%.

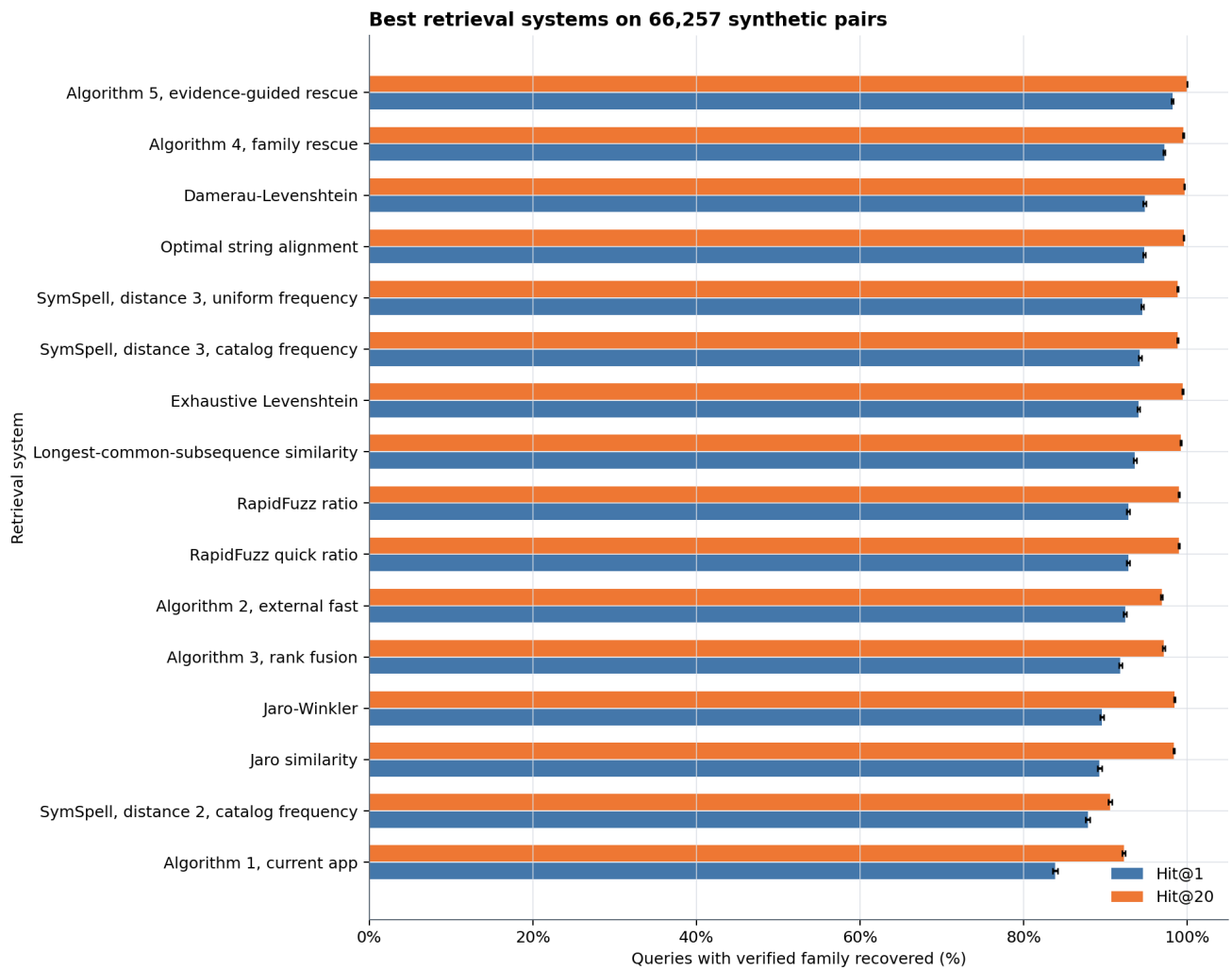


Figure 32: Bars report Hit@1 and Hit@20 on 66,257 synthetic clean-core pairs. The x-axis is recovery percentage and the y-axis lists systems. Whiskers are 95% Wilson confidence intervals.

Figure insights

1. Algorithm 5 reaches 98.19% Hit@1 and 100.00% Hit@20.
2. The best external Hit@1 result is Damerau-Levenshtein at 94.78%.
3. The best external Hit@20 result is Damerau-Levenshtein at 99.64%.

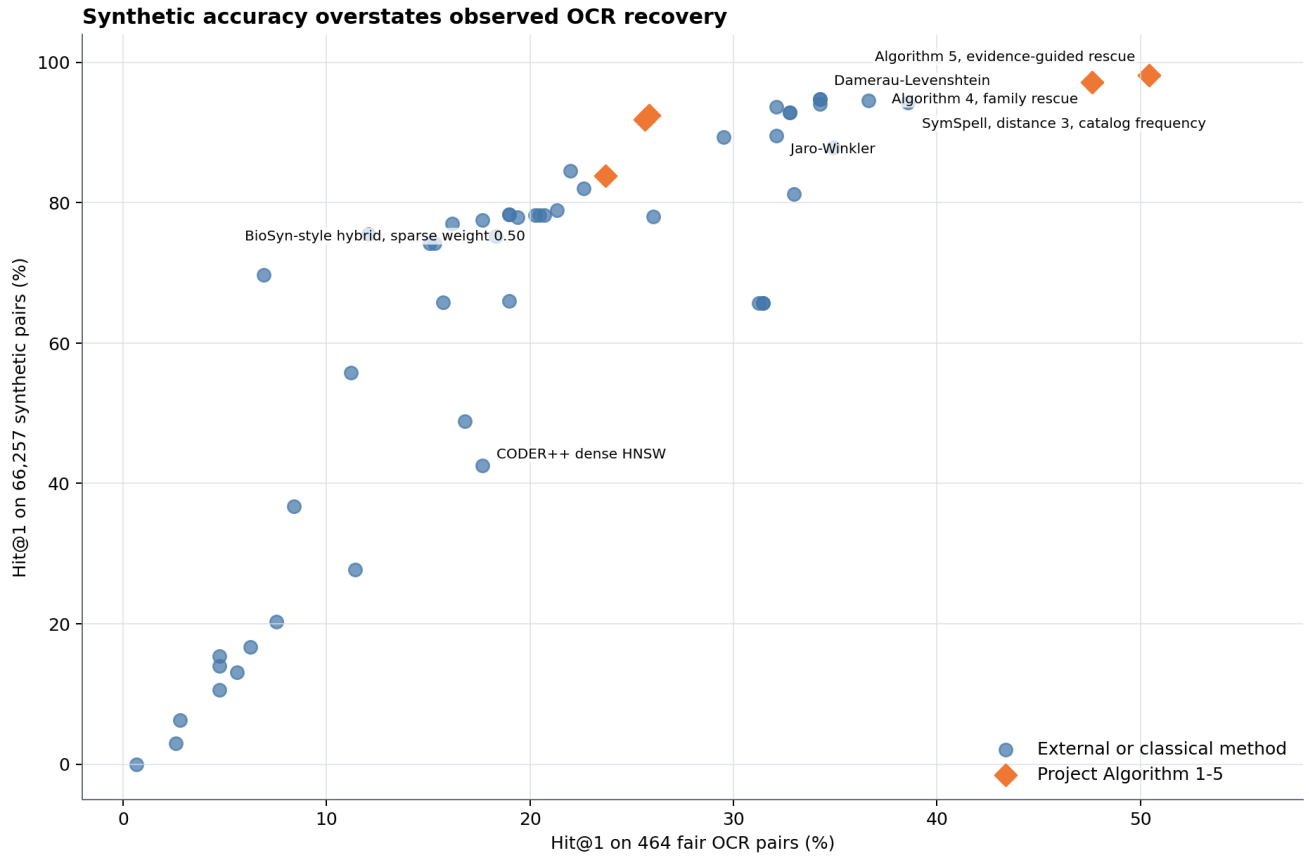


Figure 33: Each point is one system. The x-axis is Hit@1 on real OCR pairs and the y-axis is Hit@1 on synthetic pairs. Upper-left points perform well on generated edits but poorly on observed OCR corruption.

Figure insights

1. System ordering transfers strongly but not perfectly (Spearman rank correlation $\rho = 0.886$), while absolute synthetic scores remain much higher.
2. Damerau-Levenshtein is strong on controlled edits, while Algorithm 5 retains the best observed OCR result.

Table 26: Every paired system comparison against Algorithm 5 on 464 fair OCR pairs. A5-only and comparison-only counts are the discordant cases. Exact McNemar p-values are Holm-adjusted across the 52 comparisons; deltas and paired 95% intervals are in percentage points.

Comparison	Comp. H@1	A5-comp. H@1 (95% CI)	A5-only 1	Comp.-only 1	Holm p1	Comp. H@20	A5-comp. H@20 (95% CI)	A5-only 20	Comp.-only 20	Holm p20
Algorithm 4, family rescue	47.63%	+2.80 [+1.08, +4.53]	15	2	0.00235	71.12%	+1.72 [+0.22, +3.23]	11	3	0.05737
SymSpell, distance 3, catalog frequency	38.58%	+11.85 [+8.19, +15.52]	68	13	7.617e-10	55.82%	+17.03 [+13.58, +20.69]	81	2	1.298e-20
SymSpell, distance 3, uniform frequency	36.64%	+13.79 [+10.34, +17.46]	71	7	5.8e-14	56.25%	+16.59 [+13.15, +20.04]	79	2	4.122e-20
SymSpell, distance 2, catalog frequency	34.91%	+15.52 [+11.85, +19.40]	83	11	3.202e-14	41.59%	+31.25 [+27.16, +35.56]	145	0	1.704e-42
Damerau-Levenshtein	34.27%	+16.16 [+12.93, +19.61]	75	0	7.412e-22	59.27%	+13.58 [+10.34, +16.81]	65	2	3.706e-16
Optimal string alignment	34.27%	+16.16 [+12.93, +19.61]	75	0	7.412e-22	59.27%	+13.58 [+10.34, +16.81]	65	2	3.706e-16
Exhaustive Levenshtein	34.27%	+16.16 [+12.93, +19.61]	75	0	7.412e-22	59.27%	+13.58 [+10.34, +16.81]	65	2	3.706e-16
Fuse.js default Bitap	32.97%	+17.46 [+13.79, +21.12]	86	5	3.186e-19	58.84%	+14.01 [+10.78, +17.46]	68	3	4.552e-16
RapidFuzz ratio	32.76%	+17.67 [+13.79, +21.55]	90	8	6.525e-18	60.34%	+12.50 [+9.48, +15.73]	61	3	2.371e-14
RapidFuzz quick ratio	32.76%	+17.67 [+13.79, +21.55]	90	8	6.525e-18	60.34%	+12.50 [+9.48, +15.73]	61	3	2.371e-14
Jaro-Winkler	32.11%	+18.32 [+14.44, +22.41]	94	9	3.822e-18	67.46%	+5.39 [+2.37, +8.41]	39	14	0.001605
Longest-common-subsequence similarity	32.11%	+18.32 [+14.66, +21.98]	87	2	2.071e-22	56.68%	+16.16 [+12.93, +19.61]	77	2	1.464e-19
RapidFuzz token ratio	31.47%	+18.97 [+15.09, +23.06]	96	8	2.765e-19	59.05%	+13.79 [+10.56, +17.24]	67	3	7.756e-16
RapidFuzz token-sort ratio	31.47%	+18.97 [+15.09, +23.06]	96	8	2.765e-19	59.05%	+13.79 [+10.56, +17.24]	67	3	7.756e-16
RapidFuzz token-set ratio	31.25%	+19.18 [+15.09, +23.28]	97	8	1.645e-19	59.05%	+13.79 [+10.56, +17.24]	67	3	7.756e-16
Jaro similarity	29.53%	+20.91 [+16.81, +25.00]	104	7	4.202e-22	60.56%	+12.28 [+9.05, +15.73]	64	7	3.779e-12
Fuse.js token search	26.08%	+24.35 [+20.26, +28.45]	117	4	1.124e-28	51.29%	+21.55 [+17.67, +25.43]	102	2	1.346e-26
Algorithm 2, external fast	25.86%	+24.57 [+20.69, +28.66]	116	2	8.03e-31	51.29%	+21.55 [+17.67, +25.43]	103	3	1.175e-25
Algorithm 3, rank fusion	25.65%	+24.78 [+20.91, +28.88]	117	2	4.298e-31	53.02%	+19.83 [+16.16, +23.71]	95	3	2.179e-23
Algorithm 1, current app	23.71%	+26.72 [+22.63, +30.82]	125	1	7.464e-35	38.15%	+34.70 [+30.39, +39.01]	161	0	2.668e-47
BM25 Okapi over character bigrams	22.63%	+27.80 [+23.49, +32.11]	137	8	3.41e-30	50.43%	+22.41 [+18.53, +26.51]	109	5	3.405e-25
Character-bigram Dice	21.98%	+28.45 [+24.14, +32.76]	138	6	2.193e-32	53.02%	+19.83 [+15.95, +23.92]	99	7	1.226e-20
BM25L over character trigrams	21.34%	+29.09 [+24.78, +33.41]	140	5	5.094e-34	46.77%	+26.08 [+21.98, +30.17]	125	4	1.002e-30
BM25 Okapi over character trigrams	20.69%	+29.74 [+25.43, +34.05]	143	5	7.7e-35	47.41%	+25.43 [+21.34, +29.53]	122	4	7.051e-30
BioSyn-style hybrid, sparse weight 0.50	20.47%	+29.96 [+25.43, +34.48]	145	6	2.501e-34	53.02%	+19.83 [+15.73, +23.92]	100	8	3.766e-20
BM25Plus over character trigrams	20.26%	+30.17 [+25.86, +34.48]	144	4	3.097e-36	47.41%	+25.43 [+21.34, +29.53]	122	4	7.051e-30
BioSyn-style hybrid, sparse weight 0.75	19.40%	+31.03 [+26.51, +35.56]	149	5	1.776e-36	50.86%	+21.98 [+17.89, +26.08]	109	7	2.516e-23
BioSyn-style hybrid, sparse weight 0.25	18.97%	+31.47 [+26.94, +35.99]	153	7	1.737e-35	52.16%	+20.69 [+16.59, +25.00]	106	10	3.678e-20
QuickUMLS-style 3-gram Dice	18.97%	+31.47 [+27.16, +35.99]	150	4	6.283e-38	46.12%	+26.72 [+22.63, +30.82]	126	2	1.65e-33
QuickUMLS-style 3-gram Jaccard	18.97%	+31.47 [+27.16, +35.99]	150	4	6.283e-38	46.12%	+26.72 [+22.63, +30.82]	126	2	1.65e-33
Character 3-gram TF-IDF	18.32%	+32.11 [+27.80, +36.42]	152	3	9.243e-40	47.41%	+25.43 [+21.34, +29.74]	123	5	4.211e-29
QuickUMLS-style 3-gram cosine	17.67%	+32.76 [+28.45, +37.28]	156	4	1.218e-39	45.91%	+26.94 [+22.84, +31.03]	128	3	8.811e-33
CODER++ dense HNSW	17.67%	+32.76 [+28.23, +37.50]	161	9	1.006e-35	32.33%	+40.52 [+35.99, +45.04]	190	2	2.421e-52
xMEN-style SapBERT plus TF-IDF RRF	16.81%	+33.62 [+29.09, +38.15]	162	6	5.066e-39	52.37%	+20.47 [+16.59, +24.57]	102	7	1.969e-21
NYSIIS plus edit ranking	16.16%	+34.27 [+29.74, +38.79]	162	3	1.153e-42	39.22%	+33.62 [+29.31, +38.15]	160	4	9.171e-41
Match Rating Codex plus edit ranking	15.73%	+34.70 [+30.17, +39.22]	165	4	3.143e-42	46.34%	+26.51 [+22.20, +30.82]	129	6	9.76e-30
preon without token matching	15.30%	+35.13 [+30.82, +39.44]	163	0	6.5e-48	16.16%	+56.68 [+52.16, +61.21]	263	0	6.072e-78
preon default drug normalizer	15.09%	+35.34 [+31.03, +39.66]	164	0	3.336e-48	15.95%	+56.90 [+52.37, +61.42]	264	0	3.104e-78
Metaphone plus edit ranking	12.07%	+38.36 [+33.84, +42.89]	180	2	2.173e-49	34.05%	+38.79 [+34.27, +43.32]	182	2	5.553e-50
SapBERT dense HNSW	11.42%	+39.01 [+34.48, +43.75]	186	5	4.867e-47	38.58%	+34.27 [+29.74, +38.79]	167	8	2.846e-38
RapidFuzz weighted ratio	11.21%	+39.22 [+34.70, +43.75]	183	1	6.187e-52	56.25%	+16.59 [+12.93, +20.47]	84	7	9.252e-17
QuickUMLS-style 3-gram overlap	8.41%	+42.03 [+37.50, +46.55]	196	1	8.28e-56	37.93%	+34.91 [+30.39, +39.44]	167	5	1.507e-41
FlexSearch tolerant LatinAdvanced	7.54%	+42.89 [+38.36, +47.41]	200	1	5.531e-57	9.70%	+63.15 [+58.84, +67.46]	293	0	5.907e-87
Phonetic baseline	6.90%	+43.53 [+39.01, +48.06]	202	0	1.431e-59	23.28%	+49.57 [+45.04, +54.09]	230	0	4.984e-68
RapidFuzz partial ratio	6.25%	+44.18 [+39.44, +48.92]	208	3	4.092e-56	44.40%	+28.45 [+24.14, +32.76]	138	6	3.237e-32
Soundex plus edit ranking	5.60%	+44.83 [+40.09, +49.35]	210	2	3.087e-58	24.35%	+48.49 [+43.75, +53.02]	227	2	2.564e-63
FlexSearch full LatinAdvanced	4.74%	+45.69 [+41.16, +50.22]	212	0	1.519e-62	7.54%	+65.30 [+60.99, +69.61]	303	0	5.891e-90
FlexSearch tolerant default	4.74%	+45.69 [+41.16, +50.22]	212	0	1.519e-62	4.74%	+68.10 [+63.79, +72.20]	316	0	7.491e-94
FlexSearch tolerant LatinExtra	4.74%	+45.69 [+40.95, +50.22]	213	1	7.676e-61	18.75%	+54.09 [+49.57, +58.62]	251	0	2.432e-74
BiomedBERT dense HNSW	2.80%	+47.63 [+42.89, +52.16]	223	2	4.527e-62	7.33%	+65.52 [+61.21, +69.83]	304	0	3.007e-90
Exact or prefix match	2.59%	+47.84 [+43.10, +52.37]	223	1	8.513e-64	3.02%	+69.83 [+65.73, +74.14]	325	1	2.44e-94
preon without partial edit matching	0.65%	+49.78 [+45.26, +54.31]	231	0	3.014e-68	0.65%	+72.20 [+68.10, +76.29]	335	0	1.486e-99

Table 27: Every paired system comparison against Algorithm 5 on 66,257 synthetic pairs. A5-only and comparison-only counts are the discordant cases. Exact McNemar p-values are Holm-adjusted across the 52 comparisons; deltas and paired 95% intervals are in percentage points.

Comparison	Comp. H@1	A5-comp. H@1 (95% CI)	A5-only 1	Comp.-only 1	Holm p1	Comp. H@20	A5-comp. H@20 (95% CI)	A5-only 20	Comp.-only 20	Holm p20
Algorithm 4, family rescue	97.19%	+1.00 [+0.92, +1.09]	715	50	1.08e-151	99.52%	+0.48 [+0.43, +0.54]	321	1	2.268e-94
Damerau-Levenshtein	94.78%	+3.41 [+3.26, +3.56]	2440	183	0	99.64%	+0.36 [+0.32, +0.40]	239	1	2.728e-70
Optimal string alignment	94.75%	+3.44 [+3.29, +3.59]	2455	175	0	99.58%	+0.42 [+0.37, +0.47]	281	1	1.457e-82
SymSpell, distance 3, uniform frequency	94.51%	+3.68 [+3.52, +3.83]	2642	205	0	98.86%	+1.14 [+1.06, +1.22]	758	1	4.01e-225
SymSpell, distance 3, catalog frequency	94.22%	+3.97 [+3.81, +4.13]	2802	173	0	98.80%	+1.20 [+1.12, +1.28]	795	1	3.442e-236
Exhaustive Levenshtein	94.05%	+4.14 [+3.98, +4.30]	2917	174	0	99.44%	+0.56 [+0.50, +0.61]	370	1	6.187e-109
Longest-common-subsequence similarity	93.60%	+4.59 [+4.42, +4.76]	3226	184	0	99.21%	+0.78 [+0.72, +0.85]	521	1	3.809e-154
RapidFuzz ratio	92.79%	+5.40 [+5.22, +5.58]	3729	154	0	99.00%	+0.99 [+0.92, +1.07]	660	1	9.686e-196
RapidFuzz quick ratio	92.79%	+5.40 [+5.22, +5.58]	3729	154	0	99.00%	+0.99 [+0.92, +1.07]	660	1	9.686e-196
Algorithm 2, external fast	92.41%	+5.78 [+5.60, +5.97]	4061	229	0	96.88%	+3.12 [+2.99, +3.25]	2067	0	0
Algorithm 3, rank fusion	91.84%	+6.35 [+6.16, +6.55]	4432	223	0	97.12%	+2.88 [+2.75, +3.01]	1907	0	0
Jaro-Winkler	89.57%	+8.62 [+8.40, +8.84]	5868	157	0	98.43%	+1.57 [+1.48, +1.67]	1041	0	8.488e-313
Jaro similarity	89.29%	+8.90 [+8.67, +9.12]	6063	168	0	98.38%	+1.62 [+1.53, +1.72]	1075	0	0
SymSpell, distance 2, catalog frequency	87.84%	+10.35 [+10.11, +10.59]	7004	146	0	90.55%	+9.44 [+9.22, +9.67]	6258	1	0
Character-bigram Dice	84.54%	+13.65 [+13.38, +13.92]	9151	106	0	95.56%	+4.44 [+4.28, +4.59]	2940	0	0
Algorithm 1, current app	83.86%	+14.33 [+14.06, +14.60]	9650	155	0	92.23%	+7.77 [+7.56, +7.97]	5145	0	0
BM25 Okapi over character bigrams	82.03%	+16.16 [+15.88, +16.45]	10806	97	0	94.40%	+5.60 [+5.43, +5.77]	3711	0	0
Fuse.js default Bitap	81.22%	+16.97 [+16.68, +17.26]	11328	86	0	95.00%	+5.00 [+4.83, +5.16]	3310	0	0
BM25L over character trigrams	78.96%	+19.23 [+18.92, +19.53]	12832	91	0	91.82%	+8.18 [+7.97, +8.39]	5420	0	0
QuickUMLS-style 3-gram Dice	78.34%	+19.84 [+19.53, +20.15]	13246	98	0	92.52%	+7.48 [+7.28, +7.67]	4953	0	0
QuickUMLS-style 3-gram Jaccard	78.34%	+19.84 [+19.53, +20.15]	13246	98	0	92.52%	+7.48 [+7.28, +7.67]	4953	0	0
BM25 Okapi over character trigrams	78.25%	+19.94 [+19.63, +20.25]	13306	93	0	92.17%	+7.83 [+7.62, +8.03]	5185	0	0
BM25Plus over character trigrams	78.19%	+19.99 [+19.69, +20.30]	13340	92	0	92.17%	+7.83 [+7.63, +8.04]	5189	0	0
BioSyn-style hybrid, sparse weight 0.50	78.17%	+20.02 [+19.71, +20.33]	13371	107	0	93.78%	+6.22 [+6.03, +6.40]	4118	0	0
Fuse.js token search	78.06%	+20.13 [+19.82, +20.44]	13396	56	0	93.34%	+6.66 [+6.47, +6.85]	4412	0	0
BioSyn-style hybrid, sparse weight 0.75	77.88%	+20.31 [+20.00, +20.62]	13555	99	0	93.40%	+6.60 [+6.41, +6.79]	4374	0	0
QuickUMLS-style 3-gram cosine	77.46%	+20.73 [+20.42, +21.04]	13828	93	0	92.40%	+7.60 [+7.40, +7.80]	5037	0	0
NYSIIS plus edit ranking	77.01%	+21.18 [+20.86, +21.50]	14185	150	0	93.90%	+6.10 [+5.91, +6.28]	4040	1	0
Metaphone plus edit ranking	75.55%	+22.63 [+22.31, +22.96]	15233	236	0	94.62%	+5.38 [+5.21, +5.55]	3563	0	0
Character 3-gram TF-IDF	75.21%	+22.98 [+22.66, +23.31]	15319	92	0	91.66%	+8.34 [+8.13, +8.55]	5524	0	0
preon default drug normalizer	74.16%	+24.03 [+23.71, +24.36]	15950	26	0	74.37%	+25.63 [+25.30, +25.96]	16982	0	0
preon without token matching	74.16%	+24.03 [+23.71, +24.36]	15950	26	0	74.37%	+25.63 [+25.30, +25.96]	16982	0	0
Phonetic baseline	69.66%	+28.53 [+28.18, +28.88]	19068	163	0	92.07%	+7.93 [+7.72, +8.13]	5253	0	0
BioSyn-style hybrid, sparse weight 0.25	66.01%	+32.18 [+31.82, +32.54]	21423	101	0	90.20%	+9.80 [+9.57, +10.02]	6491	1	0
Match Rating Codex plus edit ranking	65.82%	+32.37 [+32.01, +32.73]	21622	175	0	94.58%	+5.42 [+5.25, +5.59]	3591	0	0
RapidFuzz token ratio	65.67%	+32.52 [+32.15, +32.88]	21678	133	0	85.47%	+14.53 [+14.26, +14.80]	9626	1	0
RapidFuzz token-set ratio	65.66%	+32.53 [+32.17, +32.90]	21688	133	0	85.46%	+14.54 [+14.28, +14.81]	9637	1	0
RapidFuzz token-sort ratio	65.64%	+32.54 [+32.18, +32.91]	21696	133	0	85.46%	+14.54 [+14.27, +14.81]	9634	1	0
RapidFuzz weighted ratio	55.73%	+42.46 [+42.08, +42.84]	28175	44	0	97.20%	+2.80 [+2.68, +2.93]	1856	0	0
xMEN-style SapBERT plus TF-IDF RRF	48.85%	+49.34 [+48.95, +49.72]	32784	94	0	92.30%	+7.70 [+7.50, +7.91]	5104	1	0
CODER++ dense HNSW	42.56%	+55.63 [+55.25, +56.01]	36951	91	0	66.26%	+33.73 [+33.38, +34.09]	22351	0	0
QuickUMLS-style 3-gram overlap	36.74%	+61.45 [+61.07, +61.82]	40759	46	0	88.37%	+11.63 [+11.39, +11.87]	7705	0	0
SapBERT dense HNSW	27.67%	+70.52 [+70.17, +70.87]	46793	69	0	56.45%	+43.54 [+43.17, +43.92]	28852	1	0
FlexSearch tolerant LatinAdvanced	20.32%	+77.87 [+77.55, +78.19]	51687	95	0	21.89%	+78.11 [+77.79, +78.42]	51751	0	0
RapidFuzz partial ratio	16.68%	+81.51 [+81.21, +81.81]	54036	31	0	92.90%	+7.10 [+6.90, +7.29]	4704	0	0
FlexSearch tolerant LatinExtra	15.43%	+82.76 [+82.47, +83.05]	54899	63	0	29.51%	+70.49 [+70.15, +70.83]	46706	0	0
FlexSearch full LatinAdvanced	13.94%	+84.25 [+83.97, +84.53]	55841	21	0	15.75%	+84.25 [+83.97, +84.53]	55822	0	0
Soundex plus edit ranking	13.09%	+85.10 [+84.83, +85.38]	56471	86	0	67.73%	+32.27 [+31.92, +32.62]	21382	0	0
FlexSearch tolerant default	10.56%	+87.63 [+87.38, +87.88]	58072	10	0	10.91%	+89.09 [+88.85, +89.32]	59026	0	0
BiomedBERT dense HNSW	6.22%	+91.97 [+91.76, +92.17]	60957	23	0	13.01%	+86.99 [+86.74, +87.25]	57639	0	0
Exact or prefix match	3.00%	+95.19 [+95.02, +95.35]	63078	10	0	3.01%	+96.99 [+96.86, +97.12]	64260	0	0
preon without partial edit matching	0.00%	+98.19 [+98.09, +98.29]	65057	0	0	0.00%	+100.00 [+100.00, +100.00]	66256	0	0

Table 28: Concrete paired OCR outcomes for six representative competitors. Two rows are shown per comparison and outcome class; rank 999 means the verified family is outside the top 20. The complete example set remains in `paired_ocr_examples.csv`.

Comparison	Paired outcome	Input	Expected	Edits Operation	Bigrams A5 rank A5 top 1	Comp. rank Comp. top 1
Algorithm 4, family rescue	Algorithm 5 only recovers within top 20 A BAA G LAR		ABASAGLAR	1 mixed operations	61 ABASAGLAR CARTRIDGES	999 EFFACLAR H
Algorithm 4, family rescue	Algorithm 5 only recovers within top 20 LANFUS		LANTUS	1 substitutions only	31 LANTUS CARTRIDGES	999 LANVIS
Algorithm 4, family rescue	Both miss top 20	BSLX	LASIX	3 mixed operations	0999 PELOX	999 none
Algorithm 4, family rescue	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 LANDO
Algorithm 4, family rescue	Competitor only recovers within top 20 TEFA		TELFAST	3 missing characters only	2999 TERA	13 TERA
Algorithm 4, family rescue	Competitor only recovers within top 20 PETAWOLIN		PETHIDINE	5 mixed operations	3999 PATADOLINE	20 PATADOLINE
Damerau-Levenshtein	Algorithm 5 only recovers within top 20 ABA SAGLAR		ABASAGLAR	0 extra characters only	81 ABASAGLAR CARTRIDGES	999 ADALAT LA
Damerau-Levenshtein	Algorithm 5 only recovers within top 20 A BAA G LAR		ABASAGLAR	1 mixed operations	61 ABASAGLAR CARTRIDGES	999 ADALAT LA
Damerau-Levenshtein	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 AZHA
Damerau-Levenshtein	Both miss top 20	LANLTA	LANTUS	3 substitutions only	2999 LANETAL	999 LANETAL
Damerau-Levenshtein	Competitor only recovers within top 20 LORRIX		LASIX	3 mixed operations	1999 LORNOX	17 LORNOX
Damerau-Levenshtein	Competitor only recovers within top 20 LACTULOSE		LACTO	4 extra characters only	3999 LACTULOSE HEK	13 ACTILYSE
Jaro similarity	Algorithm 5 only recovers within top 20 LANTUY		LANTUS	1 substitutions only	41 LANTUS CARTRIDGES	999 FLAGENTYL
Jaro similarity	Algorithm 5 only recovers within top 20 ABDSAGLOR		ABASAGLAR	2 substitutions only	41 ABASAGLAR CARTRIDGES	999 ADAGEL
Jaro similarity	Both miss top 20	BSLX	LASIX	3 mixed operations	0999 PELOX	999 SALEX
Jaro similarity	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 LANTHAREX
Jaro similarity	Competitor only recovers within top 20 TEFA		TELFAST	3 missing characters only	2999 TERA	1 TELFAST
Jaro similarity	Competitor only recovers within top 20 LACTULOSE		LACTO	4 extra characters only	3999 LACTULOSE HEK	3 LACTULOSE HEK
Jaro-Winkler	Algorithm 5 only recovers within top 20 ABDSAGLOR		ABASAGLAR	2 substitutions only	41 ABASAGLAR CARTRIDGES	999 ADAGEL
Jaro-Winkler	Algorithm 5 only recovers within top 20 ABDSAQLAR		ABASAGLAR	2 substitutions only	41 ABASAGLAR CARTRIDGES	999 ASMALAIR
Jaro-Winkler	Both miss top 20	BSLX	LASIX	3 mixed operations	0999 PELOX	999 SALEX
Jaro-Winkler	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 LANTHAREX
Jaro-Winkler	Competitor only recovers within top 20 LANLTA		LANTUS	3 substitutions only	2999 LANETAL	18 LANETAL
Jaro-Winkler	Competitor only recovers within top 20 OSCAR		OSTOCAL	3 mixed operations	2999 ASCOR	14 EMO SCAR
SymSpell, distance 3, catalog frequency	Algorithm 5 only recovers within top 20 ABA SAGLAR		ABASAGLAR	0 extra characters only	81 ABASAGLAR CARTRIDGES	999 none
SymSpell, distance 3, catalog frequency	Algorithm 5 only recovers within top 20 A BAA G LAR		ABASAGLAR	1 mixed operations	61 ABASAGLAR CARTRIDGES	999 none
SymSpell, distance 3, catalog frequency	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 AZHA
SymSpell, distance 3, catalog frequency	Both miss top 20	LANLTA	LANTUS	3 substitutions only	2999 LANETAL	999 LANETAL
SymSpell, distance 3, catalog frequency	Competitor only recovers within top 20 BSLX		LASIX	3 mixed operations	0999 PELOX	14 ISLA
SymSpell, distance 3, catalog frequency	Competitor only recovers within top 20 LORRIX		LASIX	3 mixed operations	1999 LORNOX	16 LORNOX
preon default drug normalizer	Algorithm 5 only recovers within top 20 ABA SAGLAR		ABASAGLAR	0 extra characters only	81 ABASAGLAR CARTRIDGES	999 none
preon default drug normalizer	Algorithm 5 only recovers within top 20 A BAA G LAR		ABASAGLAR	1 mixed operations	61 ABASAGLAR CARTRIDGES	999 none
preon default drug normalizer	Both miss top 20	LANHA	LANTUS	3 mixed operations	2999 LAMA	999 none
preon default drug normalizer	Both miss top 20	LANLTA	LANTUS	3 substitutions only	2999 LANETAL	999 none

Table 29: Failure-driver profile for every system on 464 fair OCR pairs. A bigram is two adjacent characters, for example AB, BA, and AS in ABAS. Short means at most four compact characters; long means at least eight; mixed combines two or more edit mechanisms; extreme means compact edit distance divided by compact target length exceeds 0.60. Spearman rho reports the monotonic association between edit count and binary Hit@20; negative values mean recovery tends to fall as edits increase.

System	d<=3	H@20	d>=4	H@20	rho(d,H@20)	0 bigrams	2+ bigrams	short query	long query	mixed	ops.	extreme
Algorithm 5, evidence-guided rescue	97.35%	27.16%	-0.726	1.56%	90.86%	36.96%	78.95%	56.45%	7.96%			
Algorithm 4, family rescue	95.36%	25.93%	-0.698	1.56%	88.79%	36.96%	77.37%	53.63%	7.96%			
Jaro-Winkler	91.39%	22.84%	-0.691	0.00%	85.84%	41.30%	73.16%	50.00%	12.39%			
BioSyn-style hybrid, sparse weight 0.50	71.52%	18.52%	-0.560	0.00%	72.27%	19.57%	65.26%	38.71%	6.19%			
xMEN-style SapBERT plus TF-IDF RRF	70.86%	17.90%	-0.563	0.00%	70.80%	21.74%	64.21%	37.90%	5.31%			
BioSyn-style hybrid, sparse weight 0.25	70.53%	17.90%	-0.578	0.00%	69.91%	30.43%	59.47%	36.69%	6.19%			
Fuse.js default Bitap	81.79%	16.05%	-0.689	0.00%	78.17%	26.09%	71.58%	39.52%	2.65%			
BioSyn-style hybrid, sparse weight 0.75	69.54%	16.05%	-0.560	0.00%	69.32%	19.57%	64.21%	36.69%	5.31%			
Jaro similarity	85.10%	14.81%	-0.708	0.00%	77.88%	36.96%	63.68%	44.35%	8.85%			
Exhaustive Levenshtein	83.77%	13.58%	-0.634	1.56%	74.04%	32.61%	61.58%	41.13%	2.65%			
Damerau-Levenshtein	83.77%	13.58%	-0.634	1.56%	74.04%	32.61%	61.58%	41.13%	2.65%			
Optimal string alignment	83.77%	13.58%	-0.634	1.56%	74.04%	32.61%	61.58%	41.13%	2.65%			
Character 3-gram TF-IDF	65.89%	12.96%	-0.544	0.00%	64.90%	15.22%	60.53%	34.68%	4.42%			
RapidFuzz ratio	86.42%	11.73%	-0.711	0.00%	77.58%	34.78%	65.26%	43.55%	3.54%			
RapidFuzz quick ratio	86.42%	11.73%	-0.711	0.00%	77.58%	34.78%	65.26%	43.55%	3.54%			
RapidFuzz token ratio	84.44%	11.73%	-0.669	0.00%	75.81%	32.61%	63.68%	42.74%	3.54%			
RapidFuzz token-sort ratio	84.44%	11.73%	-0.669	0.00%	75.81%	32.61%	63.68%	42.74%	3.54%			
RapidFuzz token-set ratio	84.44%	11.73%	-0.669	0.00%	75.81%	32.61%	63.68%	42.74%	3.54%			
Character-bigram Dice	75.17%	11.73%	-0.647	0.00%	71.09%	32.61%	62.11%	36.29%	5.31%			
Fuse.js token search	72.52%	11.73%	-0.628	0.00%	69.91%	2.17%	68.42%	33.47%	1.77%			
Match Rating Codex plus edit ranking	64.90%	11.73%	-0.606	3.12%	59.88%	19.57%	53.68%	29.03%	3.54%			
SapBERT dense HNSW	52.98%	11.73%	-0.517	0.00%	51.03%	23.91%	38.95%	24.60%	3.54%			
BM25 Okapi over character trigrams	66.89%	11.11%	-0.565	0.00%	64.90%	15.22%	58.95%	33.87%	4.42%			
BM25Plus over character trigrams	66.89%	11.11%	-0.565	0.00%	64.90%	15.22%	58.95%	33.87%	4.42%			
BM25L over character trigrams	65.89%	11.11%	-0.560	0.00%	64.01%	15.22%	58.95%	33.47%	3.54%			
Algorithm 3, rank fusion	75.50%	11.11%	-0.697	0.00%	70.50%	30.43%	61.05%	32.66%	2.65%			
BM25 Okapi over character bigrams	72.19%	9.88%	-0.619	0.00%	69.03%	17.39%	62.63%	33.47%	4.42%			
Algorithm 2, external fast	73.51%	9.88%	-0.704	0.00%	68.44%	30.43%	59.47%	31.45%	1.77%			
Longest-common-subsequence similarity	82.12%	9.26%	-0.661	0.00%	72.27%	34.78%	57.89%	38.71%	1.77%			
QuickUMLS-style 3-gram overlap	53.31%	9.26%	-0.502	0.00%	51.92%	4.35%	50.53%	25.81%	4.42%			
RapidFuzz weighted ratio	81.79%	8.64%	-0.713	0.00%	75.22%	21.74%	63.68%	37.50%	3.54%			
QuickUMLS-style 3-gram cosine	65.89%	8.64%	-0.606	0.00%	62.83%	15.22%	55.26%	31.05%	3.54%			
RapidFuzz partial ratio	63.58%	8.64%	-0.613	0.00%	60.77%	6.52%	56.32%	27.02%	3.54%			
QuickUMLS-style 3-gram Jaccard	66.56%	8.02%	-0.618	0.00%	63.13%	15.22%	54.74%	31.05%	2.65%			
QuickUMLS-style 3-gram Dice	66.56%	8.02%	-0.618	0.00%	63.13%	15.22%	54.74%	31.05%	2.65%			
NYSIIS plus edit ranking	55.96%	8.02%	-0.536	3.12%	49.56%	17.39%	42.63%	22.18%	1.77%			
CODER++ dense HNSW	46.36%	6.17%	-0.490	0.00%	42.48%	17.39%	29.47%	20.16%	3.54%			
Metaphone plus edit ranking	49.67%	4.94%	-0.541	1.56%	44.25%	17.39%	36.84%	17.34%	0.88%			
Soundex plus edit ranking	35.10%	4.32%	-0.454	0.00%	32.15%	13.04%	29.47%	11.69%	0.00%			
Phonetic baseline	33.77%	3.70%	-0.460	0.00%	31.27%	10.87%	30.53%	12.10%	0.00%			
Algorithm 1, current app	57.28%	2.47%	-0.660	0.00%	50.44%	21.74%	45.26%	16.94%	0.00%			
FlexSearch tolerant LatinExtra	27.81%	1.85%	-0.395	0.00%	23.89%	8.70%	16.84%	7.66%	0.00%			
FlexSearch full LatinAdvanced	10.60%	1.85%	-0.280	0.00%	10.03%	2.17%	8.42%	3.63%	0.00%			
BiomedBERT dense HNSW	10.60%	1.23%	-0.263	0.00%	10.03%	0.00%	7.37%	1.21%	1.77%			
Exact or prefix match	4.30%	0.62%	-0.202	0.00%	4.13%	8.70%	1.05%	0.00%	0.88%			
SymSpell, distance 3, uniform frequency	86.42%	0.00%	-0.692	3.12%	70.80%	34.78%	50.53%	37.10%	0.00%			
SymSpell, distance 3, catalog frequency	85.76%	0.00%	-0.696	3.12%	69.91%	34.78%	50.53%	36.29%	0.00%			
SymSpell, distance 2, catalog frequency	63.91%	0.00%	-0.765	0.00%	52.21%	32.61%	34.21%	16.13%	0.00%			
FlexSearch tolerant LatinAdvanced	14.90%	0.00%	-0.386	0.00%	11.21%	15.22%	5.79%	0.81%	0.00%			
preon default drug normalizer	24.50%	0.00%	-0.577	0.00%	21.83%	2.17%	15.26%	0.40%	0.00%			
preon without token matching	24.83%	0.00%	-0.581	0.00%	22.12%	2.17%	15.26%	0.40%	0.00%			
preon without partial edit matching	0.99%	0.00%	-0.141	0.00%	0.88%	2.17%	0.00%	0.00%	0.00%			
FlexSearch tolerant default	7.28%	0.00%	-0.291	0.00%	6.19%	8.70%	1.58%	0.00%	0.00%			

Table 30: Worst sufficiently populated slice for every system on 464 fair OCR pairs. The search considers mistake category, error type, operation family and profile, edit distance, query length, and shared-bigram evidence. A group needs at least 10 cases before it can be called a system’s weakest slice.

System	Worst dimension	Worst supported group	Cases	Hit@1	Hit@20
Algorithm 1, current app	category	extreme distance prediction	113	0.00%	0.00%
Algorithm 2, external fast	distance band	6 plus edits	69	0.00%	0.00%
Algorithm 3, rank fusion	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BM25 Okapi over character bigrams	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BM25 Okapi over character trigrams	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BM25L over character trigrams	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BM25Plus over character trigrams	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BioSyn-style hybrid, sparse weight 0.25	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BioSyn-style hybrid, sparse weight 0.50	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BioSyn-style hybrid, sparse weight 0.75	shared bigram band	0 shared bigrams	64	0.00%	0.00%
BiomedBERT dense HNSW	distance band	6 plus edits	69	0.00%	0.00%
CODER++ dense HNSW	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Character 3-gram TF-IDF	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Character-bigram Dice	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Damerau-Levenshtein	distance band	6 plus edits	69	0.00%	0.00%
Exact or prefix match	operation profile	mixed operations	248	0.00%	0.00%
Exhaustive Levenshtein	distance band	6 plus edits	69	0.00%	0.00%
FlexSearch full LatinAdvanced	category	extreme distance prediction	113	0.00%	0.00%
FlexSearch tolerant LatinAdvanced	error type	multi edit ocr error	161	0.00%	0.00%
FlexSearch tolerant LatinExtra	category	extreme distance prediction	113	0.00%	0.00%
FlexSearch tolerant default	operation profile	mixed operations	248	0.00%	0.00%
Fuse.js default Bitap	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Fuse.js token search	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Jaro similarity	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Jaro-Winkler	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Longest-common-subsequence similarity	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Match Rating Codex plus edit ranking	distance band	6 plus edits	69	0.00%	0.00%
Metaphone plus edit ranking	distance band	6 plus edits	69	0.00%	0.00%
NYSIIS plus edit ranking	distance band	6 plus edits	69	0.00%	0.00%
Optimal string alignment	distance band	6 plus edits	69	0.00%	0.00%
Phonetic baseline	category	extreme distance prediction	113	0.00%	0.00%
QuickUMLS-style 3-gram Dice	shared bigram band	0 shared bigrams	64	0.00%	0.00%
QuickUMLS-style 3-gram Jaccard	shared bigram band	0 shared bigrams	64	0.00%	0.00%
QuickUMLS-style 3-gram cosine	shared bigram band	0 shared bigrams	64	0.00%	0.00%
QuickUMLS-style 3-gram overlap	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz partial ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz quick ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz token ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz token-set ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz token-sort ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
RapidFuzz weighted ratio	shared bigram band	0 shared bigrams	64	0.00%	0.00%
SapBERT dense HNSW	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Soundex plus edit ranking	category	extreme distance prediction	113	0.00%	0.00%
SymSpell, distance 2, catalog frequency	error type	multi edit ocr error	161	0.00%	0.00%
SymSpell, distance 3, catalog frequency	error type	multi edit ocr error	161	0.00%	0.00%
SymSpell, distance 3, uniform frequency	error type	multi edit ocr error	161	0.00%	0.00%
preon default drug normalizer	distance band	2 3 edits	186	0.00%	0.00%
preon without partial edit matching	category	standard ocr error	250	0.00%	0.00%
preon without token matching	distance band	2 3 edits	186	0.00%	0.00%
xMEN-style SapBERT plus TF-IDF RRF	shared bigram band	0 shared bigrams	64	0.00%	0.00%
Algorithm 4, family rescue	shared bigram band	0 shared bigrams	64	1.56%	1.56%
Algorithm 5, evidence-guided rescue	shared bigram band	0 shared bigrams	64	1.56%	1.56%

Table 31: Failure-driver profile for every system on 66,257 synthetic pairs. A bigram is two adjacent characters, for example AB, BA, and AS in ABAS. Short means at most four compact characters; long means at least eight; mixed combines two or more edit mechanisms; extreme means compact edit distance divided by compact target length exceeds 0.60. Spearman rho reports the monotonic association between edit count and binary Hit@20; negative values mean recovery tends to fall as edits increase.

System	d<=3	H@20	d>=4	H@20	rho(d,H@20)	0 bigrams	2+ bigrams	short query	long query	mixed	ops.	extreme
Algorithm 5, evidence-guided rescue	100.00%	100.00%	-0.004	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Damerau-Levenshtein	99.72%	92.23%	-0.106	70.49%	99.92%	91.49%	99.99%	98.73%	65.85%			
Jaro similarity	98.50%	86.92%	-0.196	49.00%	99.04%	85.82%	99.47%	95.16%	43.09%			
Optimal string alignment	99.72%	86.51%	-0.117	68.48%	99.89%	91.49%	99.98%	98.50%	53.66%			
Jaro-Winkler	98.57%	85.69%	-0.191	50.43%	99.06%	85.70%	99.46%	95.37%	40.65%			
Algorithm 4, family rescue	99.67%	85.29%	-0.095	72.49%	99.92%	86.21%	99.92%	98.58%	31.71%			
RapidFuzz ratio	99.16%	84.88%	-0.166	47.28%	99.64%	89.82%	99.85%	96.85%	36.59%			
RapidFuzz quick ratio	99.16%	84.88%	-0.166	47.28%	99.64%	89.82%	99.85%	96.85%	36.59%			
Longest-common-subsequence similarity	99.39%	83.65%	-0.156	55.30%	99.72%	91.88%	99.85%	97.45%	27.64%			
Exhaustive Levenshtein	99.67%	78.75%	-0.136	59.60%	99.85%	89.69%	99.92%	98.01%	26.83%			
RapidFuzz weighted ratio	97.51%	69.35%	-0.213	15.47%	98.54%	43.17%	99.37%	92.39%	12.20%			
Metaphone plus edit ranking	94.91%	68.80%	-0.163	69.05%	95.25%	70.75%	97.12%	90.28%	23.58%			
NYSIIS plus edit ranking	94.19%	68.12%	-0.165	47.28%	94.68%	62.76%	97.25%	87.37%	28.46%			
Phonetic baseline	92.35%	67.03%	-0.109	69.34%	92.59%	68.69%	95.32%	90.76%	20.33%			
Match Rating Codex plus edit ranking	94.91%	65.40%	-0.241	68.19%	95.15%	78.99%	95.93%	88.53%	34.96%			
RapidFuzz token ratio	85.70%	65.26%	-0.091	46.13%	85.88%	86.47%	84.27%	81.90%	32.52%			
RapidFuzz token-sort ratio	85.69%	65.26%	-0.090	46.13%	85.87%	86.60%	84.25%	81.88%	32.52%			
RapidFuzz token-set ratio	85.68%	65.26%	-0.091	46.13%	85.87%	86.47%	84.25%	81.87%	32.52%			
Fuse.js default Bitap	95.35%	64.31%	-0.244	12.89%	96.43%	47.42%	97.84%	86.95%	11.38%			
Fuse.js token search	93.72%	59.40%	-0.256	4.30%	95.07%	29.25%	97.30%	83.43%	7.32%			
Character-bigram Dice	95.98%	58.58%	-0.292	0.00%	97.49%	63.40%	98.67%	87.53%	13.82%			
BM25 Okapi over character bigrams	94.80%	58.31%	-0.295	0.00%	96.36%	52.19%	98.28%	84.58%	13.01%			
BioSyn-style hybrid, sparse weight 0.50	94.20%	56.27%	-0.306	13.75%	95.62%	56.06%	97.43%	83.09%	17.07%			
BioSyn-style hybrid, sparse weight 0.75	93.84%	53.68%	-0.311	12.03%	95.30%	54.38%	97.36%	82.16%	15.45%			
RapidFuzz partial ratio	93.35%	52.45%	-0.318	5.16%	94.78%	37.89%	97.15%	81.11%	18.70%			
xMEN-style SapBERT plus TF-IDF RRF	92.74%	52.45%	-0.311	15.47%	94.00%	57.99%	96.13%	80.21%	18.70%			
BM25 Okapi over character trigrams	92.64%	50.68%	-0.321	0.00%	94.30%	39.30%	96.94%	79.39%	8.94%			
BM25Plus over character trigrams	92.63%	50.68%	-0.321	0.00%	94.29%	39.30%	96.94%	79.37%	8.94%			
BM25L over character trigrams	92.28%	50.54%	-0.321	0.00%	93.93%	38.53%	96.83%	78.75%	8.94%			
QuickUMLS-style 3-gram Jaccard	92.99%	50.41%	-0.331	0.00%	94.66%	40.72%	96.91%	79.94%	8.13%			
QuickUMLS-style 3-gram Dice	92.99%	50.41%	-0.331	0.00%	94.66%	40.72%	96.91%	79.94%	8.13%			
QuickUMLS-style 3-gram cosine	92.87%	50.27%	-0.333	0.00%	94.53%	40.46%	96.85%	79.63%	8.13%			
Character 3-gram TF-IDF	92.14%	49.18%	-0.329	0.00%	93.77%	39.56%	96.57%	78.11%	8.13%			
Algorithm 3, rank fusion	97.68%	47.55%	-0.246	49.57%	98.15%	69.46%	98.60%	90.82%	12.20%			
Algorithm 2, external fast	97.43%	47.55%	-0.263	48.71%	97.95%	70.62%	98.46%	89.89%	8.94%			
BioSyn-style hybrid, sparse weight 0.25	90.71%	45.37%	-0.356	16.91%	91.81%	59.15%	93.95%	75.02%	18.70%			
QuickUMLS-style 3-gram overlap	88.90%	40.60%	-0.332	0.00%	90.41%	22.16%	95.06%	72.47%	7.32%			
Soundex plus edit ranking	68.13%	31.74%	-0.190	54.73%	68.01%	60.05%	69.27%	60.42%	16.26%			
CODER++ dense HNSW	66.73%	25.07%	-0.265	20.06%	67.23%	47.42%	68.26%	47.75%	13.82%			
SapBERT dense HNSW	56.85%	21.39%	-0.247	18.91%	57.24%	47.81%	57.47%	38.64%	20.33%			
FlexSearch tolerant LatinExtra	29.69%	13.49%	-0.097	38.11%	29.34%	26.03%	25.57%	29.14%	6.50%			
SymSpell, distance 3, uniform frequency	99.84%	11.04%	-0.207	77.36%	99.13%	94.97%	98.99%	95.91%	21.14%			
SymSpell, distance 3, catalog frequency	99.78%	11.04%	-0.211	71.63%	99.12%	93.56%	98.99%	95.72%	19.51%			
Algorithm 1, current app	93.15%	10.76%	-0.350	46.70%	93.21%	59.79%	94.56%	79.55%	5.69%			
preon default drug normalizer	75.13%	5.99%	-0.650	0.00%	76.08%	0.00%	80.58%	43.04%	0.00%			
preon without token matching	75.13%	5.99%	-0.650	0.00%	76.08%	0.00%	80.58%	43.04%	0.00%			
BiomedBERT dense HNSW	13.13%	2.18%	-0.122	2.87%	13.21%	13.02%	12.50%	7.76%	2.44%			
FlexSearch tolerant LatinAdvanced	22.13%	1.09%	-0.208	9.74%	22.04%	42.91%	18.21%	15.83%	3.25%			
FlexSearch tolerant default	11.03%	0.68%	-0.182	2.87%	11.10%	26.68%	9.00%	5.52%	2.44%			
FlexSearch full LatinAdvanced	15.92%	0.41%	-0.270	0.86%	16.04%	10.70%	14.19%	4.18%	0.81%			
SymSpell, distance 2, catalog frequency	91.57%	0.00%	-0.567	33.81%	91.36%	91.24%	90.64%	73.37%	9.76%			
Exact or prefix match	3.05%	0.00%	-0.145	0.00%	3.08%	11.21%	2.23%	0.03%	0.00%			
preon without partial edit matching	0.00%	0.00%	-	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			

Table 32: Worst sufficiently populated slice for every system on 66,257 synthetic pairs. The search considers mistake category, error type, operation family and profile, edit distance, query length, and shared-bigram evidence. A group needs at least 10 cases before it can be called a system’s weakest slice.

System	Worst dimension	Worst supported group	Cases	Hit@1	Hit@20
BM25 Okapi over character bigrams	shared bigram band	0 shared bigrams	349	0.00%	0.00%
BM25 Okapi over character trigrams	shared bigram band	1 shared bigram	1144	0.00%	0.00%
BM25L over character trigrams	shared bigram band	1 shared bigram	1144	0.00%	0.00%
BM25Plus over character trigrams	shared bigram band	1 shared bigram	1144	0.00%	0.00%
BiomedBERT dense HNSW	error type	deletion last char pos 18	41	0.00%	0.00%
Character 3-gram TF-IDF	shared bigram band	1 shared bigram	1144	0.00%	0.00%
Character-bigram Dice	shared bigram band	0 shared bigrams	349	0.00%	0.00%
Exact or prefix match	distance band	2 3 edits	26859	0.00%	0.00%
FlexSearch full LatinAdvanced	error type	transposition plus deletion variant 1 effective damerau 2	357	0.00%	0.00%
FlexSearch tolerant LatinAdvanced	error type	ligature m to rn pos 0	177	0.00%	0.00%
FlexSearch tolerant LatinExtra	error type	ligature m to rn pos 0	177	0.00%	0.00%
FlexSearch tolerant default	error type	vowel plus phonetic variant 0 effective damerau 2	398	0.00%	0.00%
Fuse.js default Bitap	error type	transposition middle pos 1	11	0.00%	0.00%
Fuse.js token search	error type	transposition middle pos 1	11	0.00%	0.00%
Match Rating Codex plus edit ranking	error type	multi phonetic ph to f pos 0	19	0.00%	0.00%
QuickUMLS-style 3-gram Dice	shared bigram band	1 shared bigram	1144	0.00%	0.00%
QuickUMLS-style 3-gram Jaccard	shared bigram band	1 shared bigram	1144	0.00%	0.00%
QuickUMLS-style 3-gram cosine	shared bigram band	1 shared bigram	1144	0.00%	0.00%
QuickUMLS-style 3-gram overlap	shared bigram band	1 shared bigram	1144	0.00%	0.00%
SapBERT dense HNSW	error type	visual F to L pos 0	10	0.00%	0.00%
Soundex plus edit ranking	error type	ligature m to rn pos 0	177	0.00%	0.00%
SymSpell, distance 2, catalog frequency	category	combined errors effective 3	4346	0.00%	0.00%
SymSpell, distance 3, catalog frequency	category	combined errors effective 4 plus	563	0.00%	0.00%
SymSpell, distance 3, uniform frequency	category	combined errors effective 4 plus	563	0.00%	0.00%
preon default drug normalizer	shared bigram band	1 shared bigram	1144	0.00%	0.00%
preon without partial edit matching	shared bigram band	4 plus shared bigrams	58023	0.00%	0.00%
preon without token matching	shared bigram band	1 shared bigram	1144	0.00%	0.00%
Algorithm 1, current app	error type	ligature vowel transpose variant 0 effective damerau 4	177	1.69%	2.82%
RapidFuzz partial ratio	shared bigram band	0 shared bigrams	349	0.00%	5.16%
Phonetic baseline	error type	multi phonetic th to s pos 0	15	6.67%	6.67%
BioSyn-style hybrid, sparse weight 0.75	shared bigram band	1 shared bigram	1144	1.14%	10.66%
BioSyn-style hybrid, sparse weight 0.50	shared bigram band	0 shared bigrams	349	0.57%	13.75%
CODER++ dense HNSW	error type	keyboard vowel delete variant 0 effective damerau 3	324	4.63%	13.89%
RapidFuzz weighted ratio	shared bigram band	0 shared bigrams	349	3.15%	15.47%
xMEN-style SapBERT plus TF-IDF RRF	shared bigram band	0 shared bigrams	349	1.15%	15.47%
BioSyn-style hybrid, sparse weight 0.25	shared bigram band	0 shared bigrams	349	2.58%	16.91%
RapidFuzz token ratio	error type	multi phonetic f to ph pos 18	11	27.27%	36.36%
RapidFuzz token-set ratio	error type	multi phonetic f to ph pos 18	11	27.27%	36.36%
RapidFuzz token-sort ratio	error type	multi phonetic f to ph pos 18	11	27.27%	36.36%
Metaphone plus edit ranking	error type	ligature cl to d pos 4	16	18.75%	37.50%
NYSIIS plus edit ranking	error type	keyboard adjacent R to E sampled pos 2	13	7.69%	38.46%
Algorithm 2, external fast	error type	ligature vowel transpose variant 2 effective damerau 4	204	39.22%	45.10%
Algorithm 3, rank fusion	error type	ligature vowel transpose variant 2 effective damerau 4	204	39.22%	45.10%
RapidFuzz quick ratio	shared bigram band	0 shared bigrams	349	16.05%	47.28%
RapidFuzz ratio	shared bigram band	0 shared bigrams	349	16.05%	47.28%
Jaro similarity	shared bigram band	0 shared bigrams	349	17.77%	49.00%
Jaro-Winkler	shared bigram band	0 shared bigrams	349	18.05%	50.43%
Longest-common-subsequence similarity	shared bigram band	0 shared bigrams	349	19.48%	55.30%
Exhaustive Levenshtein	shared bigram band	0 shared bigrams	349	9.74%	59.60%
Algorithm 4, family rescue	error type	transposition middle pos 1	11	54.55%	63.64%
Optimal string alignment	shared bigram band	0 shared bigrams	349	15.76%	68.48%
Damerau-Levenshtein	shared bigram band	0 shared bigrams	349	16.05%	70.49%
Algorithm 5, evidence-guided rescue	error type	ligature d to cl pos 1	16	93.75%	93.75%

11.6 Algorithm 5 Failure Taxonomy

Failure rate is misses divided by rows in one subgroup. Failure share is subgroup misses divided by all Algorithm 5 top-20 misses in that dataset.

Table 33: Complete Algorithm 5 mistake and evidence taxonomy on 464 fair OCR pairs. The report retains every measured. The full row-level taxonomy, including all 4,184 synthetic error labels, remains in `a5_failure_taxonomy.csv`.

Dimension	Group	Cases	H@1	H@20	Misses	Failure rate	Failure share
category	extreme distance prediction	113	0.00%	7.96%	104	92.04%	82.54%
category	high distance prediction	88	18.18%	75.00%	22	25.00%	17.46%
category	normalized exact match	4	100.00%	100.00%	0	0.00%	0.00%
category	standard ocr error	250	82.80%	100.00%	0	0.00%	0.00%
category	visible name fragment	9	77.78%	100.00%	0	0.00%	0.00%
distance band	6 plus edits	69	0.00%	4.35%	66	95.65%	52.38%
distance band	4 5 edits	93	10.75%	44.09%	52	55.91%	41.27%
distance band	2 3 edits	186	59.14%	95.70%	8	4.30%	6.35%
distance band	0 exact after normalization	4	100.00%	100.00%	0	0.00%	0.00%
distance band	1 single edit	112	98.21%	100.00%	0	0.00%	0.00%
error type	multi edit ocr error	161	6.21%	27.33%	117	72.67%	92.86%
error type	two or three edit ocr error	184	59.78%	95.65%	8	4.35%	6.35%
error type	visible prefix or suffix fragment	14	78.57%	92.86%	1	7.14%	0.79%
error type	single edit ocr error	105	98.10%	100.00%	0	0.00%	0.00%
operation family	multi edit ocr error	161	6.21%	27.33%	117	72.67%	92.86%
operation family	two or three edit ocr error	184	59.78%	95.65%	8	4.35%	6.35%
operation family	visible prefix or suffix fragment	14	78.57%	92.86%	1	7.14%	0.79%
operation family	single edit ocr error	105	98.10%	100.00%	0	0.00%	0.00%
operation profile	mixed operations	248	27.42%	56.45%	108	43.55%	85.71%
operation profile	substitutions only	158	74.05%	91.14%	14	8.86%	11.11%
operation profile	extra characters only	28	92.86%	92.86%	2	7.14%	1.59%
operation profile	missing characters only	30	76.67%	93.33%	2	6.67%	1.59%
query length band	6 7 characters	165	46.67%	76.97%	38	23.03%	30.16%
query length band	4 5 characters	90	44.44%	64.44%	32	35.56%	25.40%
query length band	8 9 characters	133	63.91%	81.95%	24	18.05%	19.05%
query length band	1 3 characters	19	10.53%	15.79%	16	84.21%	12.70%
query length band	10 plus characters	57	52.63%	71.93%	16	28.07%	12.70%
shared bigram band	0 shared bigrams	64	1.56%	1.56%	63	98.44%	50.00%
shared bigram band	1 shared bigram	61	14.75%	47.54%	32	52.46%	25.40%
shared bigram band	2 3 shared bigrams	168	47.62%	82.74%	29	17.26%	23.02%
shared bigram band	4 plus shared bigrams	171	84.21%	98.83%	2	1.17%	1.59%

Table 34: Complete Algorithm 5 mistake and evidence taxonomy on 66,257 synthetic pairs. The report retains all broader groups and the 50 error labels with the largest top-20 failure impact. The full row-level taxonomy, including all 4,184 synthetic error labels, remains in `a5_failure_taxonomy.csv`.

Dimension	Group	Cases	H@1	H@20	Misses	Failure rate	Failure share
category	ligature confusion	3991	97.97%	99.97%	1	0.03%	100.00%
category	combined errors effective 1	496	99.80%	100.00%	0	0.00%	0.00%
category	combined errors effective 2	7463	96.17%	100.00%	0	0.00%	0.00%
category	combined errors effective 3	4346	86.68%	100.00%	0	0.00%	0.00%
category	combined errors effective 4 plus	563	92.54%	100.00%	0	0.00%	0.00%
category	consonant frame wrong vowels heavy	68	97.06%	100.00%	0	0.00%	0.00%
category	dangerous name pair gold closer	139	97.84%	100.00%	0	0.00%	0.00%
category	double letter reduction expansion	2986	100.00%	100.00%	0	0.00%	0.00%
category	keyboard adjacent sampled	1915	98.54%	100.00%	0	0.00%	0.00%
category	multi char phonetic confusion	4985	99.78%	100.00%	0	0.00%	0.00%
category	ocr letter digit confusion	3998	99.80%	100.00%	0	0.00%	0.00%
category	ocr plus error effective 1	2	100.00%	100.00%	0	0.00%	0.00%
category	ocr plus error effective 2	2997	99.23%	100.00%	0	0.00%	0.00%
category	single char deletion position weighted	4935	99.45%	100.00%	0	0.00%	0.00%
category	single char insertion	2974	99.76%	100.00%	0	0.00%	0.00%
category	single letter phonetic confusion	5932	99.83%	100.00%	0	0.00%	0.00%
category	single letter visual confusion	7799	99.65%	100.00%	0	0.00%	0.00%
category	speed typing errors	2932	99.86%	100.00%	0	0.00%	0.00%
category	transposition position weighted	2992	99.87%	100.00%	0	0.00%	0.00%
category	wrong vowels relabelled by actual changes	4744	98.80%	100.00%	0	0.00%	0.00%
distance band	2 3 edits	26859	96.39%	100.00%	1	0.00%	100.00%
distance band	1 edit	38664	99.67%	100.00%	0	0.00%	0.00%
distance band	4 5 edits	734	85.97%	100.00%	0	0.00%	0.00%
operation family	visual or ocr	15777	99.26%	99.99%	1	0.01%	100.00%
operation family	deletion	4013	99.30%	100.00%	0	0.00%	0.00%
operation family	insertion	2563	99.69%	100.00%	0	0.00%	0.00%
operation family	keyboard	1914	98.54%	100.00%	0	0.00%	0.00%
operation family	mixed operations	18432	94.93%	100.00%	0	0.00%	0.00%
operation family	other	4785	100.00%	100.00%	0	0.00%	0.00%
operation family	phonetic	10928	99.80%	100.00%	0	0.00%	0.00%
operation family	transposition	3000	99.87%	100.00%	0	0.00%	0.00%
operation family	vowel	4845	98.76%	100.00%	0	0.00%	0.00%
operation profile	visual or ocr	15777	99.26%	99.99%	1	0.01%	100.00%
operation profile	deletion	4013	99.30%	100.00%	0	0.00%	0.00%
operation profile	insertion	2563	99.69%	100.00%	0	0.00%	0.00%
operation profile	keyboard	1914	98.54%	100.00%	0	0.00%	0.00%
operation profile	mixed operations	18432	94.93%	100.00%	0	0.00%	0.00%
operation profile	other	4785	100.00%	100.00%	0	0.00%	0.00%
operation profile	phonetic	10928	99.80%	100.00%	0	0.00%	0.00%
operation profile	transposition	3000	99.87%	100.00%	0	0.00%	0.00%
operation profile	vowel	4845	98.76%	100.00%	0	0.00%	0.00%
query length band	5 7 characters	11282	92.69%	99.99%	1	0.01%	100.00%
query length band	11 plus characters	32611	99.91%	100.00%	0	0.00%	0.00%
query length band	1 4 characters	776	77.84%	100.00%	0	0.00%	0.00%
query length band	8 10 characters	21588	99.20%	100.00%	0	0.00%	0.00%
shared bigram band	1 shared bigram	1144	74.13%	99.91%	1	0.09%	100.00%
shared bigram band	0 shared bigrams	349	56.45%	100.00%	0	0.00%	0.00%
shared bigram band	2 3 shared bigrams	6741	92.00%	100.00%	0	0.00%	0.00%
shared bigram band	4 plus shared bigrams	58023	99.63%	100.00%	0	0.00%	0.00%
error type	ligature d to cl pos 1	16	93.75%	93.75%	1	6.25%	100.00%
error type	wrong 1 vowel changes	2207	99.41%	100.00%	0	0.00%	0.00%
error type	wrong 2 vowel changes	1806	98.67%	100.00%	0	0.00%	0.00%
error type	wrong 3 vowel changes	731	97.26%	100.00%	0	0.00%	0.00%
error type	double first letter	635	100.00%	100.00%	0	0.00%	0.00%
error type	stutter prefix 2	617	100.00%	100.00%	0	0.00%	0.00%
error type	stutter prefix 3	576	100.00%	100.00%	0	0.00%	0.00%
error type	repeat last letter	564	99.65%	100.00%	0	0.00%	0.00%
error type	skip second letter	543	99.63%	100.00%	0	0.00%	0.00%
error type	deletion first char pos 0	496	98.39%	100.00%	0	0.00%	0.00%
error type	vowel plus phonetic variant 3 effective damerau 2	435	98.62%	100.00%	0	0.00%	0.00%
error type	deletion plus phonetic variant 1 effective damerau 2	423	95.74%	100.00%	0	0.00%	0.00%
error type	deletion plus phonetic variant 3 effective damerau 2	409	94.13%	100.00%	0	0.00%	0.00%
error type	vowel plus phonetic variant 2 effective damerau 2	404	97.28%	100.00%	0	0.00%	0.00%
error type	vowel plus phonetic variant 0 effective damerau 2	398	98.74%	100.00%	0	0.00%	0.00%

Dimension	Group	Cases	H@1	H@20	Misses	Failure rate	Failure share
error type	visual plus phonetic variant 1 effective damerau 2	387	98.19%	100.00%	0	0.00%	0.00%
error type	deletion plus phonetic variant 2 effective damerau 2	383	95.56%	100.00%	0	0.00%	0.00%
error type	vowel plus phonetic variant 1 effective damerau 2	379	98.68%	100.00%	0	0.00%	0.00%
error type	transposition plus deletion variant 0 effective damerau 2	378	95.50%	100.00%	0	0.00%	0.00%
error type	visual plus phonetic variant 3 effective damerau 2	372	95.97%	100.00%	0	0.00%	0.00%
error type	deletion plus phonetic variant 0 effective damerau 2	370	97.03%	100.00%	0	0.00%	0.00%
error type	transposition plus deletion variant 1 effective damerau 2	357	98.60%	100.00%	0	0.00%	0.00%
error type	visual visual phonetic variant 1 effective damerau 3	344	89.83%	100.00%	0	0.00%	0.00%
error type	visual plus phonetic variant 2 effective damerau 2	343	94.17%	100.00%	0	0.00%	0.00%
error type	visual visual phonetic variant 2 effective damerau 3	329	85.41%	100.00%	0	0.00%	0.00%
error type	vowel phonetic deletion variant 2 effective damerau 3	327	86.85%	100.00%	0	0.00%	0.00%
error type	vowel phonetic deletion variant 0 effective damerau 3	326	86.50%	100.00%	0	0.00%	0.00%
error type	visual plus phonetic variant 0 effective damerau 2	325	96.31%	100.00%	0	0.00%	0.00%
error type	keyboard vowel delete variant 0 effective damerau 3	324	76.85%	100.00%	0	0.00%	0.00%
error type	transposition plus deletion variant 2 effective damerau 2	312	97.44%	100.00%	0	0.00%	0.00%
error type	deletion last char pos 7	311	99.68%	100.00%	0	0.00%	0.00%
error type	visual visual phonetic variant 0 effective damerau 3	304	86.51%	100.00%	0	0.00%	0.00%
error type	ligature plus vowel variant 1 effective damerau 3	292	93.84%	100.00%	0	0.00%	0.00%
error type	multi phonetic i to y pos 1	277	100.00%	100.00%	0	0.00%	0.00%
error type	transposition plus deletion variant 3 effective damerau 2	270	98.52%	100.00%	0	0.00%	0.00%
error type	deletion last char pos 8	265	99.62%	100.00%	0	0.00%	0.00%
error type	ligature plus vowel variant 2 effective damerau 3	262	93.51%	100.00%	0	0.00%	0.00%
error type	keyboard vowel delete variant 2 effective damerau 3	251	83.27%	100.00%	0	0.00%	0.00%
error type	ligature plus vowel variant 3 effective damerau 3	250	92.00%	100.00%	0	0.00%	0.00%
error type	deletion last char pos 6	243	98.77%	100.00%	0	0.00%	0.00%
error type	ligature plus vowel variant 0 effective damerau 3	239	92.47%	100.00%	0	0.00%	0.00%
error type	vowel phonetic deletion variant 1 effective damerau 3	239	89.54%	100.00%	0	0.00%	0.00%
error type	keyboard vowel delete variant 1 effective damerau 3	234	86.32%	100.00%	0	0.00%	0.00%
error type	multi phonetic i to y pos 3	225	100.00%	100.00%	0	0.00%	0.00%
error type	deletion middle vowel pos 1	224	99.55%	100.00%	0	0.00%	0.00%
error type	insertion adjacent key letter at pos 0	224	98.66%	100.00%	0	0.00%	0.00%
error type	multi phonetic i to y pos 5	221	100.00%	100.00%	0	0.00%	0.00%
error type	ligature vowel transpose variant 1 effective damerau 3	216	75.00%	100.00%	0	0.00%	0.00%
error type	multi phonetic i to y pos 4	211	100.00%	100.00%	0	0.00%	0.00%
error type	insertion duplicate letter at pos 0	209	100.00%	100.00%	0	0.00%	0.00%

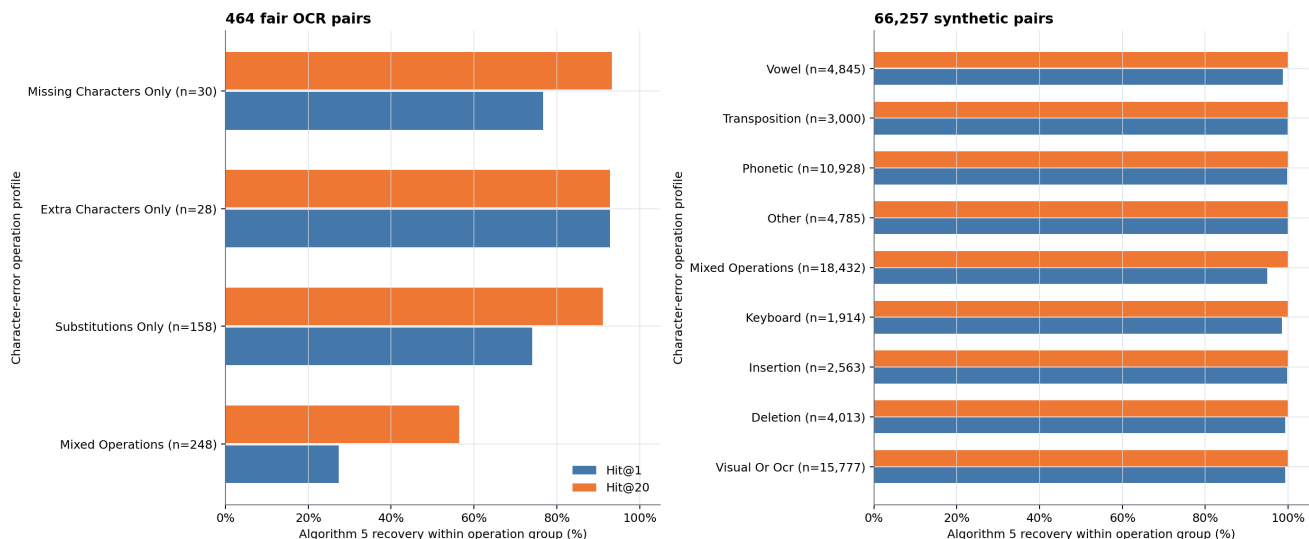


Figure 34: Blue bars report Hit@1 and orange bars report Hit@20 within each Algorithm 5 operation group. The OCR panel uses mechanical edit profiles such as substitution-only and mixed operations. The synthetic panel uses generator families such as visual/OCR, keyboard, phonetic, and vowel corruption. Each y-axis label prints its own denominator, so labels are not assumed equivalent across the two datasets.

Figure insights

1. Mixed-operation corruption is materially harder than a single isolated edit mechanism.
2. The OCR and synthetic gaps differ, so synthetic operation labels cannot replace observed OCR evaluation.

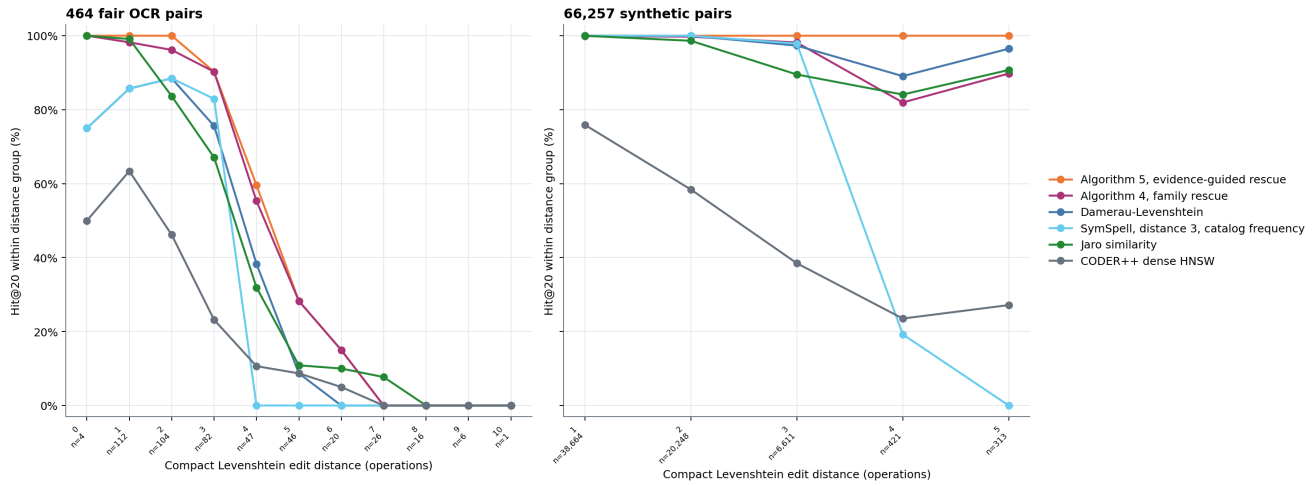


Figure 35: The x-axis is compact Levenshtein edit count and the y-axis is Hit@20 within that exact-distance group. The left panel uses OCR pairs; the right uses synthetic pairs. Each x-axis tick also gives the number of query-target pairs at that distance.

Figure insights

1. All lexical systems lose recall as edits accumulate, but the decline is much sharper on observed OCR.
2. Algorithm 5 gains are concentrated in multi-edit retrieval, not exact matching.

Table 35: Algorithm 5 rank correlations with Hit@20. Spearman rho measures whether larger values tend to occur with higher or lower recovery; it does not establish causation. OCR shared-bigram evidence is the exact count. Synthetic artifacts retain bands, encoded 0, 1, 2, and 4 for zero, one, two-to-three, and four-plus shared bigrams.

Dataset	Measured variable	Spearman rho Reading
464 fair OCR pairs	Edit distance (operations)	-0.726 higher values occur with less recovery
464 fair OCR pairs	Normalized distance (ratio)	-0.743 higher values occur with less recovery
464 fair OCR pairs	Query length (characters)	+0.188 higher values occur with more recovery
464 fair OCR pairs	Shared-bigram evidence	+0.668 higher values occur with more recovery
464 fair OCR pairs	Candidate count	+0.270 higher values occur with more recovery
464 fair OCR pairs	Latency (ms)	+0.087 higher values occur with more recovery
66,257 synthetic pairs	Edit distance (operations)	-0.004 higher values occur with less recovery
66,257 synthetic pairs	Normalized distance (ratio)	-0.007 higher values occur with less recovery
66,257 synthetic pairs	Query length (characters)	+0.006 higher values occur with more recovery
66,257 synthetic pairs	Shared-bigram evidence	+0.011 higher values occur with more recovery
66,257 synthetic pairs	Candidate count	-0.007 higher values occur with less recovery
66,257 synthetic pairs	Latency (ms)	-0.003 higher values occur with less recovery

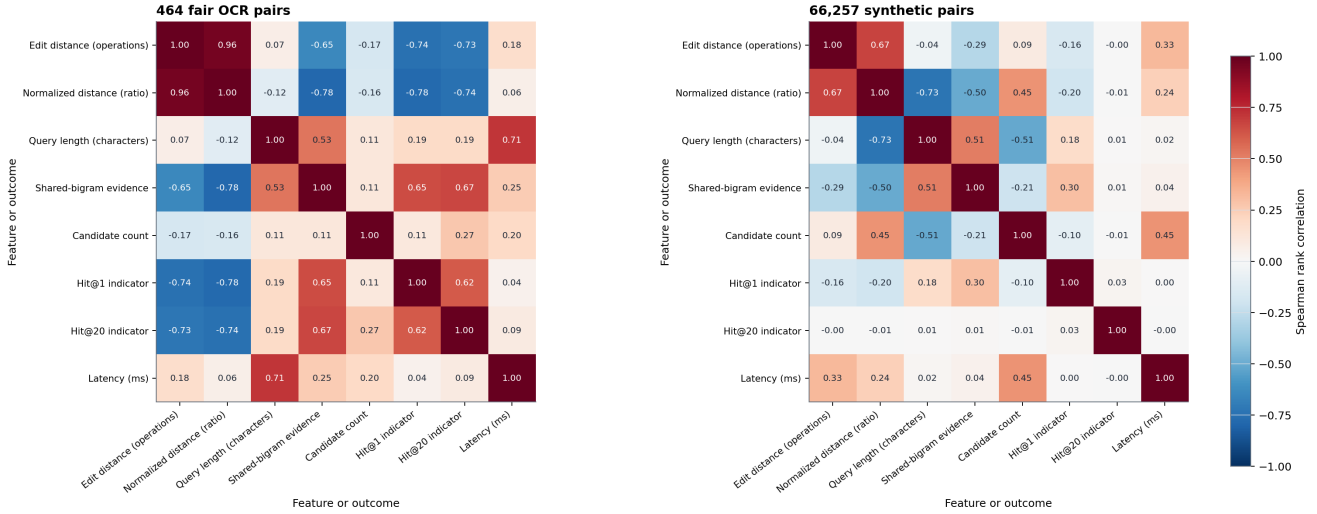


Figure 36: Each cell reports Spearman rank correlation for Algorithm 5. Rows and columns contain compact edit count, edit count divided by target length, query length, shared-bigram evidence, returned candidate count, Hit@1, Hit@20, and latency. Blue cells are positive, red cells are negative, and values near zero have little monotonic association. Synthetic shared-bigram evidence uses the ordinal band encoding defined in Table 35.

Figure insights

1. On OCR, the strongest measured monotonic relation with Hit@20 is Normalized distance (ratio) ($\rho = -0.743$).
2. Synthetic Hit@20 has only 1 miss, so its near-zero feature correlations are a ceiling effect rather than stable failure-driver estimates.

11.7 External Composite Ablations and Controlled Variants

A strict component ablation replaces a complete composite with one measured branch. Parameter sensitivity changes a radius, similarity, tokenizer, encoder, or weighting rule without claiming that the changed configuration is a component removal.

Table 36: Ablation coverage by method family. A component removal is reported only when a complete system contains separable branches; atomic scorers receive controlled variants instead of artificial ablations.

Method family	Analysis type	What is compared
RapidFuzz and character distances	Atomic scorer or controlled variant	Full, partial, token, Jaro, LCS, Levenshtein, OSA, and Damerau orderings are separate complete scorers; removing a component from one is undefined.
q-gram and BM25 retrieval	Parameter sensitivity	Two- versus three-character grams, four QuickUMLS coefficients, and Okapi versus BM25L/BM25Plus isolate token size and scoring formula.
SymSpell	Component plus radius	Uniform versus catalog frequency removes tie evidence; maximum edit radius two versus three changes candidate coverage.
Phonetic encoders	Model-family comparison	Soundex, Metaphone, NYSIIS, Match Rating Codex, and the domain phonetic baseline are atomic encoders followed by the same edit tie ordering.
Biomedical dense retrieval	Checkpoint and index audit	BiomedBERT, SapBERT, and CODER++ use the same pooling and catalog; HNSW top-20 recall is checked against exact dense search.
BioSyn and xMEN	Strict branch removal	Dense-only and sparse-only branches are compared with weighted hybrid or reciprocal-rank fusion.
preon	Strict stage removal	The token stage and normalized-Levenshtein partial stage are each removed from the published cascade.
Fuse.js	Parameter sensitivity	The default Bitap ranking is compared with the library’s token-search mode on the same catalog and queries.
FlexSearch	Tokenizer and encoder sensitivity	Tolerant versus full tokenization and default, LatinAdvanced, and LatinExtra encoders isolate index expansion and phonetic normalization.
Algorithm 5	Named and atomic removal	Forty-two named components or groups and 54 feature flags are removed on OCR, alongside one complete reference; 12 broad removals plus the complete reference are rerun on all synthetic pairs.

Table 37: Configuration sensitivity inside every measured external method family. Hit@1 spread is the best minus worst configuration in percentage points; a large spread means the implementation choice matters as much as the family label.

Dataset	Method family	Configs. Best Hit@1 system	Best H@1 Worst Hit@1 system	Worst H@1 H@1 spread	Best Hit@20 system	Best H@20
464 fair OCR pairs	delete-index spelling correction	3 SymSpell, distance 3, catalog frequency	38.58% SymSpell, distance 2, catalog frequency	34.91%	+3.66 SymSpell, distance 3, uniform frequency	56.25%
464 fair OCR pairs	edit distance	3 Exhaustive Levenshtein	34.27% Exhaustive Levenshtein	34.27%	+0.00 Exhaustive Levenshtein	59.27%
464 fair OCR pairs	client-side fuzzy search	1 Fuse.js default Bitap	32.97% Fuse.js default Bitap	32.97%	+0.00 Fuse.js default Bitap	58.84%
464 fair OCR pairs	normalized edit similarity	2 RapidFuzz quick ratio	32.76% RapidFuzz quick ratio	32.76%	+0.00 RapidFuzz quick ratio	60.34%
464 fair OCR pairs	character similarity	2 Jaro-Winkler	32.11% Jaro similarity	29.53%	+2.59 Jaro-Winkler	67.46%
464 fair OCR pairs	ordered subsequence	1 Longest-common-subsequence similarity	32.11% Longest-common-subsequence similarity	32.11%	+0.00 Longest-common-subsequence similarity	56.68%
464 fair OCR pairs	token-aware edit similarity	2 RapidFuzz token-sort ratio	31.47% RapidFuzz token-set ratio	31.25%	+0.22 RapidFuzz token-sort ratio	59.05%
464 fair OCR pairs	token-aware similarity	1 RapidFuzz token ratio	31.47% RapidFuzz token ratio	31.47%	+0.00 RapidFuzz token ratio	59.05%
464 fair OCR pairs	token-aware client-side fuzzy search	1 Fuse.js token search	26.08% Fuse.js token search	26.08%	+0.00 Fuse.js token search	51.29%
464 fair OCR pairs	sparse probabilistic retrieval	4 BM25 Okapi over character bigrams	22.63% BM25Plus over character trigrams	20.26%	+2.37 BM25 Okapi over character bigrams	50.43%
464 fair OCR pairs	q-gram set similarity	5 Character-bigram Dice	21.98% QuickUMLS-style 3-gram overlap	8.41%	+13.58 Character-bigram Dice	53.02%
464 fair OCR pairs	sparse-dense hybrid	3 BioSyn-style hybrid, sparse weight 0.50	20.47% BioSyn-style hybrid, sparse weight 0.25	18.97%	+1.51 BioSyn-style hybrid, sparse weight 0.50	53.02%
464 fair OCR pairs	sparse character retrieval	1 Character 3-gram TF-IDF	18.32% Character 3-gram TF-IDF	18.32%	+0.00 Character 3-gram TF-IDF	47.41%
464 fair OCR pairs	biomedical dense retrieval	3 CODER++ dense HNSW	17.67% BiomedBERT dense HNSW	2.80%	+14.87 SapBERT dense HNSW	38.58%
464 fair OCR pairs	rank-fusion ensemble	1 xMEN-style SapBERT plus TF-IDF RRF	16.81% xMEN-style SapBERT plus TF-IDF RRF	16.81%	+0.00 xMEN-style SapBERT plus TF-IDF RRF	52.37%
464 fair OCR pairs	phonetic encoding	4 NYSIIS plus edit ranking	16.16% Soundex plus edit ranking	5.60%	+10.56 Match Rating Codex plus edit ranking	46.34%
464 fair OCR pairs	staged drug-name normalization	3 preon without token matching	15.30% preon without partial edit matching	0.65%	+14.66 preon without token matching	16.16%
464 fair OCR pairs	adaptive edit similarity	1 RapidFuzz weighted ratio	11.21% RapidFuzz weighted ratio	11.21%	+0.00 RapidFuzz weighted ratio	56.25%
464 fair OCR pairs	indexed fuzzy and phonetic search	1 FlexSearch tolerant LatinAdvanced	7.54% FlexSearch tolerant LatinAdvanced	7.54%	+0.00 FlexSearch tolerant LatinAdvanced	9.70%
464 fair OCR pairs	phonetic	1 Phonetic baseline	6.90% Phonetic baseline	6.90%	+0.00 Phonetic baseline	23.28%
464 fair OCR pairs	substring edit similarity	1 RapidFuzz partial ratio	6.25% RapidFuzz partial ratio	6.25%	+0.00 RapidFuzz partial ratio	44.40%
464 fair OCR pairs	indexed aggressive phonetic search	1 FlexSearch tolerant LatinExtra	4.74% FlexSearch tolerant LatinExtra	4.74%	+0.00 FlexSearch tolerant LatinExtra	18.75%
464 fair OCR pairs	indexed fuzzy search	1 FlexSearch tolerant default	4.74% FlexSearch tolerant default	4.74%	+0.00 FlexSearch tolerant default	4.74%
464 fair OCR pairs	indexed substring and phonetic search	1 FlexSearch full LatinAdvanced	4.74% FlexSearch full LatinAdvanced	4.74%	+0.00 FlexSearch full LatinAdvanced	7.54%
464 fair OCR pairs	exact and prefix	1 Exact or prefix match	2.59% Exact or prefix match	2.59%	+0.00 Exact or prefix match	3.02%
66,257 synthetic pairs	edit distance	3 Damerau-Levenshtein	94.78% Exhaustive Levenshtein	94.05%	+0.73 Damerau-Levenshtein	99.64%
66,257 synthetic pairs	delete-index spelling correction	3 SymSpell, distance 3, uniform frequency	94.51% SymSpell, distance 2, catalog frequency	87.84%	+6.67 SymSpell, distance 3, uniform frequency	98.86%
66,257 synthetic pairs	ordered subsequence	1 Longest-common-subsequence similarity	93.60% Longest-common-subsequence similarity	93.60%	+0.00 Longest-common-subsequence similarity	99.21%
66,257 synthetic pairs	normalized edit similarity	2 RapidFuzz quick ratio	92.79% RapidFuzz quick ratio	92.79%	+0.00 RapidFuzz quick ratio	99.00%
66,257 synthetic pairs	character similarity	2 Jaro-Winkler	89.57% Jaro similarity	89.29%	+0.28 Jaro-Winkler	98.43%
66,257 synthetic pairs	q-gram set similarity	5 Character-bigram Dice	84.54% QuickUMLS-style 3-gram overlap	36.74%	+47.80 Character-bigram Dice	95.56%
66,257 synthetic pairs	sparse probabilistic retrieval	4 BM25 Okapi over character bigrams	82.03% BM25Plus over character trigrams	78.19%	+3.83 BM25 Okapi over character bigrams	94.40%
66,257 synthetic pairs	client-side fuzzy search	1 Fuse.js default Bitap	81.22% Fuse.js default Bitap	81.22%	+0.00 Fuse.js default Bitap	95.00%
66,257 synthetic pairs	sparse-dense hybrid	3 BioSyn-style hybrid, sparse weight 0.50	78.17% BioSyn-style hybrid, sparse weight 0.25	66.01%	+12.16 BioSyn-style hybrid, sparse weight 0.50	93.78%
66,257 synthetic pairs	token-aware client-side fuzzy search	1 Fuse.js token search	78.06% Fuse.js token search	78.06%	+0.00 Fuse.js token search	93.34%
66,257 synthetic pairs	phonetic encoding	4 NYSIIS plus edit ranking	77.01% Soundex plus edit ranking	13.09%	+63.92 Metaphone plus edit ranking	94.62%
66,257 synthetic pairs	sparse character retrieval	1 Character 3-gram TF-IDF	75.21% Character 3-gram TF-IDF	75.21%	+0.00 Character 3-gram TF-IDF	91.66%
66,257 synthetic pairs	staged drug-name normalization	3 preon default drug normalizer	74.16% preon without partial edit matching	0.00%	+74.16 preon default drug normalizer	74.37%
66,257 synthetic pairs	phonetic	1 Phonetic baseline	69.66% Phonetic baseline	69.66%	+0.00 Phonetic baseline	92.07%
66,257 synthetic pairs	token-aware similarity	1 RapidFuzz token ratio	65.67% RapidFuzz token ratio	65.67%	+0.00 RapidFuzz token ratio	85.47%
66,257 synthetic pairs	token-aware edit similarity	2 RapidFuzz token-set ratio	65.66% RapidFuzz token-set ratio	65.64%	+0.01 RapidFuzz token-set ratio	95.46%
66,257 synthetic pairs	adaptive edit similarity	1 RapidFuzz weighted ratio	55.73% RapidFuzz weighted ratio	55.73%	+0.00 RapidFuzz weighted ratio	97.20%
66,257 synthetic pairs	rank-fusion ensemble	1 xMEN-style SapBERT plus TF-IDF RRF	48.85% xMEN-style SapBERT plus TF-IDF RRF	48.85%	+0.00 xMEN-style SapBERT plus TF-IDF RRF	92.30%
66,257 synthetic pairs	biomedical dense retrieval	3 CODER++ dense HNSW	42.56% BiomedBERT dense HNSW	6.22%	+36.33 CODER++ dense HNSW	66.26%
66,257 synthetic pairs	indexed fuzzy and phonetic search	1 FlexSearch tolerant LatinAdvanced	20.32% FlexSearch tolerant LatinAdvanced	20.32%	+0.00 FlexSearch tolerant LatinAdvanced	21.89%
66,257 synthetic pairs	substring edit similarity	1 RapidFuzz partial ratio	16.68% RapidFuzz partial ratio	16.68%	+0.00 RapidFuzz partial ratio	92.90%
66,257 synthetic pairs	indexed aggressive phonetic search	1 FlexSearch tolerant LatinExtra	15.43% FlexSearch tolerant LatinExtra	15.43%	+0.00 FlexSearch tolerant LatinExtra	29.51%
66,257 synthetic pairs	indexed substring and phonetic search	1 FlexSearch full LatinAdvanced	13.94% FlexSearch full LatinAdvanced	13.94%	+0.00 FlexSearch full LatinAdvanced	15.75%
66,257 synthetic pairs	indexed fuzzy search	1 FlexSearch tolerant default	10.56% FlexSearch tolerant default	10.56%	+0.00 FlexSearch tolerant default	10.91%
66,257 synthetic pairs	exact and prefix	1 Exact or prefix match	3.00% Exact or prefix match	3.00%	+0.00 Exact or prefix match	3.01%

Table 38: Complete controlled-variant sensitivity for implementation families with at least two evaluated configurations. Deltas are relative to that family’s best measured configuration on the same dataset, not relative to Algorithm 5.

Dataset	Implementation family	Configuration	Hit@1 Delta from family best	H@1 Hit@20 Delta from family best	H@20
464 fair OCR pairs	BM25 character retrieval	BM25 Okapi over character bigrams	22.63%	+0.00 50.43%	+0.00
464 fair OCR pairs	BM25 character retrieval	BM25L over character trigrams	21.34%	-1.29 46.77%	-3.66
464 fair OCR pairs	BM25 character retrieval	BM25 Okapi over character trigrams	20.69%	-1.94 47.41%	-3.02
464 fair OCR pairs	BM25 character retrieval	BM25Plus over character trigrams	20.26%	-2.37 47.41%	-3.02
464 fair OCR pairs	Biomedical dense encoders	CODER++ dense HNSW	17.67%	+0.00 32.33%	-6.25
464 fair OCR pairs	Biomedical dense encoders	SapBERT dense HNSW	11.42%	-6.25 38.58%	+0.00
464 fair OCR pairs	Biomedical dense encoders	BiomedBERT dense HNSW	2.80%	-14.87 7.33%	-31.25
464 fair OCR pairs	Character q-gram retrieval	Character-bigram Dice	21.98%	+0.00 53.02%	+0.00
464 fair OCR pairs	Character q-gram retrieval	QuickUMLS-style 3-gram Dice	18.97%	-3.02 46.12%	-6.90
464 fair OCR pairs	Character q-gram retrieval	QuickUMLS-style 3-gram Jaccard	18.97%	-3.02 46.12%	-6.90
464 fair OCR pairs	Character q-gram retrieval	QuickUMLS-style 3-gram cosine	17.67%	-4.31 45.91%	-7.11
464 fair OCR pairs	Character q-gram retrieval	QuickUMLS-style 3-gram overlap	8.41%	-13.58 37.93%	-15.09
464 fair OCR pairs	Character-similarity scorers	Optimal string alignment	34.27%	+0.00 59.27%	-1.29
464 fair OCR pairs	Character-similarity scorers	Damerau-Levenshtein	34.27%	+0.00 59.27%	-1.29
464 fair OCR pairs	Character-similarity scorers	RapidFuzz quick ratio	32.76%	-1.51 60.34%	-0.22
464 fair OCR pairs	Character-similarity scorers	RapidFuzz ratio	32.76%	-1.51 60.34%	-0.22
464 fair OCR pairs	Character-similarity scorers	Longest-common-subsequence similarity	32.11%	-2.16 56.68%	-3.88
464 fair OCR pairs	Character-similarity scorers	RapidFuzz token-sort ratio	31.47%	-2.80 59.05%	-1.51
464 fair OCR pairs	Character-similarity scorers	RapidFuzz token-set ratio	31.25%	-3.02 59.05%	-1.51
464 fair OCR pairs	Character-similarity scorers	Jaro similarity	29.53%	-4.74 60.56%	+0.00
464 fair OCR pairs	Character-similarity scorers	RapidFuzz weighted ratio	11.21%	-23.06 56.25%	-4.31
464 fair OCR pairs	Character-similarity scorers	RapidFuzz partial ratio	6.25%	-28.02 44.40%	-16.16
464 fair OCR pairs	FlexSearch	FlexSearch tolerant LatinAdvanced	7.54%	+0.00 9.70%	-9.05
464 fair OCR pairs	FlexSearch	FlexSearch tolerant LatinExtra	4.74%	-2.80 18.75%	+0.00
464 fair OCR pairs	FlexSearch	FlexSearch full LatinAdvanced	4.74%	-2.80 7.54%	-11.21
464 fair OCR pairs	FlexSearch	FlexSearch tolerant default	4.74%	-2.80 4.74%	-14.01
464 fair OCR pairs	Fuse.js	Fuse.js default Bitap	32.97%	+0.00 58.84%	+0.00
464 fair OCR pairs	Fuse.js	Fuse.js token search	26.08%	-6.90 51.29%	-7.54
464 fair OCR pairs	Phonetic encoders	NYSIIS plus edit ranking	16.16%	+0.00 39.22%	-7.11
464 fair OCR pairs	Phonetic encoders	Match Rating Codex plus edit ranking	15.73%	-0.43 46.34%	+0.00
464 fair OCR pairs	Phonetic encoders	Metaphone plus edit ranking	12.07%	-4.09 34.05%	-12.28
464 fair OCR pairs	Phonetic encoders	Soundex plus edit ranking	5.60%	-10.56 24.35%	-21.98
464 fair OCR pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.50	20.47%	+0.00 53.02%	+0.00
464 fair OCR pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.75	19.40%	-1.08 50.86%	-2.16
464 fair OCR pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.25	18.97%	-1.51 52.16%	-0.86
464 fair OCR pairs	SymSpell delete-index retrieval	SymSpell, distance 3, catalog frequency	38.58%	+0.00 55.82%	-0.43
464 fair OCR pairs	SymSpell delete-index retrieval	SymSpell, distance 3, uniform frequency	36.64%	-1.94 56.25%	+0.00
464 fair OCR pairs	SymSpell delete-index retrieval	SymSpell, distance 2, catalog frequency	34.91%	-3.66 41.59%	-14.66
464 fair OCR pairs	preon staged normalizer	preon without token matching	15.30%	+0.00 16.16%	+0.00
464 fair OCR pairs	preon staged normalizer	preon default drug normalizer	15.09%	-0.22 15.95%	-0.22
464 fair OCR pairs	preon staged normalizer	preon without partial edit matching	0.65%	-14.66 0.65%	-15.52
66,257 synthetic pairs	BM25 character retrieval	BM25 Okapi over character bigrams	82.03%	+0.00 94.40%	+0.00
66,257 synthetic pairs	BM25 character retrieval	BM25L over character trigrams	78.96%	-3.07 91.82%	-2.58
66,257 synthetic pairs	BM25 character retrieval	BM25 Okapi over character trigrams	78.25%	-3.78 92.17%	-2.22
66,257 synthetic pairs	BM25 character retrieval	BM25Plus over character trigrams	78.19%	-3.83 92.17%	-2.23
66,257 synthetic pairs	Biomedical dense encoders	CODER++ dense HNSW	42.56%	+0.00 66.26%	+0.00
66,257 synthetic pairs	Biomedical dense encoders	SapBERT dense HNSW	27.67%	-14.89 56.45%	-9.81
66,257 synthetic pairs	Biomedical dense encoders	BiomedBERT dense HNSW	6.22%	-36.33 13.01%	-53.26
66,257 synthetic pairs	Character q-gram retrieval	Character-bigram Dice	84.54%	+0.00 95.56%	+0.00
66,257 synthetic pairs	Character q-gram retrieval	QuickUMLS-style 3-gram Dice	78.34%	-6.19 92.52%	-3.04
66,257 synthetic pairs	Character q-gram retrieval	QuickUMLS-style 3-gram Jaccard	78.34%	-6.19 92.52%	-3.04
66,257 synthetic pairs	Character q-gram retrieval	QuickUMLS-style 3-gram cosine	77.46%	-7.08 92.40%	-3.16
66,257 synthetic pairs	Character q-gram retrieval	QuickUMLS-style 3-gram overlap	36.74%	-47.80 88.37%	-7.19
66,257 synthetic pairs	Character-similarity scorers	Damerau-Levenshtein	94.78%	+0.00 99.64%	+0.00
66,257 synthetic pairs	Character-similarity scorers	Optimal string alignment	94.75%	-0.03 99.58%	-0.06
66,257 synthetic pairs	Character-similarity scorers	Longest-common-subsequence similarity	93.60%	-1.18 99.21%	-0.43
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz quick ratio	92.79%	-1.99 99.00%	-0.64
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz ratio	92.79%	-1.99 99.00%	-0.64
66,257 synthetic pairs	Character-similarity scorers	Jaro similarity	89.29%	-5.49 98.38%	-1.26
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz token-set ratio	65.66%	-29.13 85.46%	-14.18
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz token-sort ratio	65.64%	-29.14 85.46%	-14.18
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz weighted ratio	55.73%	-39.05 97.20%	-2.44
66,257 synthetic pairs	Character-similarity scorers	RapidFuzz partial ratio	16.68%	-78.10 92.90%	-6.74
66,257 synthetic pairs	FlexSearch	FlexSearch tolerant LatinAdvanced	20.32%	+0.00 21.89%	-7.61

Dataset	Implementation family	Configuration	Hit@1 Delta from family best	H@1	Hit@20 Delta from family best	H@20
66,257 synthetic pairs	FlexSearch	FlexSearch tolerant LatinExtra	15.43%	-4.90	29.51%	+0.00
66,257 synthetic pairs	FlexSearch	FlexSearch full LatinAdvanced	13.94%	-6.38	15.75%	-13.76
66,257 synthetic pairs	FlexSearch	FlexSearch tolerant default	10.56%	-9.77	10.91%	-18.59
66,257 synthetic pairs	Fuse.js	Fuse.js default Bitap	81.22%	+0.00	95.00%	+0.00
66,257 synthetic pairs	Fuse.js	Fuse.js token search	78.06%	-3.17	93.34%	-1.66
66,257 synthetic pairs	Phonetic encoders	NYSIIS plus edit ranking	77.01%	+0.00	93.90%	-0.72
66,257 synthetic pairs	Phonetic encoders	Metaphone plus edit ranking	75.55%	-1.45	94.62%	+0.00
66,257 synthetic pairs	Phonetic encoders	Match Rating Codex plus edit ranking	65.82%	-11.19	94.58%	-0.04
66,257 synthetic pairs	Phonetic encoders	Soundex plus edit ranking	13.09%	-63.92	67.73%	-26.89
66,257 synthetic pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.50	78.17%	+0.00	93.78%	+0.00
66,257 synthetic pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.75	77.88%	-0.29	93.40%	-0.39
66,257 synthetic pairs	Sparse-Dense Hybrid	BioSyn-style hybrid, sparse weight 0.25	66.01%	-12.16	90.20%	-3.58
66,257 synthetic pairs	SymSpell delete-index retrieval	SymSpell, distance 3, uniform frequency	94.51%	+0.00	98.86%	+0.00
66,257 synthetic pairs	SymSpell delete-index retrieval	SymSpell, distance 3, catalog frequency	94.22%	-0.29	98.80%	-0.06
66,257 synthetic pairs	SymSpell delete-index retrieval	SymSpell, distance 2, catalog frequency	87.84%	-6.67	90.55%	-8.30
66,257 synthetic pairs	preon staged normalizer	preon default drug normalizer	74.16%	+0.00	74.37%	+0.00
66,257 synthetic pairs	preon staged normalizer	preon without token matching	74.16%	+0.00	74.37%	+0.00
66,257 synthetic pairs	preon staged normalizer	preon without partial edit matching	0.00%	-74.16	0.00%	-74.37

Table 39: Every external composite removal and controlled parameter contrast. Deltas are changed configuration minus the stated reference.

Dataset	Study	Change	Reference H@1	Changed H@1	Delta H@1	Delta H@20
464 fair OCR pairs	BM25 Okapi bigrams	Replace character bigrams with trigrams	22.63%	20.69%	-1.94	-3.02
464 fair OCR pairs	BM25 Okapi trigrams	Replace Okapi length normalization with BM25L	20.69%	21.34%	+0.65	-0.65
464 fair OCR pairs	BM25 Okapi trigrams	Replace Okapi length normalization with BM25Plus	20.69%	20.26%	-0.43	+0.00
464 fair OCR pairs	FlexSearch full, Latin advanced	Replace full tokenization with tolerant tokenization	4.74%	7.54%	+2.80	+2.16
464 fair OCR pairs	FlexSearch tolerant, Latin advanced	Replace Latin advanced encoding with the default encoder	7.54%	4.74%	-2.80	-4.96
464 fair OCR pairs	FlexSearch tolerant, Latin extra	Replace Latin extra encoding with the default encoder	4.74%	4.74%	+0.00	-14.01
464 fair OCR pairs	Fuse.js token search	Disable token-aware matching	26.08%	32.97%	+6.90	+7.54
464 fair OCR pairs	QuickUMLS Dice	Replace Dice with Jaccard	18.97%	18.97%	+0.00	+0.00
464 fair OCR pairs	QuickUMLS Dice	Replace Dice with cosine	18.97%	17.67%	-1.29	-0.22
464 fair OCR pairs	QuickUMLS Dice	Replace Dice with overlap coefficient	18.97%	8.41%	-10.56	-8.19
464 fair OCR pairs	RapidFuzz WRatio	Use full-string ratio only	11.21%	32.76%	+21.55	+4.09
464 fair OCR pairs	RapidFuzz WRatio	Use partial-string ratio only	11.21%	6.25%	-4.96	-11.85
464 fair OCR pairs	RapidFuzz WRatio	Use token-set ratio only	11.21%	31.25%	+20.04	+2.80
464 fair OCR pairs	RapidFuzz WRatio	Use token-sort ratio only	11.21%	31.47%	+20.26	+2.80
464 fair OCR pairs	SymSpell catalog frequency	Reduce maximum edit radius from 3 to 2	38.58%	34.91%	-3.66	-14.22
464 fair OCR pairs	BioSyn 0.50	Remove dense SapBERT branch	20.47%	18.32%	-2.16	-5.60
464 fair OCR pairs	BioSyn 0.50	Remove sparse TF-IDF branch	20.47%	11.42%	-9.05	-14.44
464 fair OCR pairs	SymSpell distance 3	Remove catalog-frequency tie evidence	38.58%	36.64%	-1.94	+0.43
464 fair OCR pairs	preon default cascade	Remove exact-token stage	15.09%	15.30%	+0.22	+0.22
464 fair OCR pairs	preon default cascade	Remove normalized-Levenshtein partial stage	15.09%	0.65%	-14.44	-15.30
464 fair OCR pairs	xMEN RRF	Remove dense SapBERT branch	16.81%	18.32%	+1.51	-4.96
464 fair OCR pairs	xMEN RRF	Remove sparse TF-IDF branch	16.81%	11.42%	-5.39	-13.79
66,257 synthetic pairs	BM25 Okapi bigrams	Replace character bigrams with trigrams	82.03%	78.25%	-3.78	-2.22
66,257 synthetic pairs	BM25 Okapi trigrams	Replace Okapi length normalization with BM25L	78.25%	78.96%	+0.71	-0.35
66,257 synthetic pairs	BM25 Okapi trigrams	Replace Okapi length normalization with BM25Plus	78.25%	78.19%	-0.05	-0.01
66,257 synthetic pairs	FlexSearch full, Latin advanced	Replace full tokenization with tolerant tokenization	13.94%	20.32%	+6.38	+6.14
66,257 synthetic pairs	FlexSearch tolerant, Latin advanced	Replace Latin advanced encoding with the default encoder	20.32%	10.56%	-9.77	-10.98
66,257 synthetic pairs	FlexSearch tolerant, Latin extra	Replace Latin extra encoding with the default encoder	15.43%	10.56%	-4.87	-18.59
66,257 synthetic pairs	Fuse.js token search	Disable token-aware matching	78.06%	81.22%	+3.17	+1.66
66,257 synthetic pairs	QuickUMLS Dice	Replace Dice with Jaccard	78.34%	78.34%	+0.00	+0.00
66,257 synthetic pairs	QuickUMLS Dice	Replace Dice with cosine	78.34%	77.46%	-0.89	-0.13
66,257 synthetic pairs	QuickUMLS Dice	Replace Dice with overlap coefficient	78.34%	36.74%	-41.60	-4.15
66,257 synthetic pairs	RapidFuzz WRatio	Use full-string ratio only	55.73%	92.79%	+37.06	+1.81
66,257 synthetic pairs	RapidFuzz WRatio	Use partial-string ratio only	55.73%	16.68%	-39.05	-4.30
66,257 synthetic pairs	RapidFuzz WRatio	Use token-set ratio only	55.73%	65.66%	+9.92	-11.74
66,257 synthetic pairs	RapidFuzz WRatio	Use token-sort ratio only	55.73%	65.64%	+9.91	-11.74
66,257 synthetic pairs	SymSpell catalog frequency	Reduce maximum edit radius from 3 to 2	94.22%	87.84%	-6.38	-8.25
66,257 synthetic pairs	BioSyn 0.50	Remove dense SapBERT branch	78.17%	75.21%	-2.96	-2.12
66,257 synthetic pairs	BioSyn 0.50	Remove sparse TF-IDF branch	78.17%	27.67%	-50.50	-37.33
66,257 synthetic pairs	SymSpell distance 3	Remove catalog-frequency tie evidence	94.22%	94.51%	+0.29	+0.06
66,257 synthetic pairs	preon default cascade	Remove exact-token stage	74.16%	74.16%	+0.00	+0.00
66,257 synthetic pairs	preon default cascade	Remove normalized-Levenshtein partial stage	74.16%	0.00%	-74.16	-74.37
66,257 synthetic pairs	xMEN RRF	Remove dense SapBERT branch	48.85%	75.21%	+26.36	-0.64
66,257 synthetic pairs	xMEN RRF	Remove sparse TF-IDF branch	48.85%	27.67%	-21.18	-35.84

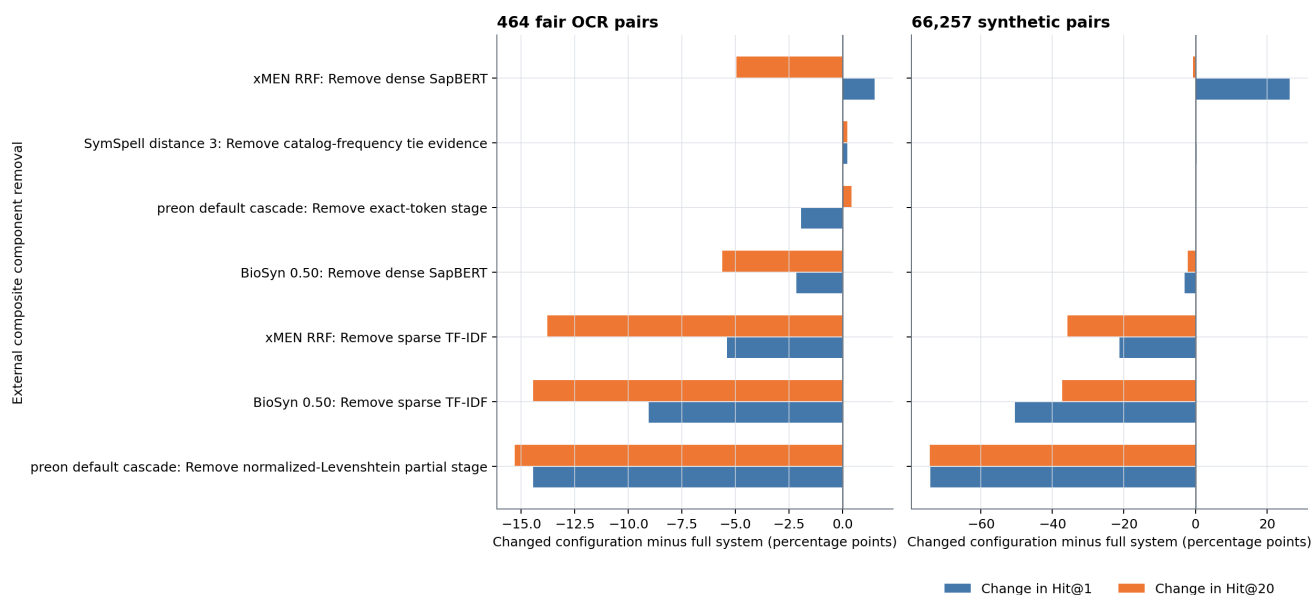


Figure 37: The x-axis is the change in Hit@1 or Hit@20 after removing one external composite branch. Negative values mean the complete composite performed better.

Figure insights

1. Sparse character evidence contributes more than dense biomedical embeddings on isolated OCR strings.
2. preon’s partial edit stage is measured separately from its token stage.

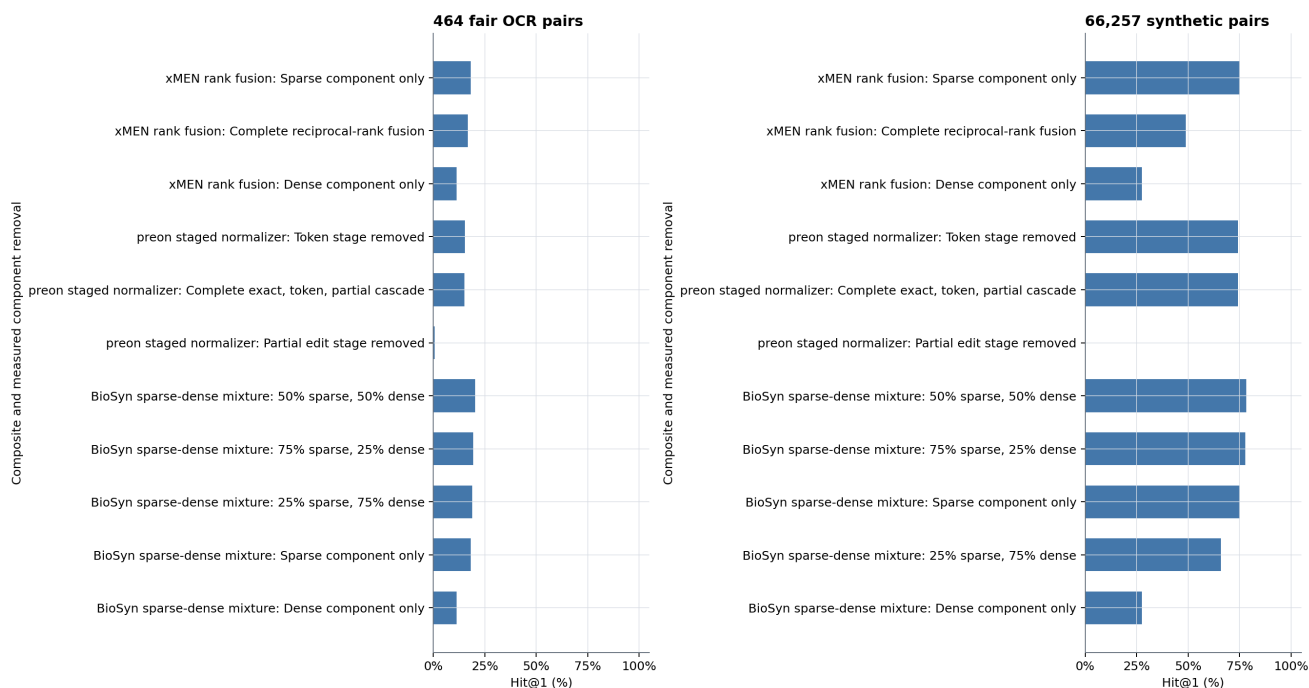


Figure 38: Each bar is Hit@1 for a complete composite or one measured branch on the named dataset. BioSyn compares sparse and dense branches with their weighted hybrid; xMEN compares reciprocal-rank fusion with each branch; preon compares its cascade with token and partial stages removed.

Figure insights

1. The sparse character branch is stronger than isolated biomedical dense embeddings for OCR strings.
2. preon's normalized-Levenshtein partial stage supplies nearly all of its retrieval coverage.

11.8 Algorithm 5 Ablation

The OCR experiment disables 43 named components or groups and 54 individual feature flags, one at a time. The synthetic confirmation reruns the 13 broad removals over all 66,257 pairs.

Table 40: Complete Algorithm 5 ablation result on all 464 fair OCR pairs. Zero delta means that switch did not alter Hit@1 or Hit@20 on this locked dataset; it may still change lower ranks or safety output. Median milliseconds are measured per warm query in the same run.

Ablation	Level	Hit@1	Delta	Hit@1 Full-only	Hit@1 Abl.-only	Hit@1	Hit@20	Delta	Hit@20 Full-only	Hit@20 Abl.-only	Hit@20	Median ms
Flag off: strict full name rerank	atomic_flag	49.57%	-0.86	4	0	72.84%	+0.00	0	23.556	0	24.001	
Flag off: contained nearest rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.231	0	23.558	
Flag off: exact chain existing augmentation	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.596	0	23.623	
Flag off: exact key pareto tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.739	0	23.675	
Flag off: exact keyboard key tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.650	0	23.634	
Flag off: exact keyboard weighted extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.945	0	23.709	
Flag off: exact ligature nearest rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.621	0	23.202	
Flag off: exact ligature rank extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.688	0	23.667	
Flag off: exact phonetic rewrite rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.543	0	23.320	
Flag off: exact transposition dual extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.726	0	23.679	
Flag off: exact transposition tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.391	0	23.924	
Flag off: exact visual edge tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.795	0	23.795	
Flag off: guarded top score dominant chain rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.666	0	23.752	
Flag off: keyboard exact key rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	24.158	0	23.807	
Flag off: keyboard vowel delete rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.775	0	23.775	
Flag off: keyboard vowel delete score tuning	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.232	0	23.672	
Flag off: ligature vowel distance extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.414	0	23.800	
Flag off: ligature vowel edge extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.597	0	23.227	
Flag off: ligature vowel in place score	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.514	0	23.202	
Flag off: ligature vowel multi chain extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.739	0	24.140	
Flag off: ligature vowel rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.139	0	23.167	
Flag off: ligature vowel transpose extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.308	0	23.287	
Flag off: ligature vowel transpose in place score	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.755	0	23.665	
Flag off: ligature vowel transpose rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.190	0	23.264	
Flag off: phonetic position tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.333	0	23.630	
Flag off: preserved top dominant nearest rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.762	0	23.467	
Flag off: preserved top higher score nearest rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.495	0	23.705	
Flag off: score dominant chain release rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	23.538	1	23.899	
Flag off: shifted edge agreement rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0	8.506	3		
Flag off: short query rescue	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: skeleton position tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: stutter prefix rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: transpose vowel delete in place score	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: transpose vowel delete rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: transpose vowel delete score tuning	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: transposition delete rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: transposition deletion exact chain	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: unique nearest rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual distance tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual multi chain rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual phonetic chain rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual phonetic dual extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual phonetic exact key extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual phonetic score tuning	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual visual delete rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: visual visual deletion chain	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: vowel phonetic delete rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: vowel phonetic multi chain extension rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: weighted edge advantage rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: weighted edge tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: weighted exact key tie rerank	atomic_flag	50.43%	+0.00	0	0	72.84%	+0.00	0				
Flag off: keyboard vowel delete in place score	atomic_flag	50.43%	+0.00	0	0	73.06%	+0.22	0				
Flag off: visual phonetic in place score	atomic_flag	50.43%	+0.00	0	0	73.06%	+0.22	0				
Flag off: strict full name evidence guard	atomic_flag	50.65%	+0.22	1	2	72.84%	+0.00	0				
Without family rescue	component	36.64%	-13.79	69	5	52.80%	-20.04	96				

Ablation	Level	Hit@1	Delta	H@1 Full-only	H@1 Abl.-only	H@1	Hit@20	Delta	H@20 Full-only	H@20 Abl.-only	H@20	Median	ms
Without external retriever	component	49.78%	-0.65	6	3	71.98%	-0.86	4	0	0	33.992		
Without context cleanup	component	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.630		
Complete Algorithm 5	reference	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.818		
Without all reranking	reranker	40.52%	-9.91	47	1	70.26%	-2.59	12	0	0	21.405		
Without candidate-pool/head rerankers	reranker	48.28%	-2.16	10	0	70.91%	-1.94	9	0	0	21.355		
Without character-evidence rerankers	reranker	48.92%	-1.51	7	0	72.84%	+0.00	0	0	0	23.432		
Without strict full-name correction	reranker	49.57%	-0.86	4	0	72.84%	+0.00	0	0	0	23.891		
Without multi-step-chain rerankers	reranker	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.563		
Without exact-key rerankers	reranker	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.260		
Without nearest-candidate rerankers	reranker	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.357		
Without weighted-edge rerankers	reranker	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.490		
Without family-head rescue	retrieval	48.06%	-2.37	12	1	70.47%	-2.37	11	0	0	23.666		
Without delete-key retrieval	retrieval	48.28%	-2.16	11	1	70.69%	-2.16	10	0	0	23.435		
Without short-edge retrieval	retrieval	49.78%	-0.65	3	0	70.04%	-2.80	14	1	1	21.731		
Without compatible-length scan	retrieval	49.78%	-0.65	3	0	71.34%	-1.51	7	0	0	21.304		
Without short visible-head rescue	retrieval	50.43%	+0.00	0	0	72.63%	-0.22	1	0	0	23.574		
Without first-character expansion	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	22.371		
Without ligature rescue	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.189		
Without short-frame rescue	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	1	1	1	23.984		
Without short-query rescue	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.843		
Without transposition rescue	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.647		
Without visual-deletion chains	retrieval	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.522		
Without all multi-step chains	retrieval	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	22.377		
Without evidence-query variants	retrieval	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	22.672		
Without keyboard rescue	retrieval	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	23.125		
Without bounded-nearest fallback	retrieval	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	19.674		
Without visual-phonetic chains	retrieval	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	23.592		
Without safety clarification gate	safety	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.737		
Without exact-chain score augmentation	scoring	50.43%	+0.00	0	0	72.84%	+0.00	0	0	0	23.461		
Without in-place chain score tuning	scoring	50.43%	+0.00	0	0	73.06%	+0.22	0	1	1	23.359		
Without character n-grams	signal	48.06%	-2.37	11	0	71.98%	-0.86	8	4	4	22.804		
Without prefix evidence	signal	48.71%	-1.72	11	3	70.91%	-1.94	9	0	0	23.586		
Without weighted confusion costs	signal	48.71%	-1.72	9	1	72.63%	-0.22	5	4	4	22.316		
Without positional evidence	signal	48.92%	-1.51	8	1	71.34%	-1.51	9	2	2	23.405		
Without raw edit similarity	signal	49.35%	-1.08	6	1	67.46%	-5.39	27	2	2	23.718		
Without weighted edit similarity	signal	49.57%	-0.86	8	4	65.73%	-7.11	35	2	2	17.988		
Without phonetic evidence	signal	49.57%	-0.86	4	0	72.84%	+0.00	0	0	0	23.257		
Without suffix evidence	signal	49.57%	-0.86	5	1	73.28%	+0.43	0	2	2	23.479		
Without consonant skeleton	signal	49.78%	-0.65	3	0	72.63%	-0.22	1	0	0	23.366		
Without subsequence evidence	signal	50.00%	-0.43	3	1	72.63%	-0.22	2	1	1	23.209		
Without length coverage	signal	50.22%	-0.22	2	1	69.83%	-3.02	15	1	1	23.611		
Without retriever agreement	signal	50.22%	-0.22	2	1	72.84%	+0.00	0	0	0	24.009		

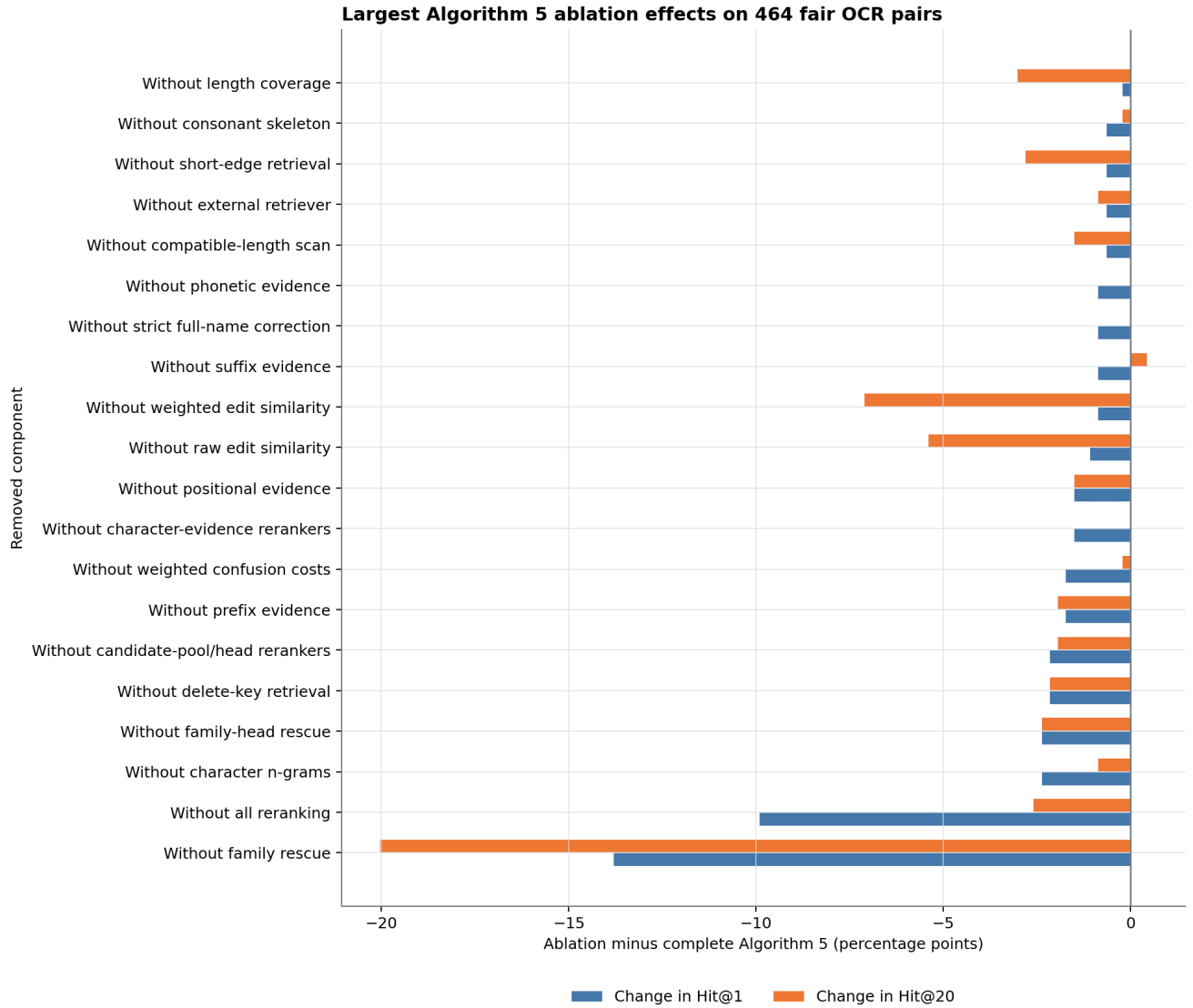


Figure 39: The x-axis is ablated minus complete Algorithm 5 in percentage points. The y-axis lists the 20 broad removals with the largest absolute effect on the 464 OCR pairs.

Figure insights

1. Family rescue supplies the largest candidate-recall gain.
2. Reranking mainly changes Hit@1, while weighted edit evidence has a larger Hit@20 effect.

Table 41: One concrete paired rank change for each of the most influential Algorithm 5 OCR removals. Rank 999 means outside the top 20.

OCR removal	Input	Expected	Full top 1	Full rank Ablated top 1	Ablated rank
Without family rescue	ABISAGLOR	ABASAGLAR	ABASAGLAR CARTRIDGES	1 MISACLOR	2
Without all reranking	ABA SAGLAR	ABASAGLAR	ABASAGLAR CARTRIDGES	1 PASCOLAR	3
Without weighted edit similarity	ANNIKAAIN	AMIKACIN	AMIKACIN	1 ANIKA HAIR	4
Without raw edit similarity	CELBON 4X	CELEBREX	CALTONEX	3 CALTONEX	4
Without length coverage	KEFOROL	KETOROLAC	CEVAROL	3 CEVAROL	2
Without short-edge retrieval	KEFOROL	KETOROLAC	CEVAROL	3 CEVAROL	2
Without family-head rescue	ABASAGLAK	ABASAGLAR	ABASAGLAR CARTRIDGES	1 ADALAT LA	2
Without character n-grams	ABACAVIR LAR	ABASAGLAR	ABASAGLAR CARTRIDGES	1 AXICA PILLS	999
Without delete-key retrieval	ABASAGLAK	ABASAGLAR	ABASAGLAR CARTRIDGES	1 ADALAT LA	2
Without candidate-pool/head rerankers	ABRASAGLAR	ABASAGLAR	ABASAGLAR CARTRIDGES	1 BROMAGAR	2
Without prefix evidence	KEFONOLAE	KETOROLAC	KETOROLAC	1 KAFINOL	3
Without weighted confusion costs	ANNIKAAIN	AMIKACIN	AMIKACIN	1 ANIKA HAIR	4

Table 42: Algorithm 5 broad-component confirmation on all 66,257 synthetic clean-core pairs; median milliseconds are measured per warm query.

Synthetic ablation	Hit@1	Delta	H@1	Full-only	H@1	Hit@20	Delta	H@20	Full-only	H@20	Median	ms
Without family rescue	95.3137%	-2.8752	2122	97.3180%	-2.6805	1776	11.087					
Without external retriever	98.0032%	-0.1856	182	99.9985%	+0.0000	0	55.300					
Without context cleanup	98.1889%	+0.0000	0	99.9985%	+0.0000	0	33.344					
Complete Algorithm 5	98.1889%	+0.0000	0	99.9985%	+0.0000	0	33.339					
Without all reranking	97.1731%	-1.0157	783	99.9940%	-0.0045	3	23.801					
Without all multi-step chains	97.6757%	-0.5132	343	99.7283%	-0.2702	179	28.264					
Without family-head rescue	98.2221%	+0.0332	5	100.0000%	+0.0015	0	31.423					
Without safety clarification gate	98.1889%	+0.0000	0	99.9985%	+0.0000	0	22.376					
Without weighted edit similarity	97.8236%	-0.3652	368	99.6785%	-0.3200	212	23.162					
Without phonetic evidence	98.0606%	-0.1283	140	99.9426%	-0.0558	38	32.832					
Without raw edit similarity	98.0832%	-0.1056	149	99.9215%	-0.0770	51	31.817					
Without character n-grams	98.1210%	-0.0679	151	99.9547%	-0.0438	30	27.842					
Without consonant skeleton	98.1511%	-0.0377	103	99.9502%	-0.0483	33	32.729					

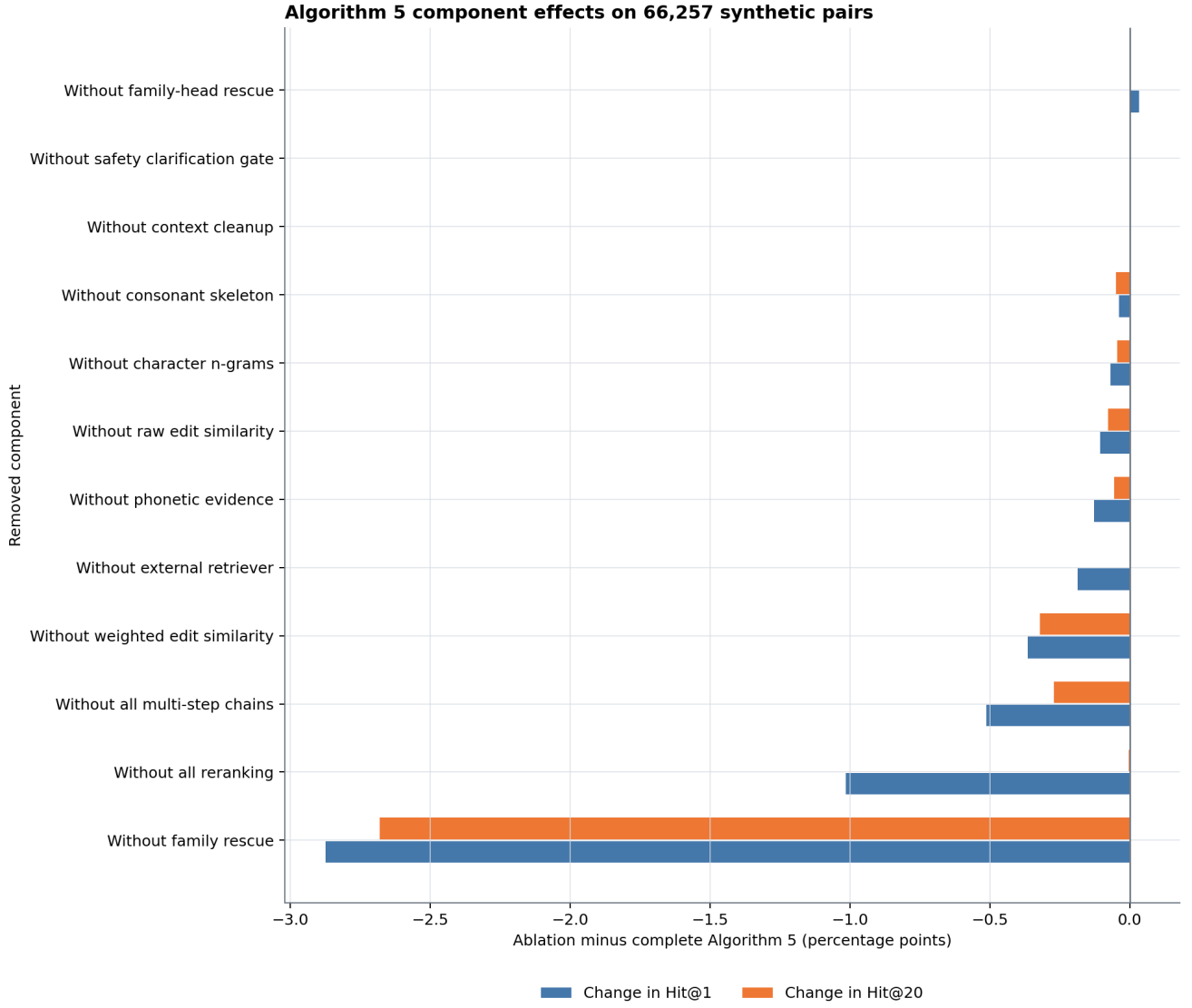


Figure 40: The x-axis is the synthetic accuracy change after one broad Algorithm 5 component is removed; the y-axis identifies the removal.

Figure insights

1. Synthetic ablations test whether OCR gains generalize across generated edit mechanisms.

Table 43: One concrete paired rank change for each influential synthetic Algorithm 5 removal. Rank 999 means outside the top 20.

Synthetic removal	Input	Expected	Full top 1	Full rank Ablated top 1	Ablated rank
Without family rescue	0ianta	OIANDA	OIANDA	1 RELANTA	999
Without all reranking	0ianta	OIANDA	OIANDA	1 SIENTA	2
Without all multi-step chains	acvnitantiicne	ACNEVIT ANTI	ACNEVIT ANTI	1 TWINIA ANTI	2
		ACNE	ACNE	ACNE	
Without weighted edit similarity	lvos	IFOS	IFOS	1 AVOSOYA	15
Without external retriever	acetosyme	ACETOTHYME	ACETOTHYME	1 ACETASEE	2
Without phonetic evidence	aclutay	ACTODAY	ACLASTA	2 ACTODAY	1
Without raw edit similarity	0stda	OSTIDA	OSTIDA	1 TDA	2
Without character n-grams	0floxan	OFLOXIN	OFLOXIN	1 CILOXAN	7
Without consonant skeleton	abnerve	EPONERVE	EPONERVE	1 ALPHANERVE	2
Without family-head rescue	ac0ne	ACZONE	ACNE STOP	2 ACZONE	1

11.9 The 0.60 Extreme OCR Cohort

For compact query q_c and compact verified target y_c , the normalized distance is

$$r = \frac{L(q_c, y_c)}{\max(|y_c|, 1)}, \quad \text{extreme when } r > 0.60.$$

Compacting removes case, spaces, and punctuation before Levenshtein distance L is calculated. The fair denominator contains 113 extreme pairs; the older inclusive 595-row view contains 121 because it retains duplicates and collision rows.

Table 44: Distribution of all 113 fair extreme OCR pairs by operation, edit band, visible length, and shared-bigram evidence.

Dimension	Group	Cases	Share	A5 H@1	A5 H@20
Operation Profile	Extra Characters Only	2	1.77%	0.00%	0.00%
Operation Profile	Missing Characters Only	1	0.88%	0.00%	0.00%
Operation Profile	Mixed Operations	99	87.61%	0.00%	8.08%
Operation Profile	Substitutions Only	11	9.73%	0.00%	9.09%
Distance Band	2 3 Edits	1	0.88%	0.00%	0.00%
Distance Band	4 5 Edits	43	38.05%	0.00%	13.95%
Distance Band	6 Plus Edits	69	61.06%	0.00%	4.35%
Query Length Band	10 Plus Characters	21	18.58%	0.00%	28.57%
Query Length Band	1 3 Characters	16	14.16%	0.00%	0.00%
Query Length Band	4 5 Characters	24	21.24%	0.00%	0.00%
Query Length Band	6 7 Characters	32	28.32%	0.00%	9.38%
Query Length Band	8 9 Characters	20	17.70%	0.00%	0.00%
Shared Bigram Band	0 Shared Bigrams	62	54.87%	0.00%	0.00%
Shared Bigram Band	1 Shared Bigram	28	24.78%	0.00%	7.14%
Shared Bigram Band	2 3 Shared Bigrams	20	17.70%	0.00%	25.00%
Shared Bigram Band	4 Plus Shared Bigrams	3	2.65%	0.00%	66.67%

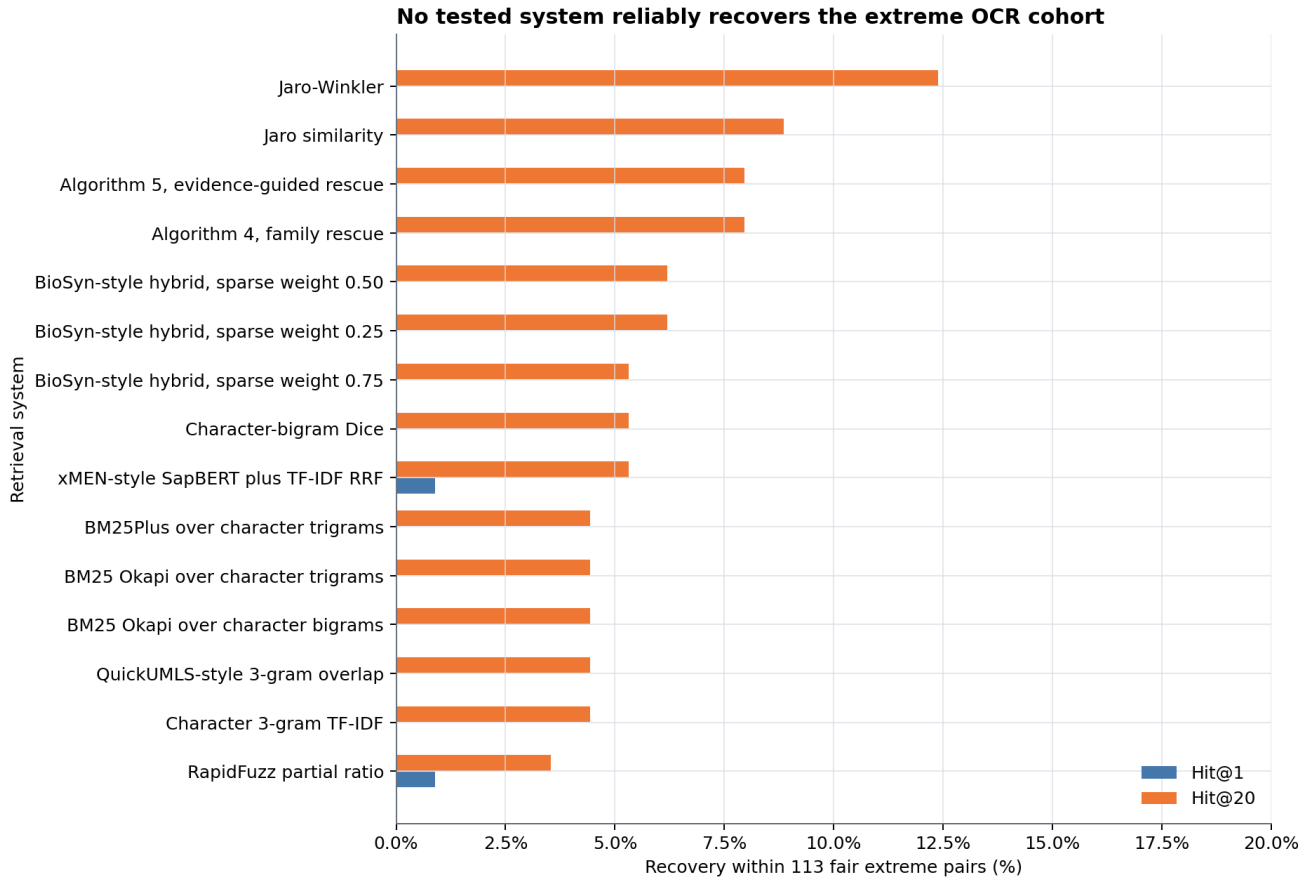


Figure 41: The x-axis is recovery within the 113 fair extreme pairs and the y-axis lists the 15 strongest systems by Hit@20. Extreme means $L(q_c, y_c)/|y_c| > 0.60$.

Figure insights

1. The extreme cohort is a low-evidence stress test, not an ordinary typo benchmark.
2. No measured method converts extreme corruption into reliable top-20 recovery.

Table 45: Concrete extreme-pair successes and failures. Rank 999 means the verified family is outside the returned top 20.

Input	Expected	A5 top 1	Operation	Edits	r Bigrams	Expected rank
RIVOFLAVIN	RIVOTRIL	RIVAFLOWVA	mixed operations	5 0.625	3	8
RIVOCICLIN	RIVOTRIL	RIVO MICRO	mixed operations	5 0.625	3	9
INDULGENCE	INDERAL	IMMULONE	mixed operations	6 0.857	2	10
KETODEXTRO	KETOROLAC	KETODERM	mixed operations	6 0.667	4	11
MYUBKL	MYOLAX	MYONIL	substitutions only	4 0.667	1	12
MYADAYS	MYOLAX	METAZ	mixed operations	4 0.667	1	13
CEFTRIAZONE	CEFIXIME	CEFTRIAZONE EVA	mixed operations	5 0.625	2	18
KETOCONAZOLE	KETOROLAC	BUTOCONAZOLE	mixed operations	6 0.667	4	18
ECOFIX	CEFIXIME	CEFIX	mixed operations	5 0.625	2	20
LACTULOSE	LACTO	LACTULOSE HEK	extra characters only	4 0.800	3	999
RIVAROXABAN	RIVO	REPATOXABAN	extra characters only	7 1.750	2	999
RIV	RIVOTRIL	RIVO	missing characters only	5 0.625	2	999
PNO	RIVO	PONO	mixed operations	3 0.750	0	999
ARDAN	RIVO	ADANT	mixed operations	4 1.000	0	999
H	RIVO	none	mixed operations	4 1.000	0	999
LABISOL	LASIX	CAPISOL	mixed operations	4 0.800	1	999
LWH	LACTO	GOTAL WHITENING	mixed operations	4 0.800	0	999
1000	TOCO	EPIAO 10000I	substitutions only	4 1.000	0	999
7000	TOCO	EPIAO 4000I	substitutions only	4 1.000	0	999
CALBOMET	CELEBREX	ALDOMET	substitutions only	5 0.625	0	999
RICOLONI	RIVOTRIL	URICOL N	substitutions only	5 0.625	1	999
VIIGAMP	VIGOREX	VIGAM S	substitutions only	5 0.714	2	999

11.10 Remaining Algorithm 5 Failures and Complementarity

Algorithm 5 has 126 top-20 OCR misses. At least one measured external method recovers 30; all measured external methods also miss 96. The first group exposes candidate rules worth studying. The second group lacks a demonstrated lexical solution in this benchmark. External recovery means that the supplied target appears in a ranking; it does not validate the label. Pairs such as LACTULOSE→LACTO require source-image and label audit before an external rule is copied into Algorithm 5.

Table 46: Oracle unions on both locked datasets answer whether methods fail on the same cases. They are diagnostic upper bounds, not deployable ranking algorithms.

Dataset	System set	Hit@20	Failures
464 fair OCR pairs	Algorithm 5	72.84%	126
464 fair OCR pairs	Any external or classical system	78.66%	99
464 fair OCR pairs	Algorithm 5 plus any external system	79.31%	96
464 fair OCR pairs	Any project Algorithm 1-5	74.14%	120
464 fair OCR pairs	Any measured system	79.31%	96
66,257 synthetic pairs	Algorithm 5	100.00%	1
66,257 synthetic pairs	Any external or classical system	99.96%	26
66,257 synthetic pairs	Algorithm 5 plus any external system	100.00%	0
66,257 synthetic pairs	Any project Algorithm 1-5	100.00%	0
66,257 synthetic pairs	Any measured system	100.00%	0

Table 47: Thirty concrete Algorithm 5 top-20 misses, split between cases recovered by at least one external method and cases missed by every measured external method. The complete 126-row audit is stored as CSV.

Input	Expected	A5 top 1	Edits	Bigrams	Operation	External recoveries	Example recovering systems	Evidence class
LACTULOSE	LACTO	LACTULOSE HEK	4	3	extra characters only	26	Exhaustive Levenshtein; Jaro-Winkler; Character 3-gram TF-IDF	recoverable by at least one measured external rule
NEUROTROPHIC	NEUROCET	NEUROTOP	7	4	mixed operations	18	Jaro-Winkler; Character 3-gram TF-IDF; BioSyn-style hybrid, sparse weight 0.25	recoverable by at least one measured external rule
TEFA	TELFAST	TERA	3	2	missing characters only	10	Jaro-Winkler; RapidFuzz token ratio; CODER++ dense HNSW	recoverable by at least one measured external rule
RIVIO	RIVOTRIL	RIVO	4	2	mixed operations	10	Jaro-Winkler; BioSyn-style hybrid, sparse weight 0.25; BioSyn-style hybrid, sparse weight 0.50	recoverable by at least one measured external rule
KETOLIDE	KETOROLAC	KETOLAC	5	4	mixed operations	10	Jaro-Winkler; Character 3-gram TF-IDF; BioSyn-style hybrid, sparse weight 0.25	recoverable by at least one measured external rule
TOCILIZUMAB	TOCO	ALIZOLAM	8	2	mixed operations	10	Character 3-gram TF-IDF; BioSyn-style hybrid, sparse weight 0.25; BioSyn-style hybrid, sparse weight 0.50	recoverable by at least one measured external rule
RIV	RIVOTRIL	RIVO	5	2	missing characters only	8	Exact or prefix match; Jaro-Winkler; BioSyn-style hybrid, sparse weight 0.25	recoverable by at least one measured external rule
LORRIX	LASIX	LORNOX	3	1	mixed operations	7	Exhaustive Levenshtein; Damerau-Levenshtein; Match Rating Codex plus edit ranking	recoverable by at least one measured external rule
KETONES	KETOROLAC	KETOKED	5	3	mixed operations	7	Character 3-gram TF-IDF; BioSyn-style hybrid, sparse weight 0.50; BioSyn-style hybrid, sparse weight 0.75	recoverable by at least one measured external rule
PETAWOLIN	PETHIDINE	PATADOLINE	5	3	mixed operations	7	BioSyn-style hybrid, sparse weight 0.25; BioSyn-style hybrid, sparse weight 0.50; BioSyn-style hybrid, sparse weight 0.75	recoverable by at least one measured external rule
LEPPPOOLAA	KETOROLAC	LIPOLEX	5	2	substitutions only	6	RapidFuzz token ratio; Longest-common-subsequence similarity; RapidFuzz quick ratio	recoverable by at least one measured external rule
ABANA	ABASAGLAR	IBAN	5	2	mixed operations	5	Fuse.js default Bitap; Fuse.js token search; QuickUMLS-style 3-gram overlap	recoverable by at least one measured external rule
RIVAROXABAN	RIVO	REPATOXABAN	7	2	extra characters only	5	Jaro-Winkler; Jaro similarity; QuickUMLS-style 3-gram overlap	recoverable by at least one measured external rule
LANLTA	LANTUS	LANETAL	3	2	substitutions only	4	Jaro-Winkler; BioSyn-style hybrid, sparse weight 0.25; RapidFuzz weighted ratio	recoverable by at least one measured external rule
INDOCIN	INDERAL	INDACIN	4	2	substitutions only	4	BioSyn-style hybrid, sparse weight 0.25; NYSIIS plus edit ranking; SapBERT dense HNSW	recoverable by at least one measured external rule
LANHA	LANTUS	LAMA	3	2	mixed operations	0	none	not recovered by any measured external rule
PNO	RIVO	PONO	3	0	mixed operations	0	none	not recovered by any measured external rule
1000	TOCO	EPIAO 10000I	4	0	substitutions only	0	none	not recovered by any measured external rule
7000	TOCO	EPIAO 4000I	4	0	substitutions only	0	none	not recovered by any measured external rule
ARDAN	RIVO	ADANT	4	0	mixed operations	0	none	not recovered by any measured external rule
H	RIVO	none	4	0	mixed operations	0	none	not recovered by any measured external rule
KEFOMAN	KATOMOR	CEFORAN	4	1	substitutions only	0	none	not recovered by any measured external rule
LABISOL	LASIX	CAPISOL	4	1	mixed operations	0	none	not recovered by any measured external rule
LWK	LASIX	SLOW K	4	0	mixed operations	0	none	not recovered by any measured external rule
MIYOKO	MYOLAX	MICO	4	1	mixed operations	0	none	not recovered by any measured external rule
PICOFIL	RIVOTRIL	PACIFIAL	4	1	mixed operations	0	none	not recovered by any measured external rule
PIROBIL	RIVOTRIL	PIROVIT	4	1	mixed operations	0	none	not recovered by any measured external rule
RVOMI	RIVOTRIL	ROVIME	4	1	mixed operations	0	none	not recovered by any measured external rule
ARYADE	RIVO	ARID	5	0	mixed operations	0	none	not recovered by any measured external rule

Input	Expected	A5 top 1	Edits Bigrams Operation		External recoveries	Example recovering systems	Evidence class
CALBOMET	CELEBEX	ALDOMET	5	0 substitutions only		0 none	not recovered by any measured external rule

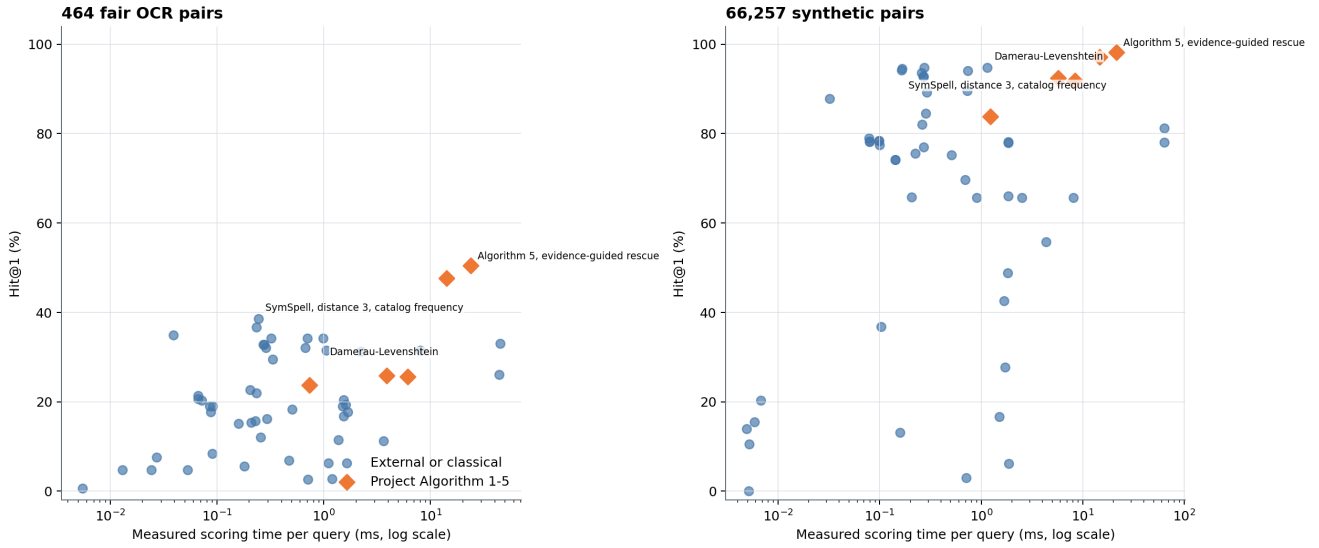


Figure 42: The x-axis is measured scoring time per query on a logarithmic scale and the y-axis is Hit@1. Separate panels show OCR and synthetic results. Vectorized methods use batch elapsed time divided by batch size; project methods time individual calls, so deployment latency still needs one isolated end-to-end harness.

Figure insights

1. No single method dominates both latency and OCR accuracy.
2. Fast synthetic edit-distance methods remain materially weaker on observed OCR.

Table 48: HNSW approximation audit against exact dense cosine ranking. OCR uses all 464 queries; synthetic uses a deterministic 1,000-query sample.

Dataset	Dense system	Queries audited	Exact top-1 agreement	Mean exact top-20 recall	Minimum recall
464 fair OCR pairs	BiomedBERT dense HNSW	464	100.00%	100.00%	100.00%
464 fair OCR pairs	SapBERT dense HNSW	464	100.00%	99.99%	95.00%
464 fair OCR pairs	CODER++ dense HNSW	464	100.00%	99.98%	95.00%
66,257 synthetic pairs	BiomedBERT dense HNSW	1000	100.00%	99.98%	95.00%
66,257 synthetic pairs	SapBERT dense HNSW	1000	100.00%	99.99%	95.00%
66,257 synthetic pairs	CODER++ dense HNSW	1000	100.00%	100.00%	100.00%

11.11 Complete External Repository Disposition

Table 49: Every repository or distinct configuration found in the competitor search, including measured, cached, represented, duplicate, and non-comparable entries.

Repository or configuration	Method family	Status	Reason
RapidFuzz ratio	normalized edit similarity	evaluated	Distinct normalized Indel similarity.
RapidFuzz partial ratio	substring edit similarity	evaluated	Tests visible fragments and partial names.
RapidFuzz token-sort ratio	token-aware edit similarity	evaluated	Tests reordered multi-word names.
RapidFuzz token-set ratio	token-aware edit similarity	evaluated	Tests partial token overlap in multi-word names.
RapidFuzz weighted ratio	adaptive edit similarity	evaluated	Adaptive combination of full, partial, and token scores.
RapidFuzz quick ratio	normalized edit similarity	evaluated	Included so equivalence with ratio can be measured rather than assumed.
Damerau-Levenshtein	edit distance	evaluated	Allows insertion, deletion, replacement, and transposition.
Optimal string alignment	edit distance	evaluated	Restricted Damerau distance used by Algorithm 5 evidence.
Jaro similarity	character similarity	evaluated	Separates Jaro from the already cached Jaro-Winkler baseline.
Longest-common-subsequence similarity	ordered subsequence	evaluated	Rewards characters retained in order.
QuickUMLS-style 3-gram Jaccard	q-gram set similarity	evaluated	QuickUMLS exposes Jaccard over character n-grams.
QuickUMLS-style 3-gram Dice	q-gram set similarity	evaluated	QuickUMLS exposes Dice over character n-grams.
QuickUMLS-style 3-gram cosine	q-gram set similarity	evaluated	QuickUMLS exposes cosine over character n-grams.
QuickUMLS-style 3-gram overlap	q-gram set similarity	evaluated	QuickUMLS exposes overlap coefficient over character n-grams.
Character-bigram Dice	q-gram set similarity	evaluated	A shorter q-gram retains evidence under heavier corruption.
BM25 Okapi over character bigrams	sparse probabilistic retrieval	evaluated	Tests BM25 with short character tokens.
BM25 Okapi over character trigrams	sparse probabilistic retrieval	evaluated	Tests BM25 with more specific character tokens.
BM25L over character trigrams	sparse probabilistic retrieval	evaluated	BM25L length-normalization variant.
BM25Plus over character trigrams	sparse probabilistic retrieval	evaluated	BM25Plus lower-bound variant.
SymSpell, distance 2, catalog frequency	delete-index spelling correction	evaluated	Delete index with catalog occurrence frequency for ties.
SymSpell, distance 3, catalog frequency	delete-index spelling correction	evaluated	Distance-radius sensitivity for the same index.
SymSpell, distance 3, uniform frequency	delete-index spelling correction	evaluated	Ablates catalog frequency from SymSpell tie ordering.
Soundex plus edit ranking	phonetic encoding	evaluated	Classic first-letter-preserving phonetic code.
Metaphone plus edit ranking	phonetic encoding	evaluated	English phonetic code designed beyond Soundex.
NYSIIS plus edit ranking	phonetic encoding	evaluated	Name-oriented phonetic normalization.
Match Rating Codex plus edit ranking	phonetic encoding	evaluated	Compact consonant-oriented phonetic code.
BiomedBERT dense HNSW	biomedical dense retrieval	evaluated	Base biomedical encoder before entity self-alignment.
SapBERT dense HNSW	biomedical dense retrieval	evaluated	Off-the-shelf biomedical entity-linking encoder.
CODER++ dense HNSW	biomedical dense retrieval	evaluated	Fine-grained medical term encoder with hard-negative training.
BioSyn-style hybrid, sparse weight 0.25	sparse-dense hybrid	evaluated	Low sparse-weight sensitivity point.
BioSyn-style hybrid, sparse weight 0.50	sparse-dense hybrid	evaluated	Equal sparse and dense contribution.
BioSyn-style hybrid, sparse weight 0.75	sparse-dense hybrid	evaluated	High sparse-weight sensitivity point.
xMEN-style SapBERT plus TF-IDF RRF	rank-fusion ensemble	evaluated	Tests the dense and n-gram ensemble exposed by xMEN.
preon default drug normalizer	staged drug-name normalization	evaluated	Directly relevant drug-name normalizer, evaluated with its published default threshold and tie behavior.
preon without token matching	staged drug-name normalization	evaluated	Component ablation measuring whether preon’s exact-token stage changes retrieval.
preon without partial edit matching	staged drug-name normalization	evaluated	Component ablation measuring the contribution of preon’s partial edit stage.
Fuse.js default Bitap	client-side fuzzy search	evaluated	Published default fuzzy ranking over the locked Egyptian family catalog.
Fuse.js token search	token-aware client-side fuzzy search	evaluated	Measures the token-search mode separately from default Bitap ranking.
FlexSearch tolerant default	indexed fuzzy search	evaluated	Published tolerant tokenizer with the default encoder and suggestions.
FlexSearch tolerant LatinAdvanced	indexed fuzzy and phonetic search	evaluated	Adds the published advanced Latin phonetic normalization to tolerant retrieval.

Repository or configuration	Method family	Status	Reason
FlexSearch tolerant LatinExtra	indexed aggressive phonetic search	evaluated	Measures the more aggressive vowel-reducing Latin encoder.
FlexSearch full LatinAdvanced	indexed substring and phonetic search	evaluated	Tests complete substring indexing with advanced Latin normalization.
Exact or prefix match	lexical baseline	cached	Existing exact/prefix result is reused; Fuse is also listed separately.
Exhaustive Levenshtein	edit distance	cached	Existing row-level result is reused.
Jaro-Winkler	character similarity	cached	Existing row-level result is reused.
Character 3-gram TF-IDF	sparse character retrieval	cached	Existing result represents the same scoring family used by scispaCy and PolyFuzz.
RapidFuzz token ratio	token-aware edit similarity	cached	Existing row-level result is reused.
Egyptian medicine phonetic baseline	phonetic encoding	cached	Existing domain phonetic result is reused; general phonetic codes are evaluated separately.
PolyFuzz	matching framework	duplicate	Its distinct underlying methods are evaluated directly, avoiding duplicate scores under a wrapper name.
scispaCy entity linker	biomedical entity linking	duplicate	Same character-trigram scoring family as the cached TF-IDF baseline; its UMLS/RxNorm catalog is not the Egyptian catalog.
SimString	q-gram retrieval engine	represented	QuickUMLS-style Dice, Jaccard, cosine, and overlap scores represent its distinct measures.
textdistance	distance library	represented	Distinct LCS, q-gram, edit, and phonetic families are evaluated through compiled implementations.
py_stringmatching	record-linkage metric library	excluded	Token-level Soft TF-IDF and Monge-Elkan collapse on compact one-token drug names and provide no scalable exact top-20 index for 66,257 queries.
OGER biomedical entity recognizer	dictionary biomedical entity recognition	excluded	OGER detects dictionary terms in documents and assigns ontology concepts; it does not rank isolated OCR strings against the locked Egyptian catalog.
OpenMed pharmaceutical NER	biomedical named-entity recognition	excluded	The pharmaceutical model detects medication spans in prose but does not normalize one OCR string to a ranked Egyptian catalog family.
Microsoft PhoneticMatching	phoneme-distance retrieval	excluded	English pronunciation dependencies and an unmaintained native build make the result environment-dependent; four reproducible phonetic encoders plus the domain phonetic baseline are tested.
KRISSBERT	contextual biomedical entity linking	excluded	Requires contextual mention prototypes and UMLS entity descriptions; this benchmark supplies isolated OCR strings and an Egyptian commercial-name catalog.
BERN2	biomedical NER and normalization	excluded	Consumes biomedical prose and normalizes to UMLS, not isolated strings against the locked Egyptian catalog.
DNorm	trained biomedical entity normalization	excluded	Requires labeled disease-normalization training data and contextual mentions; neither transfers to isolated Egyptian commercial drug names without retraining.
MedXN	clinical medication normalization	excluded	Extracts medication mentions and attributes from clinical prose and targets RxNorm, while this benchmark starts from one isolated OCR name and uses a locked Egyptian catalog.
ArboEL	trained biomedical entity linking	excluded	Requires ontology descriptions, contextual mentions, and supervised training; the locked benchmark supplies none of those inputs.
BioGenEL	generative biomedical entity linking	excluded	Requires ontology-specific model training and constrained decoding; an off-the-shelf result would not target the Egyptian commercial-name catalog.
RxNorm approximate matching	drug terminology normalization	excluded	The service ranks RxNorm concepts rather than the locked Egyptian catalog, so its output labels are not scoreable against these targets.
QuickUMLS packaged linker	UMLS concept linking	represented	The four underlying similarities are run against the correct Egyptian catalog instead of substituting UMLS.
Elasticsearch fuzzy query	search service	represented	Its method-level components are covered by edit-distance and BM25 tests; service tuning is an infrastructure experiment, not a new scoring rule.
Typesense typo-tolerant search	search service	excluded	Server ranking combines proprietary defaults and deployment settings; it is not a single reproducible algorithm ablation.
Meiliseach typo-tolerant search	search service	excluded	Server ranking combines typo tolerance with configurable ranking rules; it is outside the method-level comparison.

Repository or configuration	Method family	Status	Reason
GLINKER	zero-shot entity linking	excluded	Its published pipeline links contextual mentions through a configurable candidate store and neural disambiguator; the benchmark has isolated OCR strings, no entity descriptions, and no frozen candidate-generation contract that yields a comparable catalog-wide top 20.
Tantivy	full-text search engine library	represented	Tantivy is an indexing engine rather than one medicine-ranking rule; its distinct scoreable retrieval components are represented by the measured BM25 and edit-distance configurations.
FuzzyMatch	record matching framework	represented	Its two distinct ranking signals are measured directly as character-bigram Dice and exhaustive Levenshtein.
Abydos	string-distance and phonetic library	represented	Its scoreable method families are covered by the measured edit-distance, q-gram, Jaro, and phonetic configurations; the library wrapper adds no independent ranking rule.
Dedupe	supervised entity resolution	excluded	It requires labeled match and non-match pairs plus structured record fields; this locked benchmark supplies one OCR string per query and forbids training on evaluation labels.
BLINK	contextual entity linking	excluded	It requires contextual mentions, entity descriptions, and task-specific model preparation; those inputs do not exist for the locked Egyptian commercial-name catalog.
Ukkonen bounded edit distance	edit-distance optimization	duplicate	It computes the same Levenshtein ordering as the cached exhaustive baseline; only execution strategy changes.
StringZilla	accelerated string library	duplicate	Faster implementation of distance families already scored; it does not define a different medicine ranking.

11.12 Reproducible Artifacts

- Competitor runner: `benchmark_04_experiments/run_competitor_benchmark.py`.
- Algorithm 5 ablation runner: `benchmark_04_experiments/run_algorithm_5_ablations.py`.
- Analysis and figure generator: `benchmark_04_experiments/analyze_competitor_benchmark.py`.
- Row-level results: `benchmark_04_experiments/artifacts/06_competitor_benchmark/`.
- Statistical tables and figures: `benchmark_04_experiments/results/06_competitor_benchmark/`.

12 Algorithm 6: Evidence-Gated Consensus Retrieval

Algorithm 6 keeps Algorithm 5 as the base ranker, adds independent candidate sources, and changes the returned top 20 only when several sources support one missing candidate. It does not promote a new rank-1 result in the deployed policy.

12.1 Input, Process, and Output

Stage	Concrete row	Algorithm 6 behavior
Input	TEFA	One OCR string enters search. The evaluator hides the supplied target TELFAST.
Base ranking	Algorithm 5 top 1 is TERA; TELFAST is outside its top 20	Algorithm 5 order remains the authoritative display order.
Candidate union	Ten retrieval sources search the same catalog	Jaro-Winkler, BM25Plus character trigrams, token similarity, phonetic keys, SymSpell, and Dice bigrams add independent candidate evidence.
Bounded change	Several sources support TELFAST	One external candidate may replace rank 20; no base rank-1 candidate moves.
Output	TERA remains first; TELFAST becomes rank 20	Hit@1 remains zero for this row and Hit@20 changes from zero to one. The interface still requests confirmation.

Table 50: A complete Algorithm 6 example. The hidden target never enters retrieval, and the bounded consensus slot improves candidate coverage without claiming that TELFAST is the certain answer.

12.2 Label Audit and Evaluation Contract

The 464-pair benchmark has 371 development pairs from 18 target families and 93 retrospective holdout pairs from nine other families. Target-family overlap is zero. The five cross-fit development folds contain 76, 74, 74, 74, and 73 pairs; each fold is scored by a model trained without its target families. None of the 464 rows contains a source-image reference.

12.3 Candidate Sources

A character bigram is a two-character window. For TEFA, the bigrams are TE, EF, and FA; TELFAST shares TE and FA. A character trigram uses three-character windows. These local pieces retain evidence when one OCR error damages only part of a name.

Table 54: The ten Algorithm 6 candidate sources. Each returns catalog families; no source sees the expected label.

Source	Evidence	Role and example
Algorithm 5	Existing family rescue and deterministic ranking	MYMLAX→MYOLAX at rank 1; supplies the preserved display order.
Jaro-Winkler	Character agreement with extra prefix weight	TEFA→TELFAS at source rank 1; required for that deployed gain.

Source	Evidence	Role and example
Match Rating Approach	Compressed consonant comparison	ABASCOLOGY→ABASAGLAR at source rank 1; no isolated top- k effect.
RapidFuzz WRatio	Best normalized ratio across direct and partial token comparisons	LACTULOSE→LACTO at source rank 3; supports the candidate redundantly.
NYSIIS	Phonetic encoding	MYOLOGY→MYOLAX at source rank 1; no isolated top- k effect.
BM25Plus character 3-gram	Weighted overlap of three-character pieces	LACTULOSE→LACTO at source rank 12; required to select the deployed gain.
SymSpell, edit radius 3	Delete-key approximate retrieval	OSFOCU→OSTOCAL at source rank 1; no isolated top- k effect.
RapidFuzz token-sort ratio	Token comparison after word reordering	TEFA→TELFAST at rank 2 and LACTULOSE→LACTO at rank 5.
Soundex	Coarse first-letter and consonant sound code	TELBACE→TELFAST at source rank 1; never sufficient alone for insertion.
Dice character bigram	Twice the shared-bigram count divided by total bigrams	TEFA→TELFAST at source rank 15; supplies the third required vote.

12.4 Paired Performance and Deployment Decision

12.5 End-to-End Nested Component Ablation

This experiment replaces Algorithm 5’s base output with one internal-component removal, then reruns all other Algorithm 6 candidate sources and the same consensus gate. The values below are therefore Algorithm 6 results. They are not the earlier Algorithm 5-only ablation percentages. The complete long-form result contains 96 OCR removals and the 12 available synthetic confirmations in `results/07_algorithm_6/algorithm_6_replay_metrics.csv`.

The synthetic check exposes interactions hidden by OCR alone. Removing all multi-step chains changes no OCR top- k outcome, yet loses 343 synthetic Hit@1 cases while gaining 3 and loses 178 synthetic Hit@20 cases with no gains. Removing family-head rescue loses 11 OCR top-20 cases but gains 22 net synthetic Hit@1 cases and recovers `aclox→ADOX`, the one complete Algorithm 6 synthetic top-20 miss. The correct follow-up is a narrower guard, not deletion of the whole family-head component.

12.6 Candidate-Source Ablation

12.7 Learned OCR Confusion Evidence

Directional costs are learned only from the 211 provisional P2 development rows. For substitution count $c(o, t)$ and the largest count for observed character o , $m(o)$, the cost is

$$w(o, t) = \max \left(0.20, 1 - 0.80 \frac{c(o, t)}{m(o)} \frac{c(o, t)}{c(o, t) + 3} \right).$$

Missing- and extra-character costs use the same frequency shrinkage with scale 0.55 and floor 0.45. Unseen operations cost 1. A lower cost means the operation appeared more consistently in eligible OCR alignments; it does not mean the candidate is correct.

Component	What was built	Deployed status
Label reliability audit	Nearest-catalog conflict, visible evidence, candidate-union presence, severity, and image-reference checks for all 464 pairs	P0/P1 rows remain measurable but cannot train learned evidence; P2 is provisional, not human-adjudicated truth.
Leakage contract	Five target-family-disjoint cross-fit folds inside 371 development pairs; 93 historical holdout pairs remain family-disjoint	Holdout is called retrospective because it was visible in the earlier project lifecycle; no fresh-blind claim is allowed.
Candidate union	Algorithm 5 plus nine independent lexical, phonetic, and approximate-retrieval sources	Active for one bounded top-20 slot.
OCR confusion learning	Directional substitution, missing-character, and extra-character costs learned from 211 development P2 rows	Stored and exposed as evidence; it does not override deployed Algorithm 5 order.
Learned reranker	LightGBM candidate ranker trained with target-family cross-fitting and grouped feature ablations	Advisory only because it underperforms the deterministic deployment.
Rank-1 consensus gate	Candidate promotion based on source count, reciprocal-rank fusion, edit, Jaro-Winkler, and bigram evidence	Rejected after synthetic paired losses.
Selective correctness gate	Cross-fitted logistic model, threshold 0.63, at least seven supporting sources, and Levenshtein similarity at least 0.70	A pass means “likely match, confirmation required.” It never certifies clinical correctness.

Table 51: Algorithm 6 separates candidate coverage, rank ordering, label reliability, and user-facing certainty. Only bounded top-20 consensus retrieval changes the deployed ranking.

Executed component	Responsibility inside Algorithm 6	Actual case shape
Normalization and context cleanup	Normalize case, punctuation, and spacing; remove strength, route, or form context for a second retrieval pass.	AAPE RENEWAL SUNSCREEN SPF 60ML→AAPE RENEWAL SUNSCREEN SPF.
External base retrieval	Produce the broad candidate list used by the Algorithm 5 base.	Its removal loses ABDUSONG LAR→ABASAGLAR from top 20.
Family-rescue index	Retrieve by exact key, prefix, suffix, n-gram, skeleton, phonetic key, delete key, and compatible length.	MYMLAX→MYOLAX.
Evidence scoring	Score raw and weighted edit, edge, n-gram, phonetic, skeleton, subsequence, position, and length agreement.	TEFAXATE→TELFAST uses positional and edge evidence.
Error-variant paths	Generate bounded keyboard, ligature, transposition, visual, phonetic, deletion, and multi-step variants.	Synthetic uraranx→ORAMAX exercises multi-step retrieval.
Family-head and fragment paths	Retrieve longer variants from a visible family head while preserving ambiguity.	LANS→LANTUS.
Deterministic reranking	Correct tied or evidence-dominant candidates after merging.	LANYU→LANTUS leaves top 20 when reranking is removed.
Algorithm 5 clarification	Mark short or conflicting base results for confirmation without changing relevance labels.	RIV→RIVOTRIL conflicts with catalog family RIVO.
Nine additional retrievers	Produce independent lexical, phonetic, edit-index, and n-gram lists.	TEFA→TELFAST appears in Jaro–Winkler, token-sort, and Dice lists.
Family consensus and bounded slot	Merge lists by family, order by source count and reciprocal-rank fusion, then apply the three-source, 55% Levenshtein, and 85% Jaro gates.	TELFAST enters at rank 20 while TERA remains first.
Selective confirmation gate	Label strong top-1 evidence “likely match, confirmation required” at probability 0.63, seven sources, and 70% Levenshtein.	MYMLAX→MYOLAX has nine sources and 83.3% Levenshtein.
Advisory learned evidence	Store learned OCR operation costs and LightGBM scores for analysis.	Neither advisory output replaces the deployed deterministic order.

Table 52: Algorithm 6 is an end-to-end pipeline, not Algorithm 5 treated as one ablation row. The component study below removes nested Algorithm 5 stages while leaving the remaining Algorithm 6 sources and gates active.

Priority	Pairs	Automated meaning	Concrete example and action
P0	138	Strong conflict: another catalog family is at least two edits closer, severe corruption, one or two visible characters, a real-name conflict, or the supplied target is absent from measured unions.	CINNARIZIN→MYOLAX; catalog CINNARIZINE is one edit away. Do not learn this mapping before checking the crop and target.
P1	56	Borderline conflict: another catalog family is one edit closer, without a P0 trigger.	ABASCOLOGY→ABASAGLAR; SPASCOLON is one edit closer. Retain for audit, not automatic training.
P2	270	No automated conflict was found and the expected family is present in the selected candidate union.	MYMLAX→MYOLAX; the expected family is the nearest catalog family and is retrieved. Use provisionally for cross-fitted evidence learning.

Table 53: P0, P1, and P2 are review priorities, not human label verdicts. P2 is partly defined by retrievability, so its performance cannot be reported as an independent unbiased accuracy estimate.

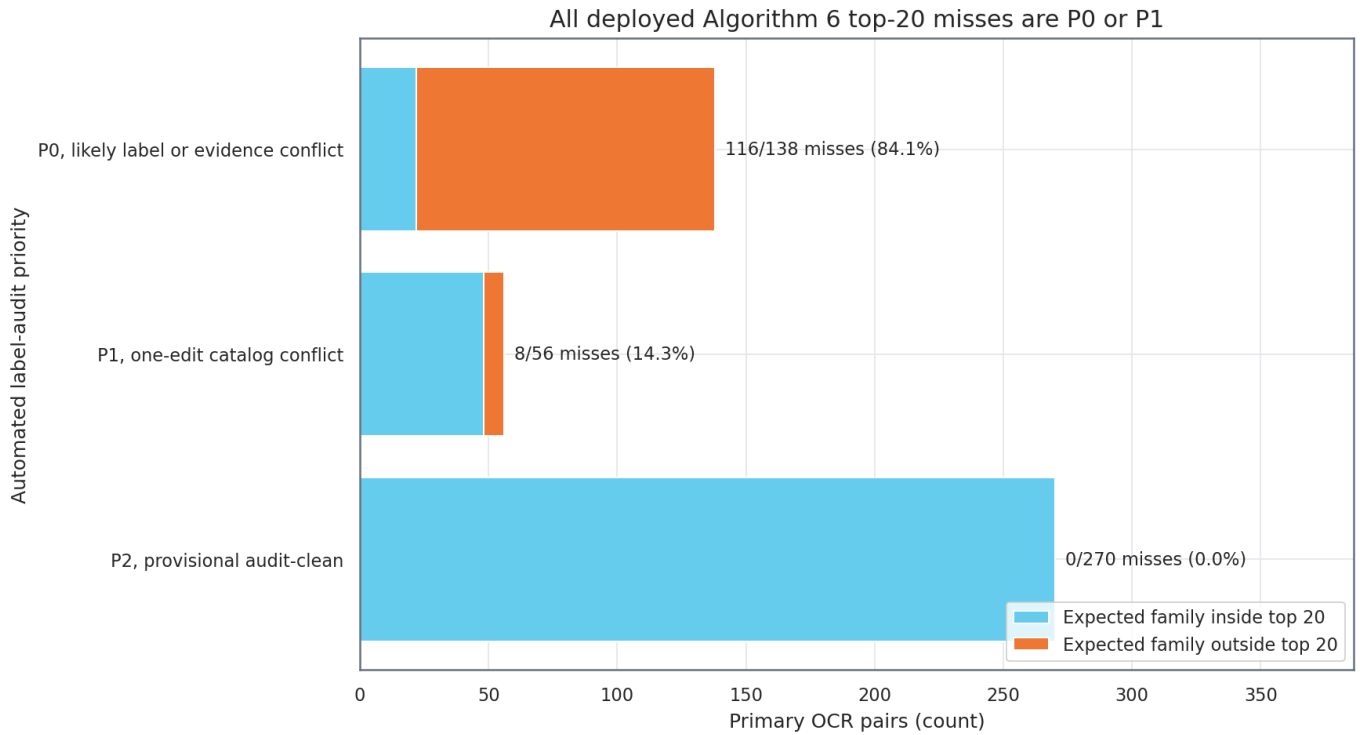


Figure 43: The y-axis shows automated audit priority; the x-axis counts primary OCR pairs. Blue is target inside Algorithm 6’s top 20 and orange is target outside it. P0 contains likely label or evidence conflicts, P1 contains a one-edit catalog conflict, and P2 has no automated conflict and requires target presence in the measured union.

Insights

1. All 124 top-20 misses are P0 or P1; P0 contains 116. The remaining benchmark error is dominated by rows that need source-image review.
2. P2 has zero top-20 misses by construction-sensitive audit rules. This supports provisional evidence learning, not a 100% independent result.

Dataset	Split	System	H@1	H@5	H@20	MRR@20	Misses
OCR	All 464	Algorithm 5	50.4310%	63.5776%	72.8448%	0.562201	126
OCR	All 464	Algorithm 6	50.4310%	63.5776%	73.2759%	0.562417	124
OCR	Development 371	Algorithm 5	50.9434%	62.5337%	71.1590%	0.559856	107
OCR	Development 371	Algorithm 6	50.9434%	62.5337%	71.6981%	0.560125	105
OCR	Retrospective holdout 93	Algorithm 5	48.3871%	67.7419%	79.5699%	0.571557	19
OCR	Retrospective holdout 93	Algorithm 6	48.3871%	67.7419%	79.5699%	0.571557	19
Synthetic	All 66,257	Algorithm 5	98.1889%	99.6529%	99.9985%	0.988413	1
Synthetic	All 66,257	Algorithm 6	98.1889%	99.6529%	99.9985%	0.988413	1

Table 55: Algorithm 6 changes candidate coverage, not rank 1. The two OCR Hit@20 gains occur in development; retrospective holdout and synthetic metrics remain unchanged.

Policy	Scope	Gains	Losses	Net	Exact paired p	Decision
Deployed bounded slot	OCR H@1	0	0	0	n/a	Base rank 1 is preserved.
Deployed bounded slot	OCR H@20	2	0	+2	0.5000	Retain as a zero-loss candidate-coverage change; do not claim significance.
Deployed bounded slot	Synthetic H@1/H@20	0	0	0	n/a	No regression across 66,257 pairs.
Exploratory rank-1 gate	OCR H@1	3	0	+3	0.2500	Insufficient alone because the same rule must transfer.
Exploratory rank-1 gate	Synthetic H@1	27	138	-111	< 0.0001	Reject; losses exceed gains by more than five to one.

Table 56: Gains and losses compare the same cases with Algorithm 5. Exact two-sided binomial tests use only discordant pairs; zero discordant pairs have no defined test.

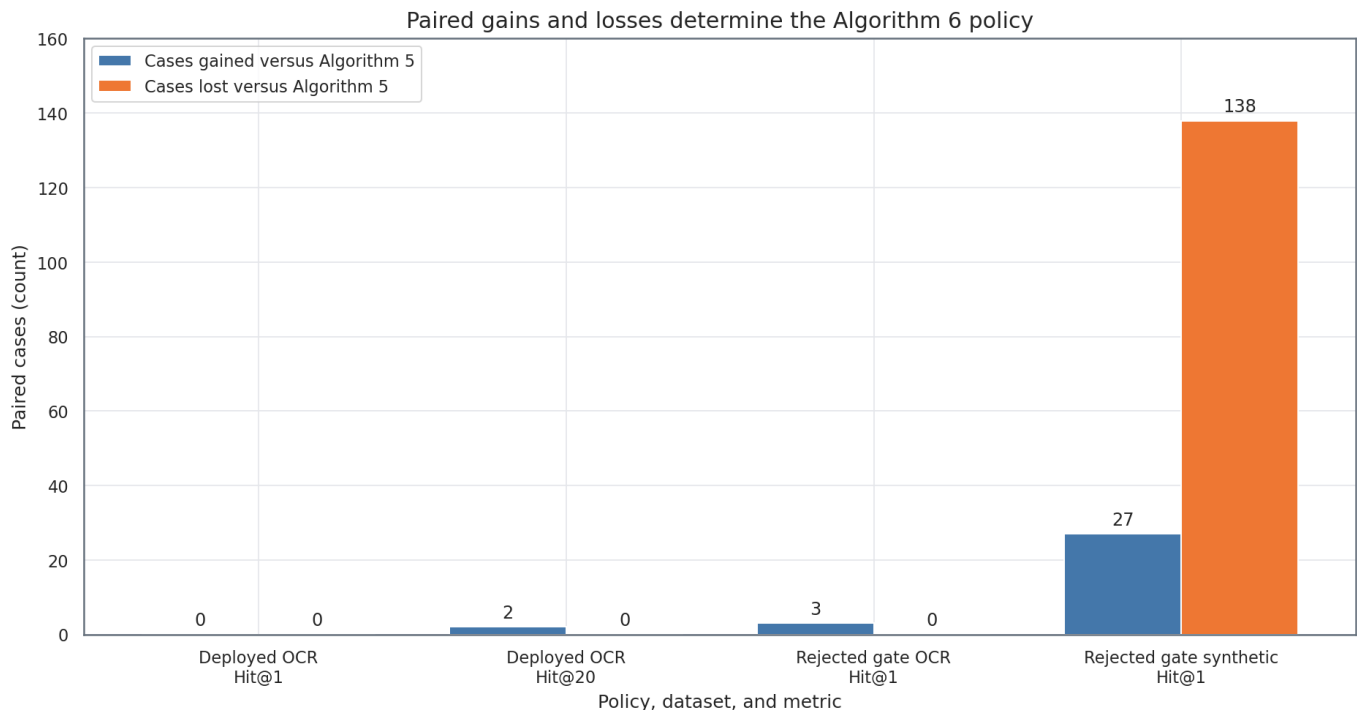


Figure 44: Each x-axis group names a policy, dataset, and metric; the y-axis counts paired gains and losses versus Algorithm 5. Blue bars are gains and orange bars are losses. The large synthetic loss bar is why the rank-1 gate is not deployed.

Insights

1. The deployed policy changes only OCR Hit@20, with two gains and zero losses.
2. The broader rank-1 rule does not transfer: 27 synthetic gains accompany 138 losses.

Input	Supplied target	Unchanged top 1	A5 rank	A6 rank	Interpretation
LACTULOSE	LACTO	LACTULOSE HEK	> 20	20	Candidate union adds the supplied family, but this is a P1 row and LACTULOSE is closer to other catalog names; crop review is required.
TEFA	TELFAS	TERA	> 20	20	Candidate union adds the supplied family, but this is P0 because TERA is two edits closer; crop review is required.

Table 57: These are the only two deployed top-20 gains. They prove retrieval change, not supplied-label validity.

Nested component removed	H@1	Δ H@1	H@20	Δ H@20	H@1 G/L	H@20 G/L	Representative effect
Family rescue	36.638%	-13.793	56.034%	-17.241	5/69	3/83	TUTORIALAE→KETOROLAC: rank 7 to > 20.
All deterministic reranking	40.517%	-9.914	70.690%	-2.586	1/47	0/12	LANYU→LANTUS: rank 2 to > 20.
Weighted edit similarity	49.569%	-0.862	66.164%	-7.112	4/8	2/35	Largest scoring-signal loss at top 20.
Raw edit similarity	49.353%	-1.078	67.888%	-5.388	1/6	2/27	Removes uniform edit scoring.
Short-edge retrieval	49.784%	-0.647	70.259%	-3.017	0/3	0/14	RICOHIL→RIVOTRIL: rank 3 to > 20.
Length coverage	50.216%	-0.216	70.259%	-3.017	1/2	1/15	Visible-length support affects candidate retention.
Family-head rescue	48.060%	-2.371	70.905%	-2.371	1/12	0/11	LANYU→LANTUS: rank 2 to > 20.
Delete-key retrieval	48.276%	-2.155	71.121%	-2.155	1/11	0/10	Missing-character candidate lookup.
Prefix evidence	48.707%	-1.724	71.336%	-1.940	3/11	0/9	RIVITAM→RIVOTRIL: rank 17 to > 20.
Positional evidence	48.922%	-1.509	71.336%	-1.940	1/8	1/10	TEFAXATE→TELFAST: rank 16 to > 20.
Character n-grams	48.060%	-2.371	71.983%	-1.293	0/11	2/8	Local pieces preserve evidence around OCR errors.
External retriever	49.784%	-0.647	72.414%	-0.862	3/6	0/4	ABDUSONG LAR→ABASAGLAR: rank 1 to > 20.
Weighted confusion costs	48.707%	-1.724	72.845%	-0.431	1/9	3/5	KETOMAN→KATOMOR: rank 14 to > 20.
Phonetic evidence	49.569%	-0.862	73.276%	0.000	0/4	0/0	ECOFIXIM→CEFIXIME: rank 1 to rank 2.
Short visible-head rescue	50.431%	0.000	73.060%	-0.216	0/0	0/1	LANS→LANTUS: rank 20 to > 20.

Table 58: Paired OCR ablation over 464 cases. G/L means ablation gains/losses against complete Algorithm 6. Positive gains do not cancel the need to report losses, and row effects are not additive because components overlap.

Configuration	OCR H@1	OCR H@20	Top-20 misses	Observed change
Complete Algorithm 6	50.4310%	73.2759%	124	Reference.
Without bounded consensus slot	50.4310%	72.8448%	126	Removes both gains and becomes Algorithm 5.
Without Jaro-Winkler	50.4310%	73.0603%	125	Loses one of two recovered rows.
Without BM25Plus character trigrams	50.4310%	73.0603%	125	Loses one of two recovered rows.
Without RapidFuzz token-sort ratio	50.4310%	73.0603%	125	Loses one of two recovered rows.
Without Dice character bigrams	50.4310%	73.0603%	125	Loses one of two recovered rows.
Without Match Rating, WRatio, NYSIIS, SymSpell, or Soundex, one at a time	50.4310%	73.2759%	124	No top-k change on these 464 pairs.

Table 59: One-source effects are not additive because the gate requires agreement. Every source removal preserves the synthetic top-k metrics; zero delta on this sample is not proof that a source is universally unnecessary.

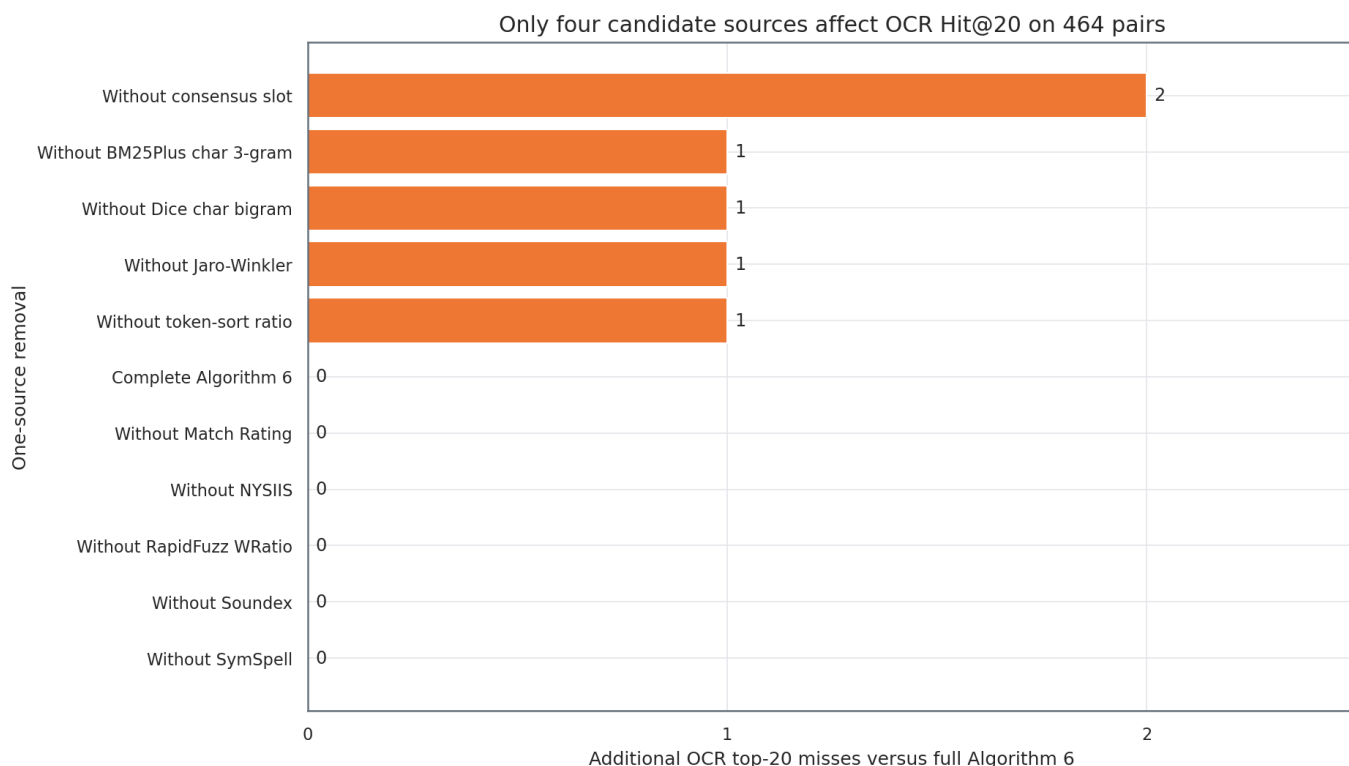


Figure 45: The y-axis names the complete system or one removed candidate source; the x-axis counts additional OCR top-20 misses relative to complete Algorithm 6. Zero means the removal does not change H@20 on these 464 pairs.

Insight

1. Jaro-Winkler, BM25Plus trigrams, token-sort ratio, and Dice bigrams each support one recovered row; removing the complete slot loses both.

Operation	OCR $\rightarrow$ target	Count	Cost	Meaning
Substitution	E $\rightarrow$ C	20	0.3043	Most repeated directional substitution in eligible development rows.
Substitution	N $\rightarrow$ R	11	0.3714	OCR N can replace target R in the aligned evidence.
Substitution	A $\rightarrow$ O	9	0.4000	Direction matters; A-to-O and O-to-A are learned separately.
Substitution	O $\rightarrow$ A	8	0.4182	Reverse of the previous row with its own frequency.
Substitution	U $\rightarrow$ O	8	0.4182	Repeated vowel-shape confusion.
Substitution	D $\rightarrow$ L	7	0.4400	Repeated shape/segmentation confusion in aligned strings.
Substitution	F $\rightarrow$ T	7	0.4400	Repeated vertical-stroke confusion.
Missing target character	empty $\rightarrow$ T	12	0.5600	OCR omitted a target T.
Extra OCR character	O $\rightarrow$ empty	11	0.5679	OCR inserted an O not present in the target.

Table 60: The table reports corpus-derived operations, not medicine-name rules. Holdout rows contribute zero counts. The learned distance is retained as advisory evidence because deploying it as a general rank override was not validated.

Configuration	Dev. H@1	Dev. H@20	Retro. H@1	Retro. H@20
Deployed deterministic Algorithm 6	50.9434%	71.6981%	48.3871%	79.5699%
Full cross-fitted LightGBM reranker	50.1348%	68.7332%	46.2366%	77.4194%
Without retriever-rank features	39.8922%	68.7332%	41.9355%	78.4946%
Without edit features	50.1348%	67.3854%	47.3118%	78.4946%
Without n-gram and edge features	49.3261%	68.1941%	47.3118%	77.4194%
Without phonetic features	49.8652%	69.0027%	47.3118%	77.4194%
Without length and position features	50.1348%	69.8113%	46.2366%	76.3441%

Table 61: The learned reranker never beats deployed Algorithm 6 on a listed split and metric. Retriever-rank features supply its largest H@1 contribution, but ablation directions vary between development and retrospective holdout. The model remains advisory.

12.8 Learned Reranker Ablation

12.9 Selective Correctness Gate

The logistic gate uses target-family cross-fitted development predictions. It accepts a result only when predicted correctness is at least 0.63, at least seven candidate sources support the first result, and top-1 Levenshtein similarity is at least 0.70. The threshold maximizes P2 development coverage subject to zero observed development errors and at least five accepted rows.

Evaluation subset	Cases	Accepted	Coverage	Correct	Errors	Selective accuracy
Cross-fit development, all labels	371	128	34.5013%	99	29	77.3438%
Cross-fit development, P2 only	211	99	46.9194%	99	0	100.0000%
Retrospective holdout, all labels	93	36	38.7097%	27	9	75.0000%
Retrospective holdout, P2 only	59	27	45.7627%	27	0	100.0000%

Table 62: The P2 Wilson 95% lower bounds are 96.2645% in development and 87.5441% in retrospective holdout. “Accepted” means likely match requiring confirmation, not automatic dispensing or clinical certification.

12.10 Remaining Failures and Runtime Fidelity

Of the 124 OCR top-20 misses, 97 supplied targets are absent from all ten source top-20 lists. Another 26 appear in only one or two sources and fail at least one similarity threshold. One target appears in four sources but fails the 55% Levenshtein gate. No missed target passes every gate and then loses the single external slot, so slot competition is not the current failure mechanism.

Table 63: Ten representative Algorithm 6 OCR top-20 failures. Source support is the number of the ten source top-20 lists containing the supplied target. The gate requires at least three sources, Levenshtein similarity at least 0.55, and Jaro–Winkler similarity at least 0.85.

OCR input	Supplied target	A6 top 1	Target support	Why target stays out	What the case shows
CINNARIZIN	MYOLAX	CINNARIZINE	0/10	Absent from every source.	Returned family is one edit away; target is nine edits away. This is likely a label conflict, not a retrieval rule to learn.
H	RIVO	no result	0/10	Absent from every source.	One visible character carries too little evidence for safe family recovery.
HOAVACH	AMIKACIN	HYABAK	0/10	Absent from every source.	Six edits separate input and target; broader lexical reranking cannot help until retrieval finds the family.

OCR input	Supplied target	A6 top 1	Target support	Why target stays out	What the case shows
LANSOPRAZOLE	LANTUS	PANTOPRAZOLE	1/10, Jaro rank 14	Source count, Lev 0.333, Jaro 0.811.	The input is two edits from PANTOPRAZOLE and nine from the supplied target, indicating a source-label conflict.
OSCAR	OSTOCAL	ASCOR	2/10, Jaro 14; SymSpell 19	Source count and Jaro 0.832; Lev 0.571 passes.	A borderline target reaches source tails, but independent agreement and prefix-sensitive similarity remain weak.
RIV	RIVOTRIL	RIVO	2/10, Jaro 12; Dice 16	Source count and Lev 0.375; Jaro 0.854 passes.	The visible fragment also exactly prefixes catalog family RIVO; the correct product needs clarification, not forced ranking.
NEUROTROPHIC	NEUROCET	NEUROTOP	4/10, best rank 4	Lev 0.417; source count and Jaro 0.850 pass.	Four sources can repeat the same weak lexical signal. The edit gate blocks consensus without enough character agreement.
ABACAVIR	ABASAGLAR	TECAVIR	1/10, Soundex rank 2	Source count, Lev 0.316, Jaro 0.653.	A coarse phonetic code alone retrieves the target, but direct spelling evidence favors other families.
LANLTA	LANTUS	LANETAL	2/10, Jaro 18; WRatio 20	Source count, Lev 0.333, Jaro 0.806.	The target exists at source-list tails but is too weak for the bounded slot. This needs stronger retrieval evidence, not a lower global gate.
TOCILIZUMAB	TOCO	ALIZOLAM	1/10, WRatio rank 3	Source count, Lev 0.273, Jaro 0.674.	The supplied target discards most visible characters; image and label review must precede algorithm tuning.

The synthetic pair `aclox`→`ADOX` remains Algorithm 6’s only top-20 miss among 66,257 pairs. Removing family-head rescue recovers it but loses 11 OCR top-20 cases, so the evidence supports a narrower family-head guard rather than deleting that component.

Surface	Verification	Current limitation
Python Algorithm 6 runtime	Exact rank, top-1, and top-20 agreement on all 464 OCR pairs versus cached replay	This is the authoritative full consensus implementation.
Stored OCR and synthetic artifacts	464 and 66,257 aligned rows with per-case Algorithm 5, Algorithm 6, and rejected-gate ranks	Historical holdout is retrospective and source crops are not linked.
Static browser application	Repeated searches, empty input, unreadable-before/middle/after states, and confirmation requirements pass the Node regression test	GitHub Pages cannot execute the Python indexes; it preserves browser Algorithm 5 ranking and exposes Algorithm 6 safety semantics, not full consensus retrieval.
Selective output	Every result requires confirmation; no result is labeled clinically certain	Pharmacist workflow value remains unmeasured until Experiment 7 is executed.

Table 64: Runtime parity applies to the Python system. The static browser fallback must not be reported as a full Algorithm 6 deployment.

Zero observed gate errors occur only in provisional P2 subsets

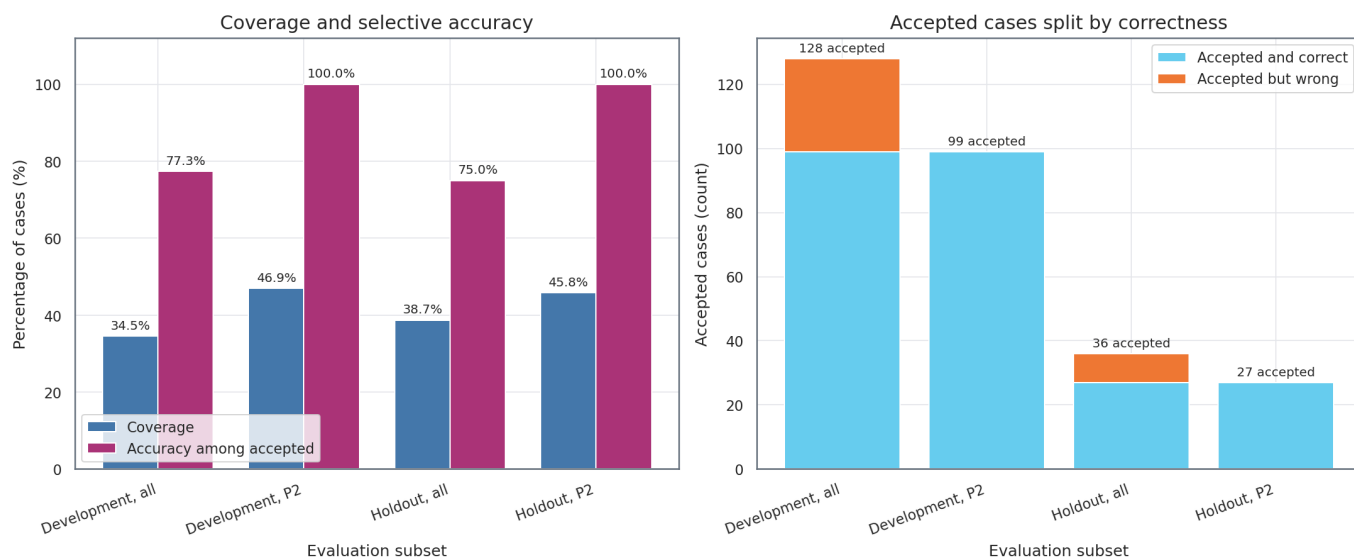


Figure 46: The left panel compares coverage and accuracy among accepted cases; the right panel separates accepted correct and wrong counts. The x-axis distinguishes all supplied labels from provisional P2 labels in cross-fit development and retrospective holdout.

Insights

1. The gate has zero observed P2 errors at about 46% coverage in both splits.
2. Accuracy falls to 75–77% when P0/P1 labels are included. Label audit changes the meaning of calibration and cannot be omitted.

The 124 remaining OCR misses are concentrated in severe corruption

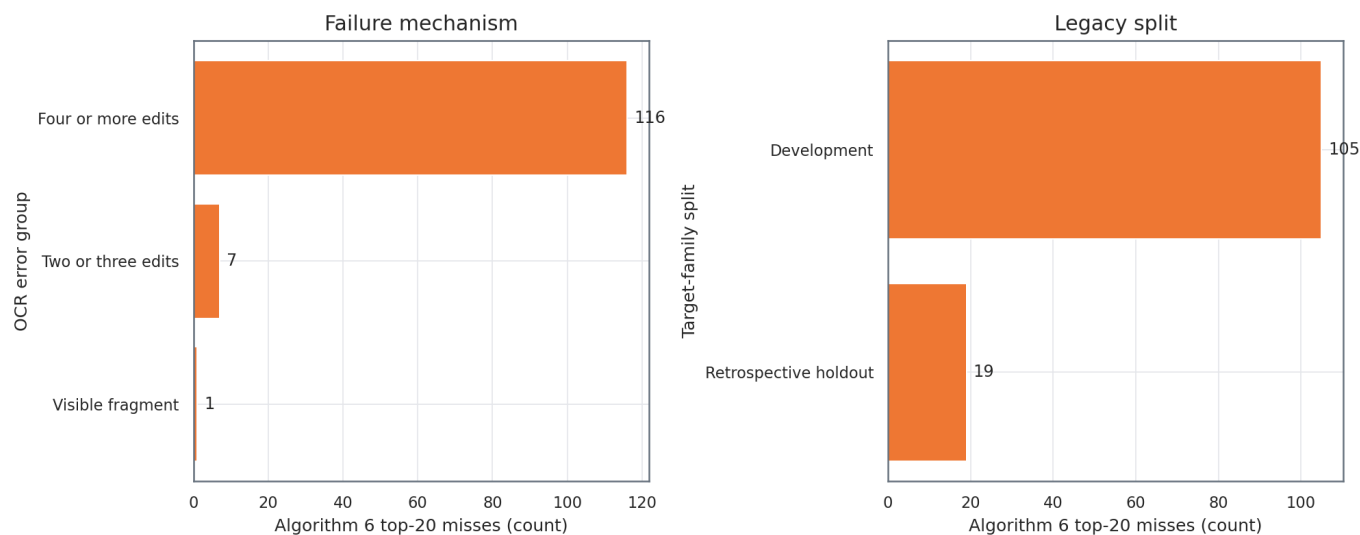


Figure 47: Both panels use Algorithm 6's 124 OCR top-20 misses. The left groups misses by compact OCR error mechanism; the right groups them by the historical target-family split. Bar length is miss count, not failure rate.

Insights

1. Four-or-more-edit corruption accounts for 116 of 124 misses; seven are two-to-three-edit rows and one is a visible fragment.
2. Development contains 105 misses and retrospective holdout contains 19; the split difference reflects target and severity mix, not only model quality.

13 Experiment 7: Pharmacist Study Status

The within-subject study is fully assigned but remains a protocol, not an outcome experiment.

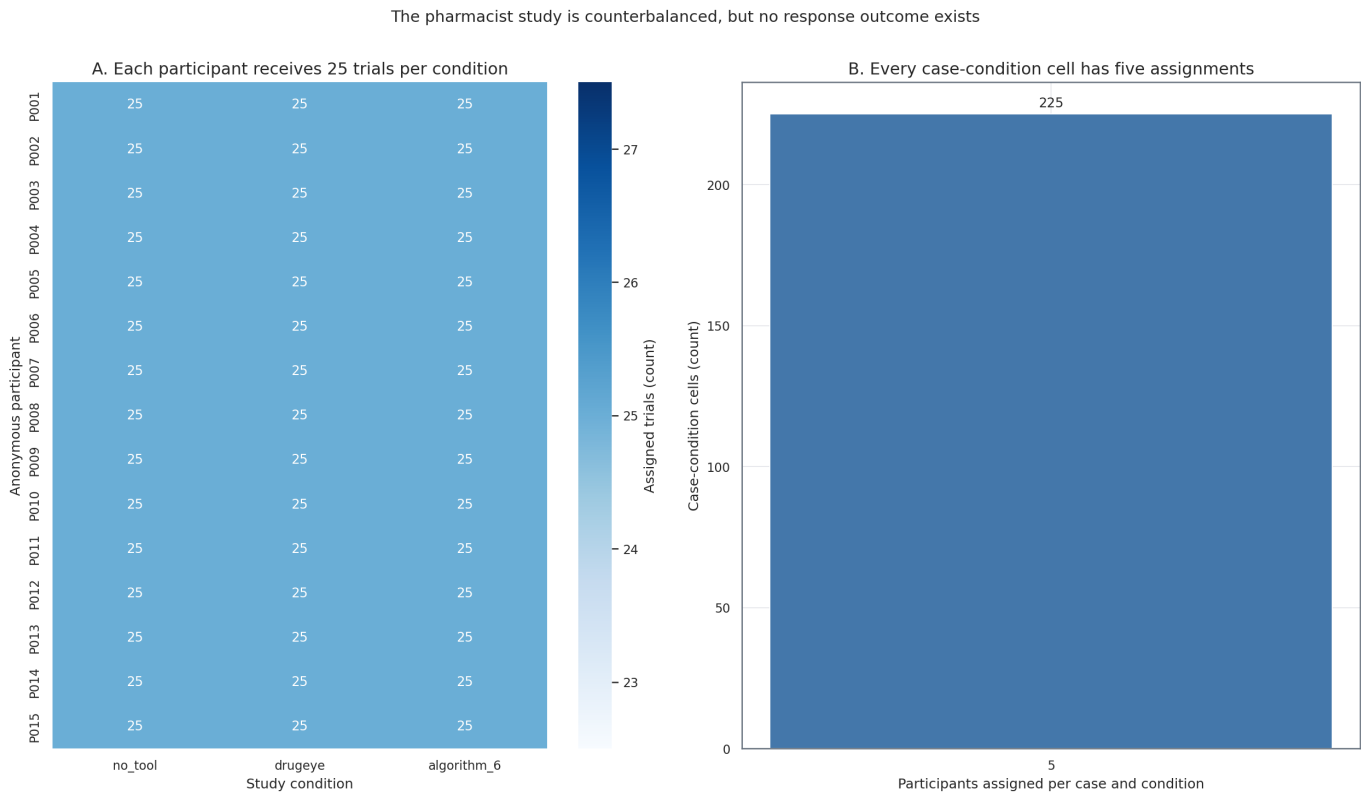


Figure 48: The left heatmap counts trials per participant and condition; the right counts presentations per case and condition. Conditions are no tool, DrugEye, and Algorithm 6. Colors encode assignment count, not performance; the validated schedule contains no participant outcomes.

Insights

1. Each of 15 participants receives 25 cases in each condition, 75 total. (The schedule contributes 375 decisions per condition and 1,125 overall.)
2. Each of 75 crops appears five times per condition and only once per participant. (The design balances condition exposure while avoiding a participant seeing the same crop twice.)
3. No accuracy, time, confidence, or unsafe-selection result exists. (Execution still requires de-identified crops, two-pharmacist target adjudication, ethics review, frozen DrugEye captures, and responses.)

Study element	Locked Experiment 7 design
Participants	15 pharmacists or senior pharmacy students, inside the planned 10–20 range.
Cases	75 deidentified prescription word crops, inside the planned 50–100 range; two independent pharmacists must adjudicate each answer before execution.
Conditions	No tool/current practice; frozen DrugEye; frozen Algorithm 6. Tool commit, catalog hash, deployment URL, and image checksum fields are part of the response contract.
Within-subject exposure	Every participant completes all three conditions, 25 different crops per condition, and never sees one crop twice.
Randomization	Case order is randomized per participant; condition assignment is counterbalanced so each crop receives five decisions per condition.
Valid actions	Select medicine; <code>cannot_decide</code> ; <code>call_doctor</code> . The last two are safe actions and the interface cannot force a medicine.
Primary outcomes	Correct selection rate and unsafe wrong-selection rate.
Secondary outcomes	Safe deferral rate, decision time, confidence, and usability.
Analysis unit	Participant-case-condition decision with participant and case effects retained; condition comparisons are paired within participant.
Current status	1,125 balanced Algorithm 6 assignments pass preflight validation; zero participant responses. Ethics approval or exemption, deidentified crops, answer adjudication, and frozen tool captures remain prerequisites.

Table 65: Experiment 7 is a complete protocol, not a completed user study. The three conditions must use the same interface and candidate-display capacity so the search source is the controlled difference.

14 Failures and Dead Ends

These findings identify work that failed, remains unsupported, or would produce misleading claims if reported as success.

- **The remaining OCR misses are not clean optimization targets.** Algorithm 6 has 124 top-20 misses; 116 are P0 and eight are P1. (No P2 row is missed, but P2 selection itself uses candidate-union presence, so this is an audit result rather than unbiased accuracy.)
- **Both Algorithm 6 gains still need crop review.** LACTULOSE→LACTO is P1 and TEFA→TELFAS is P0. (The system demonstrably retrieves the supplied target, but the absent images prevent a claim that the supplied target is valid.)
- **The learned ranker does not beat deterministic Algorithm 6.** Its cross-fit development Hit@20 is 68.7332% versus 71.6981%, and its retrospective holdout Hit@20 is 77.4194% versus 79.5699%. (It remains an analysis model, not a deployed override.)
- **The rank-1 consensus gate fails transfer.** It gains 27 and loses 138 synthetic rank-1 cases. (Three zero-loss OCR gains are not enough to justify a rule that regresses the larger locked set.)
- **Synthetic accuracy overstates observed-OCR transfer.** Edit-distance variants exceed 94% Hit@1 on generated clean-core edits but remain below 39% on the fair OCR pairs. (Controlled mutation rules do not reproduce the complete observed corruption distribution.)
- **Autocorrect artifacts contain insufficient signal.** Algorithm 4 reaches 0% Hit@1 and 3.65% Hit@20. (Mappings such as a medicine to an unrelated English word require image evidence, a language model with constraints, or human clarification, not ordinary fuzzy matching.)
- **Keyboard-shift results are feature-specific.** Algorithm 4 reaches 0.8% Hit@1 and 2.0% Hit@20. (Algorithms 1 and 3 score much higher because they contain a whole-keyboard shift reversal; the category tests that decoder, not general spelling robustness.)
- **Generic equal-distance rules did not transfer.** Edge-first improved development by 0.54 points but reduced holdout by 3.23 points. (No policy replaced current ordering.)
- **Most atomic switches are unproven on 464 pairs.** Only four of 54 atomic Algorithm 5 flags change Hit@1 or Hit@20 on the locked OCR denominator. (Zero top-k delta can still hide lower-rank or safety changes, so these switches require larger targeted tests before removal.)
- **Raw OCR-model ranking is confounded.** Models did not process a matched image set. (Case severity, target distribution, and row count differ by model.)

- **DrugEye has no matched human-study outcome.** Historical web probes and cached sample results do not replace randomized participant decisions. (No clinical or workflow superiority claim is made.)
- **Training dynamics are not defined for this run.** The study evaluates frozen string scorers, indexes, and published dense checkpoints without fine-tuning. (There are no epochs, losses, gradients, learning-rate schedules, seeds, or early-stopping events to plot.)
- **The selective gate is not clinical calibration.** It models top-1 correctness under historical supplied labels and reaches zero observed errors only on provisional P2 rows. (Its output remains “likely match, confirmation required”; probability calibration across all systems or clinical outcomes is still unsupported.)
- **The static browser is not the full Python Algorithm 6.** GitHub Pages cannot execute its consensus indexes. (The browser retains Algorithm 5 ordering with Algorithm 6 safety states until a backend or exported equivalent is deployed.)
- **GPU-hour and peak-memory efficiency were not logged.** The reproducible efficiency fields are one-time preparation time and warm-query latency on the recorded machine. (A deployment benchmark must add process-level CPU, memory, energy, and concurrent-throughput instrumentation.)

15 Open Questions and Next Steps

1. Recover or relink the original crops for all 194 P0/P1 pairs, then have two pharmacists adjudicate target family, visible evidence, and valid ambiguity before any further ranking rule is learned from them.
2. Freeze a genuinely fresh target-family-disjoint crop set. Do not reuse the 93-pair retrospective holdout for another deployment decision.
3. Execute the prepared within-subject pharmacist study after ethics approval or exemption, answer adjudication, frozen DrugEye captures, and frozen Algorithm 6 deployment metadata.
4. Expose the authoritative Python Algorithm 6 through a backend or export an exactly validated browser implementation; rerun the 464-case parity test against the deployed surface.
5. Expand matched OCR data before reconsidering the LightGBM reranker or zero-delta candidate sources. Require paired gains on fresh labels, synthetic non-regression, and no unsafe-confirmation regression.
6. Run all 14 OCR systems on the same deidentified crop set with one preprocessing contract, then report OCR transcription accuracy and downstream Algorithm 6 recovery as separate outcomes.

16 Analysis Coverage and Reproducibility

The final audit distinguishes completed analyses from analyses the available files cannot support.

- The analysis script reads every designated source in full, validates 595 OCR observations, 464 primary pairs, 66,257 clean-core pairs, the preserved 115,000-row diagnostic benchmark, and zero target-family split overlap.
- It regenerates the source inventory, 356-entry data dictionary, aggregate metrics, paired statistics, interaction tables, overlap tables, retrieval error profiles, Algorithm 6 audit and calibration views, fair-data views, and every report figure from stored artifacts.
- The PDF is compiled twice so table-of-contents entries, references, and float numbering resolve before delivery.

Requested analysis family	Status	Evidence or reason
Performance summary and ranking	Complete	Fifty-three comparison systems plus paired Algorithm 6 on 464 OCR and 66,257 synthetic pairs, plus the preserved 115,000-row diagnostic benchmark.
Paired significance and effects	Complete	Bootstrap intervals, exact McNemar tests with Holm correction, gains/losses, and matched odds ratios.
Training dynamics	Unsupported	Algorithms are rule-based retrieval systems; no epochs, losses, gradients, seeds, or learning-rate logs exist.
Controlled comparisons	Complete	Algorithm 4 and Algorithm 5 ablations, external composite removals, parameter sensitivity, and equal-distance policies.
Distribution and error analysis	Complete	Distance, operations, evidence, length, cohorts, models, targets, categories, and worst examples.
Selective correctness calibration	Partial	Algorithm 6 has target-family cross-fitted logistic scores and a locked threshold, but no fresh human-adjudicated calibration set or clinical outcome calibration.
Confusion matrices	Unsupported	Retrieval emits ranked families over a large catalog rather than one shared fixed classification output for every compared system.
Ablation and sensitivity	Complete	Ninety-seven Algorithm 5 OCR configurations, 13 synthetic confirmations, 24 Algorithm 4 configurations, external composite branches, ten Algorithm 6 source removals, and five learned-ranker feature-group removals.
Correlation and interactions	Complete	Spearman matrix and edit-distance by bigram interaction heatmap.
Compute efficiency	Partial	Warm latency, preparation time, latency distributions, Pareto frontier, package versions, and host CPU/memory are recorded; no per-query memory, GPU-energy, or power trace exists.
Human outcomes	Unsupported	Balanced assignments exist, participant responses do not.
Leakage and split checks	Complete with limitation	Zero verified-family overlap, five target-family cross-fit folds, and aligned synthetic rows; the historical holdout is explicitly retrospective, not fresh blind.
Label reliability	Automated audit complete	All 464 pairs receive P0/P1/P2 priority and nearest-catalog evidence; zero source images are linked, so human adjudication is pending.

Table 66: Unsupported analyses are omitted rather than filled with mock values or inferred outcomes. The report generator fails if key row counts or split contracts do not reconcile.